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HANDHELD SURFACE CLEANER

Abstract

Various illustrative handheld surface cleaners and methods of using handheld surface cleaners are provided. In an exemplary implementation, the handheld surface cleaner is configured to use suction to draw debris into the handheld surface cleaner. The suction also causes air to be drawn into the handheld surface cleaner. The handheld surface cleaner is configured to separate the air and the debris on board the handheld surface cleaner to allow the air to exit the handheld surface cleaner and to allow the debris to be collected in the handheld surface cleaner for later disposal.

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS [0001] The present application is a continuation of PCT Patent Application No. PCT/CN2024/134016 entitled “Handheld Surface Cleaner” filed Nov. 25, 2024, which claims priority to U.S. Provisional Patent Application No. 63/560,916 entitled “Cleaning Apparatus” filed Mar. 4, 2024, which are hereby incorporated by reference in their entireties.

FIELD

[0002] The present disclosure generally relates to handheld surface cleaners.

BACKGROUND

[0003] Surface cleaning apparatuses (also referred to herein as “surface cleaners”) are configured to be maneuvered over a surface to be cleaned (e.g., a floor, upholstery, etc.). While being maneuvered over the surface to be cleaned, the surface cleaning apparatus is configured to collect in the surface cleaning apparatus debris deposited on and/or in the surface to be cleaned. The debris can include liquid, hair, dirt, and/or other matter. The debris can later be disposed of as waste from the surface cleaning apparatus.

[0004] Water and/or other cleaning liquid can be applied to the surface to be cleaned to help clean the surface using the surface cleaning apparatus. The debris collected by the surface cleaning apparatus can include excess water and/or other cleaning liquid that was applied to the surface.

[0005] The surface cleaning apparatus can be configured to use suction in drawing the debris into the surface treatment apparatus. The suction may also help free, dislodge, or otherwise facilitate collection of the debris by the surface treatment apparatus.

[0006] In instances in which the surface cleaning apparatus applies suction and collects liquid, both air and liquid can be drawn into the surface cleaning apparatus in addition to any other debris. The air drawn into the surface cleaning apparatus should be allowed to escape from the surface cleaning apparatus while the surface cleaning apparatus is in use since it is not feasible to contain all the drawn-in air within the surface cleaning apparatus. However, it can be difficult to allow for air to be released from the surface cleaning apparatus while the surface cleaning apparatus is in use without debris also being released from the surface cleaning apparatus, which is messy and undesirable.

SUMMARY

[0007] In general, handheld surface cleaners and methods of using handheld surface cleaners are provided.

[0008] In one aspect, a cleaning system is provided that in one implementation includes a handheld surface cleaner. The handheld surface cleaner includes an inlet through which debris and air are allowed to enter the handheld surface cleaner, a main body, a tank, a separator, a separator motor, and a suction motor. Debris and air are configured to enter the handheld surface cleaner through the inlet. The tank is configured to contain and store therein the debris. The separator is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The separator has an open proximal end through which the air exits the separator. The separator has a plurality of holes in a sidewall of the separator through which the debris exits the separator. The separator motor is configured to drive rotation of the separator. The rotation of the separator is configured to direct the debris through the plurality of holes. The suction motor is configured to provide a suction force to draw the debris and air into the inlet and into the separator.

[0009] The cleaning system can have any number of variations. For example, the plurality of holes can be configured to be open with the separator rotating, and the plurality of holes can be configured to be closed with the separator not rotating. Further, each of the plurality of holes can have an associated cover that is configured to move automatically from an active configuration, in which the covers do not cover their associated holes, and a resting configuration, in which the covers cover their associated holes; and/or the handheld surface cleaner can also include a centrifugal seal upstream of the separator. Further, the rotation of the separator can be configured to cause the covers to move from the resting configuration to the active configuration, and the separator stopping rotating can be configured to cause the covers to move from the active configuration to the resting configuration.

[0010] For another example, the plurality of holes can be open regardless of whether the separator is rotating. Further, the handheld surface cleaner can also include a duckbill valve upstream of the separator.

[0011] For yet another example, the handheld surface cleaner can also include a first filter upstream of the separator, and the handheld surface cleaner can also include a second filter downstream of the separator.

[0012] For still another example, the handheld surface cleaner can also include a controller configured to control the suction motor and the separator motor based on a measured current of the separator motor. Further, the controller can be configured to cause the suction motor and the separator motor to be powered off in response to the measured current being greater than a predetermined threshold current value. Further, the controller can be configured to cause the suction motor to be powered off before the separator motor is powered off in response to the measured current being greater than a predetermined threshold current value. Further, the handheld surface cleaner can also include a current sensor operably coupled to the controller and configured to measure the current of the separator motor.

[0013] For yet another example, the handheld surface cleaner can also include a shaft operatively coupling the separator and the separator motor with an outer bearing and an inner plug, and the outer bearing and the inner plug can be keyed. Further, the inner plug can be formed of rubber.

[0014] For another example, the tank can be configured to releasably couple to the main body; with the tank releasably coupled to the main body, the suction motor and the separator motor can be allowed to be turned on; and with the tank not being releasably coupled to the main body, the suction motor and the separator motor can be disabled from being turned on. Further, the handheld surface cleaner can also include a switch that is configured to be activated and allow power to be provided to the suction motor and the separator motor with the tank releasably coupled to the main body and that is configured to be deactivated and prevent power from being provided to the suction motor and the separator motor with the tank not releasably coupled to the main body; and/or the handheld surface cleaner can also include a controller; the controller can be configured to determine whether the tank is releasably coupled to the main body; the controller can be configured to, in response to determining that the tank is releasably coupled to the main body, allow the suction motor and the separator motor to be powered on; and the controller can be configured to, in response to determining that the tank is not releasably coupled to the main body, prevent the suction motor and the separator motor from being powered on.

[0015] For yet another example, a separator assembly of the handheld surface cleaner can include the separator and the separator motor, and the separator assembly can be configured to be removable from the main body with the tank.

[0016] For another example, a separator assembly of the handheld surface cleaner can include the separator and the separator motor, and the separator motor can be configured to be removable from the tank with the main body with the separator remaining coupled to the tank.

[0017] For yet another example, the handheld surface cleaner can include a rechargeable power supply located at the main body. Further, the rechargeable power supply can include a rechargeable

battery; and/or the cleaning system can also include a tray configured to seat the handheld surface cleaner for storage, and, with the handheld surface cleaner seated in the tray, the rechargeable power supply can be configured to be recharged using a charger.

[0018] For another example, the cleaning system can also include a tray configured to seat the handheld surface cleaner for storage.

[0019] For still another example, the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the cleaning system can also include a tray configured to seat simultaneously the handheld surface cleaner and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0020] For another example, the tank can be configured to releasably couple to the main body.

[0021] For still another example, the cleaning system can also include a tray configured to seat the handheld surface cleaner. Further, the handheld surface cleaner can include a rechargeable power supply located at the main body, and the rechargeable power supply can be configured to be recharged with the tray seating the handheld surface cleaner. Further, the rechargeable power supply can include a rechargeable battery, and/or the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the tray can be configured to seat simultaneously the handheld surface cleaner and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0022] For yet another example, the cleaning system can also include a spray bottle including a spray head and a dual cleaning solution container configured to be releasably coupled to the spray head, and the spray head can be configured to expel a liquid cleaning solution to a surface to be cleaned by the handheld surface cleaner. Further, the cleaning system can also include a tray configured to seat simultaneously the handheld surface cleaner and the spray bottle. Further, the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the tray can be configured to seat simultaneously the handheld surface cleaner, the spray bottle, and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0023] In another implementation, a cleaning system is provided that includes a handheld surface cleaner that includes an inlet, a tank, a separator, a separator motor, a suction motor, and a controller. Debris and air are allowed to enter the handheld surface cleaner through the inlet. The tank is configured to contain and store therein the debris. The separator is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The separator motor is configured to drive rotation of the separator. The rotation of the separator is configured to direct the debris toward the tank. The suction motor is configured to provide a suction force to draw the debris and air into the inlet and into the separator. The controller is configured to control the suction motor and the separator motor based on a measured current of the separator motor.

[0024] The cleaning system can include any number of variations. For example, the controller can be configured to cause the suction motor and the separator motor to be powered off in response to the measured current being greater than a predetermined threshold current value. Further, the controller can be configured to cause the suction motor to be powered off before the separator motor is powered off in response to the measured current being greater than the predetermined threshold current value, or the controller can be configured to cause the suction motor and the separator motor to be powered off simultaneously in response to the measured current being greater than the predetermined threshold current value.

[0025] For another example, the handheld surface cleaner can also include a current sensor operably coupled to the controller and configured to measure the current of the separator motor.

[0026] For another example, the separator can have an open proximal end through which the air exits the separator, and the separator can have a plurality of holes in a sidewall of the separator through which the debris exits the separator. In some implementations, the plurality of holes can be open with the separator rotating, and the plurality of holes can be closed with the separator not rotating. Further, each of the plurality of holes can have an associated cover that is configured to move automatically from an active configuration, in which the covers do not cover their associated holes, and a resting configuration, in which the covers cover their associated holes; and/or the handheld surface cleaner can also include a centrifugal seal upstream of the separator. Further, the rotation of the separator can be configured to cause the covers to move from the resting configuration to the active configuration, and the separator stopping rotating can be configured to cause the covers to move from the active configuration to the resting configuration. In some implementations, the plurality of holes can be open regardless of whether the separator is rotating. Further, the handheld surface cleaner can also include a duckbill valve upstream of the separator.

[0027] For yet another example, the handheld surface cleaner can also include a first filter upstream of the separator, and the handheld surface cleaner can also include a second filter downstream of the separator.

[0028] For another example, the handheld surface cleaner can also include a main body; the controller can be configured to determine whether the tank is releasably coupled to the main body; the controller can be configured to, in response to determining that the tank is releasably coupled to the main body, allow the suction motor and the separator motor to be powered on; and the controller can be configured to, in response to determining that the tank is not releasably coupled to the main body, prevent the suction motor and the separator motor from being powered on.

[0029] For yet another example, a separator assembly of the handheld surface cleaner can include the separator and the separator motor, and the separator assembly can be configured to be removable from the tank.

[0030] For another example, the handheld surface cleaner can also include a main body, a separator assembly of the handheld surface cleaner can include the separator and the separator motor, and the separator motor can be configured to be removable from the tank with the main body, the separator remaining coupled to the tank.

[0031] For yet another example, the handheld surface cleaner can include a power supply. Further, the power supply can include a battery. Further, the battery can include a rechargeable battery.

[0032] For still another example, the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the cleaning system can also include a tray configured to seat simultaneously the handheld surface cleaner and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0033] For another example, the tank can be configured to releasably couple to the main body.

[0034] For still another example, the cleaning system can also include a tray configured to seat the handheld surface cleaner. Further, the handheld surface cleaner can include a rechargeable power supply, and the rechargeable power supply can be configured to be recharged with the tray seating the handheld surface cleaner. Further, the rechargeable power supply can include a rechargeable battery, and/or the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the tray can be configured to seat simultaneously the handheld surface cleaner and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0035] For yet another example, the cleaning system can also include a spray bottle including a

spray head and a dual cleaning solution container configured to be releasably coupled to the spray head, and the spray head can be configured to expel a liquid cleaning solution to a surface to be cleaned by the handheld surface cleaner. Further, the cleaning system can also include a tray configured to seat simultaneously the handheld surface cleaner and the spray bottle. Further, the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the tray can be configured to seat simultaneously the handheld surface cleaner, the spray bottle, and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0036] In another implementation, a cleaning system includes a body of a surface cleaner, a suction motor, a tank, a separator, and a lid. The suction motor is configured to provide a suction force to draw debris and air into the surface cleaner. The tank is coupled to the body and includes a cavity configured to receive and store the debris drawn into the surface cleaner. The separator is configured to rotate to separate the debris from the air drawn into the surface cleaner by directing the debris radially outward into the cavity of the tank. The lid of the tank defines an air flow path along which the air separated from the debris flows from the separator toward an air exit hole formed in the body of the surface cleaner. The lid in a closed position closes an opening of the tank, and the lid in an open position allows the debris to exit the cavity through the opening. The tank is configured to be removed from the body with the lid in the closed position.

[0037] The cleaning system can have any number of variations. For example, the cleaning system can also include a separator motor configured to drive the rotation of the separator, and the separator motor can be configured to be removed from the body with the tank.

[0038] For another example, the cleaning system can also include a separator motor configured to drive the rotation of the separator, and the tank can be configured to be removed from the body without the separator motor.

[0039] For yet another example, the tank can be configured to be removed from the body with the separator in the tank, and the lid moving from the closed position to the open position can be configured to cause the separator to be removed from the tank.

[0040] For still another example, the cleaning system can also include a separator motor configured to drive the rotation of the separator, and the lid moving from the closed position to the open position can be configured to cause the separator motor and the separator to be removed from the tank.

[0041] For another example, the separator can have an open distal end through which the air enters the separator, the separator can have an open proximal end through which the air exits the separator toward the air flow path, and the separator can have a plurality of holes in a sidewall of the separator through which the debris exits the separator.

[0042] For still another example, the cleaning system can also include a centrifugal seal upstream of the separator, and the centrifugal seal can be configured to rotate with the separator. Further, the separator can have a plurality of holes in a sidewall of the separator through which the debris exits the separator, each of the plurality of holes can have an associated cover that is configured to move automatically from an active configuration, in which the covers do not cover their associated holes, and a resting configuration, in which the covers cover their associated holes, the rotation of the separator can be configured to cause the covers to move from the resting configuration to the active configuration, and the separator stopping rotating can be configured to cause the covers to move from the active configuration to the resting configuration; or the separator can have a plurality of holes in a sidewall of the separator through which the debris exits the separator, and the plurality of holes can be open regardless of whether the separator is rotating; and/or the cleaning system can include a valve upstream of the separator, and the valve can be configured to allow the debris and air drawn into the surface cleaner to flow proximally toward the separator and can be configured to prevent the debris and air drawn into the surface cleaner from flowing distally.

[0043] For yet another example, the cleaning system can also include a valve upstream of the separator, and the valve can be configured to allow the debris and air drawn into the surface cleaner to flow proximally toward the separator and can be configured to prevent the debris and air drawn into the surface cleaner from flowing distally.

[0044] For still another example, the cleaning system can also include a first filter upstream of the separator, and a second filter downstream of the separator.

[0045] For yet another example, the tank can be configured to releasably couple to the body, and the cleaning system can also include a controller configured to: determine whether the tank is releasably coupled to the body, in response to determining that the tank is releasably coupled to the body, allow the suction motor to be powered on, and, in response to determining that the tank is not releasably coupled to the body, prevent the suction motor from being powered on.

[0046] For still another example, the cleaning system can also include a rechargeable power supply. Further, the rechargeable power supply can include a rechargeable battery.

[0047] For another example, the cleaning system can also include a tray configured to seat the surface cleaner for storage.

[0048] For yet another example, the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to an inlet of the surface cleaner, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the cleaning system can also include a tray configured to seat simultaneously for storage the surface cleaner and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0049] For another example, the tank can be configured to be removed from the body with the separator in the tank.

[0050] For yet another example, the tank can be configured to be removed from the body with the separator remaining coupled to the body.

[0051] In another implementation, a cleaning system includes a handheld surface cleaner that includes a distal inlet through which debris and air are allowed to enter the handheld surface cleaner, a tank, a separator, a first filter, a first seal, a barrier, and a second filter. The separator is configured to rotate to separate the debris from the air by directing the debris radially outward into the cavity of the tank and directing the air to an air flow path along which the air flows toward an air exit hole of the handheld surface cleaner. The first filter is downstream of the inlet and upstream of the separator, and the first filter is configured to filter debris above a first predetermined size from flowing to the separator. The first seal is downstream of the first filter and upstream of the separator, and the first seal is configured to prevent the debris contained and stored in the tank from reaching the first filter. The barrier is downstream of the inlet and upstream of the air exit hole, and the barrier is configured to prevent the debris contained and stored in the tank from reaching the air flow path. The second filter is downstream of the inlet and upstream of the air exit hole, and the second filter is configured to filter debris above a second predetermined size from flowing to the air exit hole.

[0052] The cleaning system can vary in any number of ways. For example, the barrier can include one of a second seal and a bracket that defines the air flow path.

[0053] For another example, the barrier can include a second seal and can include a bracket that defines the air flow path, and the second seal can be sandwiched between the separator and the bracket.

[0054] For yet another example, the separator and the first seal can be configured to rotate together relative to the first filter, the second filter, and the barrier.

[0055] For still another example, the handheld surface cleaner can also include a hosing in which the first filter is at least partially disposed, the separator and the first seal can be configured to rotate together relative to the hosing, and the rotation of the first seal can be configured to cause the first seal to move a distance proximally relative to the hosing. Further, the first seal can remain at

least partially disposed within the hosing with the first seal rotating, or the first seal can be configured to abut the hosing with the first seal not rotating and configured to not abut the hosing with the first seal rotating.

[0056] For yet another example, the handheld surface cleaner can also include a valve downstream of the inlet and upstream of the separator, and the valve can be configured to allow the debris and air to flow proximally toward the separator and configured to prevent the debris and air drawn into the surface cleaner from flowing distally toward the inlet. Further, the valve can be located entirely distal to the first seal, or the valve can extend proximally into the first seal.

[0057] For another example, the separator can have a plurality of holes in a sidewall of the separator through which the debris exits the separator, each of the plurality of holes can have an associated cover that is configured to move automatically from an active configuration, in which the covers do not cover their associated holes, and a resting configuration, in which the covers cover their associated holes, the rotation of the separator can be configured to cause the covers to move from the resting configuration to the active configuration. Further, the separator stopping rotating can be configured to cause the covers to move from the active configuration to the resting configuration; or the separator can have a plurality of holes in a sidewall of the separator through which the debris exits the separator, and the plurality of holes can be open regardless of whether the separator is rotating.

[0058] For still another example, the handheld surface cleaner can also include a rechargeable power supply. Further, the rechargeable power supply can include a rechargeable battery.

[0059] For another example, the cleaning system can also include a tray configured to seat the handheld surface cleaner for storage.

[0060] For yet another example, the cleaning system can also include a plurality of attachments configured to be selectively coupled one at a time to the inlet, and the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment. Further, the cleaning system can also include a tray configured to seat simultaneously for storage the handheld surface cleaner and the at least two of the stain tool attachment, the crevice tool attachment, and the pet hair tool attachment.

[0061] In another implementation, a cleaning system includes a surface cleaner that includes an inlet, a separator, and a motor. Debris and air re configured to be suctioned into the surface cleaner through the inlet. The separator is proximal to the inlet, has an outlet opening, has a plurality of holes formed in a sidewall of the separator, and has a plurality of covers. Each one of the plurality of covers is associated with at least one of the plurality of holes. The motor is configured to drive rotation of the separator. The rotation of the separator is configured to: cause the plurality of covers to move from a closed position, in which each of the plurality of covers closes the associated at least one of the plurality of holes, to an open position, in which each of the plurality of covers does not close the associated at least one of the plurality of holes; separate the debris from the air by directing the debris radially outward through the plurality of holes; and direct the air out of the separator through the outlet opening. The rotation of the separator stopping is configured to: cause the plurality of covers to move from the open position to the closed position.

[0062] The cleaning system can vary in any number of ways. For example, each of the plurality of covers in the closed position can provide a fluid seal of the associated at least one of the plurality of holes.

[0063] For another example, each of the plurality of covers can be attached to the separator along a single edge of the cover.

[0064] For yet another example, each of the plurality of covers can be flexible. Further, the separator can be rigid.

[0065] For still another example, the outlet opening can be at a proximal end of the separator, the separator can include an inlet opening at a distal end of the separator, and the inlet opening can be configured to allow the debris and air are allowed to enter the separator through the inlet opening.

[0066] For another example, the sidewall of the separator can extend from the inlet opening to the outlet opening.

[0067] For yet another example, wherein the surface cleaner can also include a tank having a cavity configured to contain and store therein the debris that is directed radially outward through the plurality of holes. Further, the surface cleaner can also include an air exit hole configured to allow the air that is directed out of the separator through the outlet opening to exit to external atmosphere.

[0068] For another example, the handheld cleaner can also include at least one of: a first filter upstream of the separator, and a second filter downstream of the separator.

[0069] For yet another example, each one of the plurality of covers can be associated with only one of the plurality of holes.

[0070] In another implementation, a cleaning system includes a surface cleaner including an inlet, a tank, and a rotatable separator. Debris and air is configured to be suctioned into the surface cleaner through the inlet. The tank is configured to contain and store therein the debris. The rotatable separator is proximal to the inlet and is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the surface cleaner. The rotatable separator includes a body and a seal. The body has a plurality of side holes through which the debris is configured to exit the separator for collection in the tank. The seal has an inner passageway through which the debris and air flows toward the body. The seal is configured to prevent the debris in the tank from passing to the inlet.

[0071] The surface cleaner can have any number of variations. For example, the body can have a plurality of side holes through which the debris exits the separator. Further, each of the plurality of holes can have an associated cover that is configured to move automatically from an active configuration, in which the covers do not cover their associated holes, and a resting configuration, in which the covers cover their associated holes, and the covers can be configured to cooperate with the seal to prevent the debris in the tank from passing to the inlet; or the plurality of side holes can be open regardless of whether the rotatable separator is rotating.

[0072] For another example, the rotatable separator and the motor can be configured to be removed as a unit from the tank.

[0073] For yet another example, the rotatable separator can be configured to be removed from the tank without the motor.

[0074] For another example, the rotatable separator can also include a first plurality of arms extend proximally from the body, and the rotatable separator can also include a second plurality of arms extend distally from the body.

[0075] For still another example, the surface cleaner can also include at least one of: a first filter upstream of the separator, and a second filter downstream of the separator.

[0076] For another example, the surface cleaner can also include a motor configured to drive the rotation of the rotatable separator.

[0077] In another implementation, a cleaning system includes a handheld surface cleaner that includes a proximal handle, a distal inlet, a tank, and a separator. The proximal handle defines a first longitudinal axis. Debris and air are allowed to enter the handheld surface cleaner through the distal inlet. The tank includes a cavity configured to contain and store therein the debris. The separator is proximal to the inlet and distal to the handle. The separator is configured to separate the debris and the air to allow the debris to be collected in the cavity and to allow the air to exit the handheld surface cleaner. The first flow path from the distal inlet to the separator defines a second longitudinal axis. The first longitudinal axis is at an angle relative to the second longitudinal axis that is in a range of about 150 degrees to about 170 degrees. The separator is located along the second longitudinal axis.

[0078] The cleaning system can have any number of variations. For example, the handheld surface cleaner can be configured to allow the air and the debris that enters the device to flow along the first flow path and diverge from the first flow path at the separator. Further, the separator can

include an inlet opening through which debris and air can be allowed to enter the separator, an outlet opening through which the air is configured to exit the separator, and a plurality of holes formed in a sidewall of the body and through which the debris is configured to exit the separator. [0079] For another example, the separator can be configured to direct the debris into the tank along a debris flow path that is different from an air flow path along which air flows downstream of the separator. Further, the air flow path downstream of the separator can define a third longitudinal axis that is transverse to the first flow path.

[0080] For yet another example, the separator can be configured to rotate to separate the debris and the air, the cleaning system can also include a motor configured to drive rotation of the separator, and the motor can be located along the second longitudinal axis. Further, the separator can include an inlet opening through which debris and air can be allowed to enter the separator, an outlet opening through which the air is configured to exit the separator, and a plurality of holes formed in a sidewall of the body and through which the debris is configured to exit the separator.

[0081] For still another example, the cleaning system can also include a motor configured to provide a suction force to draw the debris and the air into the inlet, and the motor can be offset from the first and second longitudinal axes.

[0082] For yet another example, the cleaning system can also include a body configured to releasably attach to the tank, the handle can define a proximal portion of the body, and the separator can be disposed in the tank and is configured to be released from the body with the tank.

[0083] For another example, the handle can define a pistol grip.

[0084] For yet another example, the cleaning system can also include a rechargeable battery, the rechargeable battery can be seated in an interior of the handle, the separator can be configured to rotate to separate the debris and the air, and the rechargeable battery can be configured to power rotation of the separator. Further, the separator can be configured to rotate to separate the debris and the air, the cleaning system can also include a motor configured to drive the rotation of the separator, the rechargeable battery can be configured to provide power to the motor, and the motor can be located along the second longitudinal axis; and/or the separator can include an inlet opening through which debris and air can be allowed to enter the separator, an outlet opening through which the air is configured to exit the separator, and a plurality of holes formed in a sidewall of the body and through which the debris is configured to exit the separator.

[0085] In another implementation, a cleaning system includes a handheld surface cleaner that includes an inlet, a cavity, a separator, and a filter. Debris and air are allowed to enter the handheld surface cleaner through the inlet. The cavity is configured to contain and store therein the debris. The separator is configured to separate the debris and the air to allow the debris to be collected in the cavity and to allow the air to exit the handheld surface cleaner. The filter is configured to be coupled to the handheld surface cleaner and be located upstream of the separator. The handheld surface cleaner is configured to releasably couple to a selected one of a plurality of attachments. The handheld surface cleaner is configured to prevent the attachment from being attached to the handheld surface cleaner unless the filter is coupled to the handheld surface cleaner.

[0086] The handheld surface cleaner can vary in any number of ways. For example, the handheld surface cleaner can also include a lockout member, the filter can include an arm configured to engage the lockout member with the filter coupled to the handheld surface cleaner, the lockout member not being engaged with the arm can be configured to prevent the attachment from being attached to the handheld surface cleaner, and the lockout member being engaged with the arm can be configured to allow the attachment to be attached to the handheld surface cleaner. Further, the handheld surface cleaner can also include a housing configured to be removably coupled to the handheld surface cleaner after the filter has been removably coupled to the handheld surface cleaner, and the housing can be configured to releasably couple to the attachment. Further, the distal housing can include at least one of: at least one hook configured to engage at least one groove of the attachment, and at least one groove configured to engage at least one hook of the

attachment.

[0087] For another example, the handheld surface cleaner can also include a housing configured to be removably coupled to the handheld surface cleaner only after the filter has been removably coupled to the handheld surface cleaner, and the housing can be configured to releasably couple to the attachment. For yet another example, the filter can be a singular part.

[0088] For still another example, the filter can include first and second parts movably attached to one another to allow the filter to move between open and closed configurations. Further, the filter can include a lock configured to lock the first and second parts in the closed configuration.

[0089] For yet another example, the handheld surface cleaner can also include a first seal configured to be located upstream of the separator and downstream of the filter. Further, the filter can include a second seal configured to be located upstream of the first seal. Further, the handheld surface cleaner can also include a third seal configured to be located downstream of the first seal.

[0090] For another example, the handheld surface cleaner can also include a motor configured to provide a suction force to draw the debris and air into the inlet and into the separator, and the filter can include a seal configured to move open automatically by the suction force.

[0091] For yet another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator.

[0092] For another example, the handheld surface cleaner can also include a second filter downstream of the separator.

[0093] For still another example, the handheld surface cleaner can also include a tank that has the cavity formed therein, the filter can be configured to be removably attached to the tank, and the handheld surface cleaner can also include a body configured to be removably coupled to the tank. Further, the handheld surface cleaner can also include a motor at the body configured to provide a suction force to draw the debris and air into the inlet and into the separator, and the filter can include a seal configured to move open automatically by the suction force.

[0094] For another example, the cleaning system can also include the plurality of attachments. Further, the plurality of attachments can include at least two of a stain tool attachment, a crevice tool attachment, and a pet hair tool attachment.

[0095] In another implementation, a handheld surface cleaner includes a body, an inlet, a tank, a separator, a separator motor, a power source, a controller, and a switch. Debris and air are allowed to enter the handheld surface cleaner through the inlet. The tank is configured to removably couple to the body and to contain and store therein the debris. The separator is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The separator motor is configured to drive the rotation of the separator. The controller is operatively coupled to the separator motor and is configured to control the separator motor. The switch is operatively coupled to the controller and is configured to be activated automatically in response to the tank being removably coupled to the body. The switch being activated is configured to allow the power source to supply power to the separator motor. The switch being deactivated is configured to prevent the power source from supplying power to the separator motor.

[0096] The cleaning system can vary in any number of ways. For example, the tank can include a protrusion configured to automatically press the switch in response to the tank being removably coupled to the body, and the controller can be configured to allow the separator motor to drive the rotation of the separator only with the switch activated. Further, the protrusion can extend from a latch configured to lock the tank and the body together; and/or the handheld surface cleaner can also include a housing having a recessed area with a width, the switch can be located in the recessed area of the housing, and the protrusion can be configured to enter the recessed area of the housing to activate the switch. Further, the width can be too small for a finger of a user to enter the recessed area.

[0097] For another example, the handheld surface cleaner can also include a latch configured to

move between a locked configuration, in which the latch locks the tank and the body together, and an unlocked configuration, in which the tank and the body are not locked together, and the latch moving from the unlocked configuration to the locked configuration can be configured to cause the switch to be activated. Further, the latch can include a protrusion configured to automatically press the switch in response to the tank being removably coupled to the body, and the controller can be configured to allow the separator motor to drive the rotation of the separator only with the switch activated; and/or the handheld surface cleaner can also include a housing having a recessed area with a width, the switch can be located in the recessed area of the housing, and the latch can be configured to enter the recessed area of the housing to activate the switch. Further, the width can be too small for a finger of a user to enter the recessed area.

[0098] For still another example, with the switch activated, a circuit including the power supply and the separator motor can be closed, and, with the switch deactivated, a circuit including the power supply and the separator motor can be open.

[0099] For yet another example, the body can include a first electrical connector, the tank can include a second electrical connector, and the first and second electrical connectors can be configured to engage automatically to establish an electrical connection between the separator motor and the controller in response to the tank being removably coupled to the body. Further, the electrical connection can be isolated from a flow path of the debris and the air in the handheld surface cleaner.

[0100] For still another example, the separator motor can be at the tank and can be configured to be removed from the body with the tank.

[0101] For yet another example, the separator motor can be at the body and can be configured to be removed from the tank with the body.

[0102] For another example, the tank can be configured to be removably coupled to the body using a selected one of rotational motion and translational motion. Further, the tank can be configured to be detached from the body using rotational motion.

[0103] For yet another example, the handheld surface cleaner can also include a suction motor at the body and configured to provide a suction force to draw debris and air into the handheld surface cleaner.

[0104] For another example, the separator motor can also be configured to provide a suction force to draw debris and air into the handheld surface cleaner.

[0105] In another implementation, a cleaning system includes a handheld surface cleaner that includes a body, an inlet, a tank, and a separator. Debris and air are allowed to enter the handheld surface cleaner through the inlet. The tank is configured to removably couple to the body and to contain and store therein the debris. The separator is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The tank is configured to be removably coupled to the body using a selected one of translational motion along a first longitudinal axis extending from the inlet to the separator and rotational motion about a rotational axis offset from the first longitudinal axis.

[0106] The cleaning system can have any number of variations. For example, the tank can be configured to be detached from the body using the rotational motion.

[0107] For another example, the tank can include a first attachment mechanism, the body can include a second attachment mechanism, and the first and second attachment mechanisms can be configured to be engaged throughout the rotational motion. Further, the first and second attachment mechanisms can be configured to not be engaged until an end of the translational motion, and/or one of the first and second attachment mechanisms can include a hook and the other of the first and second attachment mechanisms can include an opening. Further, the handheld surface cleaner can also include a bias element biasing the hook downwardly, the hook can be configured to be seated in the opening at a start of the rotational motion, and the hook can be configured to automatically enter the opening after a start of the translational motion.

[0108] For yet another example, one of the tank and the body can include a hook, the other of the tank and the body can include an opening, the hook can be configured to be seated in the opening at a start of the rotational motion, and the hook can be configured to automatically enter the opening after a start of the translational motion.

[0109] For still another example, the handheld surface cleaner can also include a handle at the body, and, with the tank removably coupled to the body, the handle can define a second longitudinal axis that is at a transverse angle to the longitudinal axis. Further, the first longitudinal axis can intersect the handle, and/or the handle can define a pistol grip.

[0110] For another example, the tank can include a first electrical connector, the body can include a second electrical connector, and the first and second electrical connectors can be configured to engage automatically to establish an electrical connection between the separator motor and the controller in response to the tank being removably coupled to the body. Further, the electrical connection can be isolated from a flow path of the debris and the air in the handheld surface cleaner.

[0111] For yet another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator. Further, the handheld surface cleaner can also include a controller, the handheld surface cleaner can also include a microswitch operatively coupled to the controller and configured to be activated automatically in response to the tank being removably coupled to the body, and the controller can be configured to allow the motor to drive the rotation of the separator only with the microswitch activated. Further, the tank can include a protrusion configured to automatically press the microswitch in response to the tank being removably coupled to the body.

[0112] For still another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator, the handheld surface cleaner can also include a controller, the handheld surface cleaner can also include a microswitch, the handheld surface cleaner can also include a latch configured to move between a locked configuration, in which the latch locks the tank and the body together, and an unlocked configuration, in which the tank and the body are not locked together, the latch moving from the unlocked configuration to the locked configuration can be configured to cause the microswitch to be activated, and the controller can be configured to prevent the motor from driving the rotation of the separator unless the microswitch is activated.

[0113] For yet another example, the handheld surface cleaner can also include a motor configured to provide a suction force to draw the debris and the air into the handheld surface cleaner.

[0114] In another implementation, a cleaning system includes a handheld surface cleaner that includes a body, a tank, and a separator. The body includes a first electrical connector. The tank includes a second electrical connector, and the tank is configured to removably couple to the body. The separator is configured to separate debris and air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The first and second electrical connectors are configured to engage automatically to establish an electrical connection in response to the tank being removably coupled to the body. The handheld surface cleaner can be configured to allow the debris and the air flow in the handheld surface cleaner along a flow path that is isolated from the electrical connection.

[0115] The cleaning system can vary in any number of ways. For example, the flow path can include a common flow path along which the debris and the air flow to the separator, a debris flow path along which the debris flows from the separator, and an air flow path that is different from the debris flow path and along which the air flows from the separator. Further, the common flow path can define a first longitudinal axis, and the tank can be configured to be removably coupled to the body using a selected one of translational motion along the first longitudinal axis and rotational motion about a rotational axis offset from the first longitudinal axis.

[0116] For another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator, the handheld surface cleaner can also include a power supply, and

the power supply can be configured to provide power to the motor with the electrical connection established.

[0117] For yet another example, the handheld surface cleaner can include a motor configured to drive rotation of the separator. Further, the motor can be located at the tank, the handheld surface cleaner can also include a power supply at the body, and the power supply can be configured to provide power to the motor with the electrical connection established.

[0118] For still another example, the first electrical connector can include a pair of conductive plates. Further, the second electrical connector can include a pair of pogo pins.

[0119] For another example, a separator assembly of the handheld surface cleaner can include the separator and the first electrical connector, and the separator assembly can be configured to be removably coupled with the tank and can be configured to be removably coupled to the body with the tank. Further, the separator assembly can also include a motor configured to drive rotation of the separator, the handheld surface cleaner can also include a power supply at the body, and the power supply can be configured to provide power to the motor with the electrical connection established.

[0120] For yet another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator, and the handheld surface cleaner can also include a housing that houses the motor and includes the first electrical connector. Further, the motor, the housing, and the separator can be configured to be removed as a unit from the tank and to be coupled as a unit to the tank. Further, the handheld surface cleaner can also include a power supply at the body, and the power supply can be configured to provide power to the motor with the electrical connection established.

[0121] For still another example, the handheld surface cleaner can also include a motor configured to provide a suction force to draw the debris and the air into the handheld surface cleaner.

[0122] For another example, the body can define a handle of the handheld surface cleaner.

[0123] For yet another example, the handheld surface cleaner can be configured to always maintain the flow path being isolated from the electrical connection.

[0124] In another implementation, a cleaning system includes a handheld surface cleaner that includes an inlet, a tank, a rotatable separator, and a valve. Debris and air can be allowed to enter the handheld surface cleaner through the inlet. The rotatable separator is configured to be located in the tank downstream of the inlet and is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The valve is removably coupled to the tank, is configured to be located in the tank downstream of the inlet and upstream of the rotatable separator, and is configured to allow flow of the debris and the air in only one direction from the inlet toward the separator.

[0125] The cleaning system can vary in any number of ways. For example, the handheld surface cleaner can also include a housing that houses the valve, and the housing and the valve can be configured to be removed as a unit from the tank.

[0126] For another example, the handheld surface cleaner can also include a body configured to be removably coupled to the tank. Further, the valve can be configured to be removed from the tank only with the body not being removably coupled with the body, and/or the body can define a handle of the handheld surface cleaner.

[0127] For yet another example, a first plurality of arms can extend proximally from a body of the rotatable separator and can be configured to rotate with the rotatable separator around the valve. Further, a second plurality of arms can extend distally from the body of the separator, the handheld surface cleaner can also include a housing downstream of the separator through which the air is configured to flow after exiting the separator, and the second plurality of arms can be configured to rotate with the rotatable separator around the housing. Further, the handheld surface cleaner can also include a motor configured to drive the rotation of the separator, and the motor can be located at least partially within the housing.

[0128] For another example, the handheld surface cleaner can also include a filter non-removably coupled to the tank and located upstream of the valve.

[0129] For yet another example, the handheld surface cleaner can also include a filter removably coupled to the tank and located upstream of the valve in a distal portion of the tank, and the valve can be configured to be removed from the tank only through an open proximal end of the tank. Further, the handheld surface cleaner can also include a second filter removably coupled to the tank and located downstream of the separator.

[0130] For still another example, the separator can be removably coupled to the tank, and the valve can be configured to be removed from the tank only with the separator not being removably coupled with the tank. Further, the separator can be configured to be removed from the tank through an open proximal end of the tank, and the valve can be configured to be removed from the tank through the open proximal end of the tank; and/or the handheld surface cleaner can also include a motor configured to drive the rotation of the separator, and the motor and the separator can be configured to be removed as a unit from the tank. For yet another example, the valve can be a duckbill valve.

[0131] In another implementation, a cleaning system includes a handheld surface cleaner that includes an inlet, a tank, a separator, a lid, and a cap. Debris and air can be allowed to enter the handheld surface cleaner through the inlet. The separator is configured to separate the debris and the air to allow the debris to be collected in the tank and to allow the air to exit the handheld surface cleaner. The lid is configured to be removed from the tank to allow the debris collected in the tank to be removed from the tank through an open end of the tank. The cap is separate from the lid and is configured to allow the debris collected in the tank to be removed from the tank through an opening formed in a sidewall of the tank. The cap is attached to the tank and is configured to be removed completely from the tank and replaced on the tank.

[0132] The cleaning system can have any number of variations. For example, the cap can be movable between an open position, in which the opening is not sealed, and a closed position, in which the opening is sealed, and the handheld surface cleaner can also include a bias element configured to bias the cap to the closed position.

[0133] For another example, the cap can be movable between an open position, in which the opening is not sealed, and a closed position, in which the opening is sealed, and the cap can include a lock configured to lock the cap in the closed position.

[0134] For yet another example, the inlet can be in a distal portion of the tank, the open end of the tank can be at a proximal end of the tank, and the cap can be located between the inlet and the open end.

[0135] For still another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator, and the handheld surface cleaner can also include a controller configured to control the motor based on a measured current of the motor. Further, the handheld surface cleaner can also include a suction motor configured to provide a suction force to draw the debris and air into the inlet, and the controller can also be configured to control the suction motor based on the measured current of the separator motor.

[0136] For yet another example, the separator can be non-removably coupled to the lid.

[0137] For another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator, and the motor can be non-removably coupled to the lid.

[0138] For yet another example, the handheld surface cleaner can also include a motor configured to drive rotation of the separator, the handheld surface cleaner can also include a body configured to be removably coupled to the tank, the body can define a handle of the handheld surface cleaner, and the motor can be at the body and can be configured to be removed from the tank with the body.

[0139] For still another example, the handheld surface cleaner can also include a body configured to be removably coupled to the tank, and the lid can be configured to be removable from the tank only with the body removed from the tank. Further, the handheld surface cleaner can also include a

suction motor at the body, and the suction motor can be configured to provide a suction force to draw the debris and air into the inlet.

[0140] For another example, the cap can be pivotally attached to the tank.

[0141] In another aspect, a separation apparatus is provided that in one implementation includes a separator that includes a body, an inlet opening, an outlet opening, a plurality of holes, and a plurality of covers. Debris and air can be allowed to enter the separator through the inlet opening. The air is configured to exit the separator through the outlet opening. The plurality of holes are formed in a sidewall of the body. The debris is configured to exit the separator through the plurality of holes. Each one of the plurality of covers is configured to move from a sealed configuration, in which each of the plurality of covers seals closed an associated at least one of the plurality of holes, to an unsealed configuration, in which each of the plurality of covers does not seal closed the associated at least one of the plurality of holes.

[0142] The separation apparatus can vary in any number of ways. For example, rotation of the separator can be configured to cause the plurality of covers to move from the sealed configuration to the unsealed configuration, and the rotation of the separator stopping is configured to cause the plurality of covers to move from the unsealed configuration to the sealed configuration. Further, the separator is configured to rotate about a longitudinal axis extending between the inlet opening and the outlet opening, and/or the separation apparatus can also include a motor configured to drive the rotation of the separator.

[0143] For another example, each of the plurality of covers can be attached to the body along a single edge of the cover.

[0144] For yet another example, each of the plurality of covers can be flexible.

[0145] For another example, the separator can also include a plurality of blades positioned downstream of the inlet opening and upstream of the outlet opening.

[0146] For yet another example, the body can be a singular element. Further, the body can be rigid, and each of the plurality of covers can be flexible.

[0147] For still another example, the body can include an inner body and an outer body that is attached to the inner body. Further, the separator can also include a plurality of blades positioned downstream of the inlet opening and upstream of the outlet opening, the blades can be formed on the inner body, and the holes can be formed in the outer body; and/or the inner and outer bodies can be rigid, and each of the plurality of covers can be flexible.

[0148] For another example, the separator can also include a first plurality of arms extend proximally from the body, and the separator can also include a second plurality of arms extend distally from the body.

[0149] For yet another example, the separation apparatus can also include a surface cleaner that includes the separator, the surface cleaner can also include an inlet through which the debris and the air is configured to be suctioned into the surface cleaner, and the separator can be downstream of the inlet. Further, the surface cleaner can also include a motor configured to drive rotation of the separator; the surface cleaner can also include a tank having a cavity configured to contain and store therein the debris that enters the separator, with the plurality of covers in the sealed configuration an interior of the separator can not be in fluid communication with the cavity through the plurality of holes, and with the plurality of covers in the unsealed configuration the interior of the separator can be in fluid communication with the cavity through the plurality of holes; the surface cleaner can be configured to be handheld; and/or the surface cleaner can also include at least one of a first filter upstream of the separator, and a second filter downstream of the separator. Further, the surface cleaner can also include an air exit hole configured to allow the air that is directed out of the separator through the outlet opening to exit to external atmosphere.

[0150] For another example, each one of the plurality of covers can be associated with only one of the plurality of holes.

[0151] In another aspect, a method of using a handheld surface cleaner as described herein is

provided.

[0152] In another aspect, a separator as described herein is provided. The separator can be configured to be used, for example, with a handheld surface cleaner.

[0153] In another aspect, a method of using a separator as described herein is provided. The separator can be configured to be used, for example, with a handheld surface cleaner.

[0154] In another aspect, a separator assembly as described herein is provided. The separator assembly can be configured to be used, for example, with a handheld surface cleaner.

[0155] In another aspect, a method of using a separator assembly as described herein is provided. The separator assembly can be configured to be used, for example, with a handheld surface cleaner.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0156] This disclosure will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0157] FIG. 1 is a perspective view of one implementation of a handheld surface cleaner;

[0158] FIG. 2 is a perspective cross-sectional view of the handheld surface cleaner of FIG. 1;

[0159] FIG. 3 is a perspective cross-sectional view of a distal portion of the handheld surface cleaner of FIG. 1;

[0160] FIG. 4 is a perspective view of a dirty water tank of the handheld surface cleaner of FIG. 1;

[0161] FIG. 5 is another perspective view of the dirty water tank of FIG. 4;

[0162] FIG. 6 is an exploded view of the dirty water tank of FIG. 4;

[0163] FIG. 7 is a perspective view of a portion of the dirty water tank of FIG. 4;

[0164] FIG. 8 is a perspective view of a main body of the handheld surface cleaner of FIG. 1;

[0165] FIG. 9 is an exploded view of the main body of FIG. 8;

[0166] FIG. 10 is a perspective view of a separator assembly of the handheld surface cleaner of FIG. 1;

[0167] FIG. 11 is a perspective cross-sectional view of the separator assembly of FIG. 10;

[0168] FIG. 12 is another perspective view of the separator assembly of FIG. 10;

[0169] FIG. 13 is a side view of the separator assembly of FIG. 10;

[0170] FIG. 14 is a perspective view of a separator of the handheld surface cleaner of FIG. 1;

[0171] FIG. 15 is a perspective view of an outer separator of the separator of FIG. 14;

[0172] FIG. 16 is a perspective view of an inner separator of the separator of FIG. 14;

[0173] FIG. 17 is a perspective view of a hair cage of the handheld surface cleaner of FIG. 1;

[0174] FIG. 18 is a perspective view of filter foam of the handheld surface cleaner of FIG. 1;

[0175] FIG. 19 is a perspective view of the handheld surface cleaner of FIG. 1 releasably coupled to one implementation of a stain tool attachment;

[0176] FIG. 20 is a perspective view of the stain tool attachment of FIG. 19 that is configured to releasably couple to a handheld surface cleaner such as the handheld surface cleaner of FIG. 1;

[0177] FIG. 21 is another perspective view of the stain tool attachment of FIG. 19;

[0178] FIG. 22 is yet another perspective view of the stain tool attachment of FIG. 19;

[0179] FIG. 23A is a perspective view of another implementation of a stain tool attachment configured to releasably couple to a handheld surface cleaner such as the handheld surface cleaner of FIG. 1;

[0180] FIG. 23B is another perspective view of the stain tool attachment of FIG. 23A;

[0181] FIG. 23C is yet another perspective view of the stain tool attachment of FIG. 23A;

[0182] FIG. 23D is a perspective view of another implementation of an attachment in the form of a stain tool attachment configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner of FIG. 1;

[0183] FIG. **24A** is a perspective view of an agitator of the stain tool attachment of FIG. **23A**;

[0184] FIG. **24B** is another perspective view of the agitator of FIG. **24A**;

[0185] FIG. **25A** is a perspective view of one implementation of a crevice tool attachment configured to releasably couple to a handheld surface cleaner such as the handheld surface cleaner of FIG. **1**;

[0186] FIG. **25B** is a perspective view of the crevice tool attachment of FIG. **25A** with an agitator of the crevice tool attachment detached;

[0187] FIG. **25C** is another perspective view of the crevice tool of FIG. **25A** with the agitator of the crevice tool attachment detached;

[0188] FIG. **26** is a perspective view of another implementation of a crevice tool attachment configured to releasably couple to a handheld surface cleaner such as the handheld surface cleaner of FIG. **1**;

[0189] FIG. **27** is a side view of the crevice tool attachment of FIG. **26**;

[0190] FIG. **28** is another perspective view of the crevice tool attachment of FIG. **26**;

[0191] FIG. **29A** is a perspective view of one implementation of a pet hair tool attachment configured to releasably couple to a handheld surface cleaner such as the handheld surface cleaner of FIG. **1**;

[0192] FIG. **29B** is another perspective view of the pet hair tool attachment of FIG. **29A**;

[0193] FIG. **29C** is another perspective view of the pet hair tool attachment of FIG. **29A**;

[0194] FIG. **30** is a perspective view of the pet hair tool attachment of FIG. **29A** with a debris storage container of the pet hair tool attachment detached;

[0195] FIG. **31** is a perspective view of the debris storage container of FIG. **30**;

[0196] FIG. **32** is a perspective view of another implementation of a separator assembly of a handheld surface cleaner;

[0197] FIG. **33** is a side view of another implementation of a handheld surface cleaner;

[0198] FIG. **34** is a perspective view of a dirty water tank of the handheld surface cleaner of FIG. **33**;

[0199] FIG. **35** is a side cross-sectional view of another implementation of a handheld surface cleaner;

[0200] FIG. **36** is a side cross-sectional view of a separator assembly of the handheld surface cleaner of FIG. **35**;

[0201] FIG. **37** is a perspective view of a dirty water tank of the handheld surface cleaner of FIG. **35**;

[0202] FIG. **38** is a perspective view of a main body of the handheld surface cleaner of FIG. **35**;

[0203] FIG. **39** is a side cross-sectional view of another implementation of a handheld surface cleaner;

[0204] FIG. **40** is a side cross-sectional view of a portion of a dirty water tank of the handheld surface cleaner of FIG. **39**;

[0205] FIG. **41** is a side cross-sectional view of another implementation of a handheld surface cleaner;

[0206] FIG. **42** is a side cross-sectional view of a distal portion of a dirty water tank of the handheld surface cleaner of FIG. **41**;

[0207] FIG. **43** is a perspective view of a separator assembly of the handheld surface cleaner of FIG. **41**;

[0208] FIG. **44** is a side cross-sectional view of a distal portion of the separator assembly of FIG. **43**;

[0209] FIG. **45** is a side cross-sectional view of a distal portion of another implementation of a separator assembly of a handheld surface cleaner;

[0210] FIG. **46** is a side cross-sectional view of a portion of another implementation of a separator assembly of a handheld surface cleaner;

[0211] FIG. **47** is a side cross-sectional view of a portion of yet another implementation of a separator assembly of a handheld surface cleaner;

[0212] FIG. **48** is a side cross-sectional view of a portion of still another implementation of a separator assembly of a handheld surface cleaner;

[0213] FIG. **49** is a side cross-sectional view of a portion of another implementation of a separator assembly of a handheld surface cleaner;

[0214] FIG. **50A** is a side cross-sectional view of one implementation of a separator, rim, and seal;

[0215] FIG. **50B** is a perspective view of the separator, rim, and seal of FIG. **50A**;

[0216] FIG. **50C** is an exploded perspective view of the separator, rim, and seal of FIG. **50A**;

[0217] FIG. **50D** is a perspective view of a separator assembly including the separator, rim, and seal of FIG. **50A**;

[0218] FIG. **51** is a side cross-sectional view of another implementation of a separator assembly;

[0219] FIG. **52** is a perspective view of another implementation of a separator;

[0220] FIG. **53** is a side cross-sectional view of a portion of another implementation of a dirty water tank of a handheld surface cleaner;

[0221] FIG. **54A** is a side cross-sectional view of a portion of yet another implementation of a dirty water tank of a handheld surface cleaner;

[0222] FIG. **54B** is a side cross-sectional view of a portion of still another implementation of a dirty water tank of a handheld surface cleaner;

[0223] FIG. **55** is a side cross-sectional view of a portion of another implementation of a dirty water tank of a handheld surface cleaner;

[0224] FIG. **56A** is a side cross-sectional view of another implementation of a handheld surface cleaner;

[0225] FIG. **56B** is a perspective view of one implementation of an inner plug;

[0226] FIG. **56C** is a perspective view of one implementation of an outer bearing;

[0227] FIG. **57A** is a perspective view of another implementation of a handheld surface cleaner;

[0228] FIG. **57B** is a perspective view of yet another implementation of a handheld surface cleaner;

[0229] FIG. **58** is a side cross-sectional view of the handheld surface cleaner of FIG. **57A**;

[0230] FIG. **59A** is a cross-sectional end view of the handheld surface cleaner of FIG. **57A**;

[0231] FIG. **59B** is a cross-sectional view of a separator of the handheld surface cleaner of FIG. **57A** coupled to a separator motor;

[0232] FIG. **60** is a perspective view of the separator of FIG. **59B**;

[0233] FIG. **61** is a side view of the handheld surface cleaner of FIG. **57A** having a separation assembly in a maintenance position;

[0234] FIG. **62** is a perspective view of a main body of the handheld surface cleaner of FIG. **57A**;

[0235] FIG. **63A** is perspective view of another implementation of a handheld surface cleaner;

[0236] FIG. **63B** is another perspective view of the handheld surface cleaner of FIG. **63A**;

[0237] FIG. **63C** is yet another perspective view of the handheld surface cleaner of FIG. **63A**;

[0238] FIG. **63D** is a side view of the handheld surface cleaner of FIG. **63A**;

[0239] FIG. **63E** is a side cross-sectional view of the handheld surface cleaner of FIG. **63A**;

[0240] FIG. **63F** is perspective cross-sectional view of a distal portion of the handheld surface cleaner of FIG. **63A**;

[0241] FIG. **63G** is a perspective view of a dirty water tank of the handheld surface cleaner of FIG. **63A**;

[0242] FIG. **63H** is another perspective view of the dirty water tank of FIG. **63G**;

[0243] FIG. **63I** is a perspective view of a distal portion of a main body of the handheld surface cleaner of FIG. **63A**;

[0244] FIG. **63J** is a perspective view of the main body of FIG. **63I**;

[0245] FIG. **63K** is a perspective view of a separator assembly of the handheld surface cleaner of FIG. **63A**;

[0246] FIG. 63L is another perspective view of the separator assembly of FIG. 63K;
[0247] FIG. 63M is a perspective view of a separator of the handheld surface cleaner of FIG. 63A;
[0248] FIG. 63N is another perspective view of the separator of FIG. 63M;
[0249] FIG. 63O is a perspective, partially exploded view of the separator of FIG. 63M;
[0250] FIG. 63P is a perspective view of a hair cage of the handheld surface cleaner of FIG. 63A;
[0251] FIG. 63Q is another perspective view of the hair cage of FIG. 63P;
[0252] FIG. 63R is an end view of the hair cage of FIG. 63P;
[0253] FIG. 63S is a perspective view of a latch of the handheld surface cleaner of FIG. 63A;
[0254] FIG. 63T is a perspective view of a portion of the handheld surface cleaner of FIG. 63A without the latch of FIG. 63S omitted;
[0255] FIG. 63U is a side, partial view of the handheld surface cleaner of FIG. 63A with one implementation of a tool engaging a microswitch of the handheld surface cleaner of FIG. 63A;
[0256] FIG. 63V is side cross-sectional view of a portion of the handheld surface cleaner of FIG. 63A;
[0257] FIG. 63W is a perspective view of an inner plug of the handheld surface cleaner of FIG. 63A;
[0258] FIG. 63X is a perspective view of an outer bearing of the handheld surface cleaner of FIG. 63A;
[0259] FIG. 63Y is a perspective view of a duckbill valve assembly of the handheld surface cleaner of FIG. 63A;
[0260] FIG. 63Z is another perspective view of the duckbill valve assembly of FIG. 63Y;
[0261] FIG. 63AA is side cross-sectional view of a portion of the handheld surface cleaner of FIG. 63A with one implementation of a tool engaging the duckbill valve assembly;
[0262] FIG. 63BB is a perspective view of a cap of the handheld surface cleaner of FIG. 63A;
[0263] FIG. 63CC is a perspective cross-sectional view of a portion of the handheld surface cleaner of FIG. 63A;
[0264] FIG. 63DD is a perspective view of a portion of the handheld surface cleaner of FIG. 63A with a distal housing of the handheld surface cleaner removed;
[0265] FIG. 63EE is a perspective view of the distal housing of the handheld surface cleaner of FIG. 63A;
[0266] FIG. 63FF is another perspective view of the distal housing of FIG. 63EE;
[0267] FIG. 63GG is a perspective view of another implementation of a main body of a handheld surface cleaner;
[0268] FIG. 63HH is another perspective view of the main body of FIG. 63GG;
[0269] FIG. 63II is a perspective view of a portion of a DC charger cord;
[0270] FIG. 63JJ is a perspective view and inset exploded view of another implementation of a separator;
[0271] FIG. 63KK is a perspective view and inset exploded view of yet another implementation of a separator;
[0272] FIG. 64 is a side schematic view of one implementation of a dock and a side view of the handheld vacuum cleaner of FIG. 57B seated in the dock;
[0273] FIG. 65A is a perspective view of one implementation of a tray;
[0274] FIG. 65B is another perspective view of the tray of FIG. 65A;
[0275] FIG. 66A is a perspective view of another implementation of a tray;
[0276] FIG. 66B is another perspective view of the tray of FIG. 66A;
[0277] FIG. 66C is a perspective cross-sectional view of the tray of FIG. 66A;
[0278] FIG. 67 is a perspective view of a portion of another implementation of a tray and a perspective view of one implementation of a removable cup;
[0279] FIG. 68A is a perspective view of one implementation of a charger;
[0280] FIG. 68B is another perspective view of the charger of FIG. 68A;

[0281] FIG. **69** is a perspective view of another implementation of a tray with implementations of a handheld surface cleaner, attachments, and a spray bottle seated by the tray;

[0282] FIG. **70A** is a perspective view of yet another implementation of a tray with implementations of a handheld surface cleaner and attachments seated by the tray;

[0283] FIG. **70B** is another perspective view of the tray of FIG. **70A** with the handheld surface cleaner and the attachments seated by the tray and with the charger of FIG. **68A** operatively coupled to the handheld surface cleaner;

[0284] FIG. **70C** is a perspective view of a portion of the handheld surface cleaner and portion of the tray of FIG. **70A** and a portion of the charger of FIG. **68A**;

[0285] FIG. **71** is perspective view of another implementation of a tray;

[0286] FIG. **72A** is a perspective view of yet another implementation of a tray;

[0287] FIG. **72B** is another perspective view of the tray of FIG. **72A**;

[0288] FIG. **72C** is a perspective view of one implementation of a cup configured to be removably coupled with the tray of FIG. **72A**;

[0289] FIG. **72D** is a perspective cross-sectional view of the cup of FIG. **72C**;

[0290] FIG. **73** is a perspective view of one implementation of a spray bottle;

[0291] FIG. **74** is cross-sectional view of the spray bottle of FIG. **73**;

[0292] FIG. **75** is a side view of the spray bottle of FIG. **73**;

[0293] FIG. **76** is another side view of the spray bottle of FIG. **73**;

[0294] FIG. **77** is a back end view of the spray bottle of FIG. **73**;

[0295] FIG. **78** is front end view of the spray bottle of FIG. **73**;

[0296] FIG. **79** is a perspective view of a spray head of the spray bottle of FIG. **73**;

[0297] FIG. **80** is another perspective view of the spray head of FIG. **79**;

[0298] FIG. **81** is a perspective view of a dual cleaning solution container of the spray bottle of FIG. **73**;

[0299] FIG. **82** is a side view of the dual cleaning solution container of FIG. **81**;

[0300] FIG. **83** is another side view of the dual cleaning solution container of FIG. **81**;

[0301] FIG. **84** is a top view of the dual cleaning solution container of FIG. **81**;

[0302] FIG. **85** is a bottom view of the dual cleaning solution container of FIG. **81**;

[0303] FIG. **86A** is a perspective view of another implementation of a spray bottle;

[0304] FIG. **86B** is another perspective view of the spray bottle of FIG. **86A**;

[0305] FIG. **87** is a perspective view of another implementation of a spray bottle;

[0306] FIG. **88** is another perspective view of the spray bottle of FIG. **87**;

[0307] FIG. **89** is a perspective view of a portion of the spray bottle of FIG. **87**;

[0308] FIG. **90** is another perspective view of a portion of the spray bottle of FIG. **87**;

[0309] FIG. **91** is a perspective cross-sectional view of the spray bottle of FIG. **87**;

[0310] FIG. **92A** is a perspective view of another implementation of a spray bottle;

[0311] FIG. **92B** is a perspective view of a portion of the spray bottle of FIG. **92A**;

[0312] FIG. **93** is a perspective view of another implementation of a spray bottle;

[0313] FIG. **94A** is a perspective view of yet another implementation of a spray bottle;

[0314] FIG. **94B** is a side view of the spray bottle of FIG. **94A**;

[0315] FIG. **94C** is a perspective view of a dual cleaning solution container of the spray bottle of FIG. **94A**;

[0316] FIG. **94D** is another perspective view of the dual cleaning solution container of FIG. **94C**;

and

[0317] FIG. **94E** is side view of the dual cleaning solution container of FIG. **94C**.

[0318] It is noted that the drawings are not necessarily to scale. The drawings are intended to depict only typical aspects of the subject matter disclosed herein, and therefore should not be considered as limiting the scope of the disclosure.

DETAILED DESCRIPTION

[0319] Certain embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices, systems, and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices, systems, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

[0320] Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

[0321] Various illustrative handheld surface cleaners and methods of using handheld surface cleaners are provided. In an exemplary implementation, the handheld surface cleaner is configured to use suction to draw debris into the handheld surface cleaner. The debris can include liquid, hair, dirt, and/or other matter. The suction also causes air to be drawn into the handheld surface cleaner. The handheld surface cleaner is configured to separate the air and the debris on board the handheld surface cleaner to allow the air to exit the handheld surface cleaner and to allow the debris to be collected in the handheld surface cleaner for later disposal.

[0322] Various illustrative spray bottles for spraying cleaning liquid for use with handheld surface cleaners and methods of using spray bottles for spraying cleaning liquid for use with handheld surface cleaners are also provided. In an exemplary implementation, the cleaning liquid includes a first liquid and a second liquid that are separately stored in the spray bottle and are mixed by the spray bottle. In other implementations, the cleaning liquid is a single cleaning liquid.

[0323] FIG. 1 illustrates one implementation of a handheld surface cleaner **10** configured to use suction to draw debris into the handheld surface cleaner **10** and to separate the air and the debris on board the handheld surface cleaner **10**. FIG. 2 shows a cross-section of the handheld surface cleaner **10** of FIG. 1. FIG. 3 shows a cross-section of a distal portion of the handheld surface cleaner **10** of FIG. 1.

[0324] The handheld surface cleaner **10** of FIG. 1 includes a dirty water tank (DWT) **12** and a main body **14** configured to releasably couple to the DWT **12**. The DWT **12** defines a distal portion of the handheld surface cleaner **10**. The main body **14** defines a proximal portion of the handheld surface cleaner **10**. FIGS. 4-6 show the DWT **12** as a standalone element. FIG. 7 shows a portion of the DWT **12**. FIGS. 8 and 9 show the main body **14** as a standalone element.

[0325] In some implementations the DWT **12** is configured to be completely released from the main body **14**. The DWT **12** being completely releasable from the main body **14** may facilitate emptying the DWT **12** of debris without the main body **14** getting dirty and/or getting in the way. In other implementations, the DWT is configured to be partially released from the main body **14** to allow emptying of debris from a debris collection and storage cavity **16** in the DWT **12**, e.g., by pouring the debris out of an open proximal end **22** of the DWT **12**. The DWT **12** being partially releasable from the main body **14** may help prevent loss of one of the DWT **12** and the main body **14** and thus rendering the handheld surface cleaner **10** unusable.

[0326] The debris collected in the dirty water tank **12** may or may not include dirty water and in some instances may not include water at all, depending on the matter being suctioned into the handheld surface cleaner **10**. For example, a liquid other than or in addition to water may be suctioned into the handheld surface cleaner **10** and collected in the DWT **12**. For another example,

clean water may be suctioned into the handheld surface cleaner **10** without any other debris, e.g., for cleaning purposes as discussed further below, and collected in the DWT **12**.

[0327] The DWT **12** includes a cap **18** configured to be opened manually by a user to allow debris to exit the debris collection and storage cavity **16** of the DWT **12** without releasing the DWT **12** from the main body **14**. The cap **18** plugs a hole **20** formed through a wall of a housing **26** of the DWT **12** (see FIG. 3). The hole **20** is in communication with the debris collection and storage cavity **16** defined by the housing **26** of the DWT **12**. The cap **18** also extends through an opening **28** formed in a lower cover **30** of the DWT **12** (see FIGS. 4-6). The lower cover **30** covers a lower portion of the DWT housing **26**. The opening **24** in the DWT housing **26** and the opening **28** in the DWT lower cover **30** are aligned to define a passageway that extends from inside the DWT **12** to outside the DWT **12**. The cap **18** is configured to cooperate with the DWT's housing **26** and lower cover **30** to selectively plug the passageway.

[0328] In some implementations, the lower cover **30** of the DWT **12** is omitted.

[0329] In some implementations, the cap **18** is completely removable from the DWT **12**, which may allow for a larger passageway for exiting of debris from the debris collection and storage cavity **16**. In other implementations, the cap **18** is attached to the DWT **12**, which may allow for debris to exit the debris collection and storage cavity **16** while helping prevent loss of the cap **18**. In this illustrated implementation, the cap **18** is attached to the DWT **12** and is formed of rubber and/or other flexible material to allow the cap **18** to bend for opening of the passageway while the cap **18** remains attached to the DWT **12**.

[0330] In the illustrated implementation of FIG. 1, the DWT **12** is pivotally coupled to the main body **14** to allow partial release of the DWT **12** from the main body **14**. In this illustrated implementation, as shown in FIG. 5, the DWT **12** includes a pair of pivot arms **32** operably coupled to a pivot pin **34** of the main body **14**. In other implementations, the main body **14** can include pivot arms operably coupled to a pivot pin of the DWT.

[0331] In some implementations the DWT **12** is pivotally coupled to the main body **14** to allow complete release of the DWT **12** from the main body **14**.

[0332] As shown in FIGS. 1-6, the handheld surface cleaner **10** includes a lock configured to help secure the DWT **12** to the main body **14**. The lock in this illustrated implementation is a latch **36**. The latch **36** can, as in this illustrated implementation, be configured to be manually moved by a user. The latch **36** can have a variety of configurations. The latch **36** in this illustrated implementation includes a hook **36a** connected to the DWT **12** that is configured to engage, e.g., clip into, a groove **38** formed in the main body **14** (see FIG. 8). In other implementations, the main body **14** can include the hook and the DWT **12** can include the groove.

[0333] The handheld surface cleaner **10** is configured to automatically detect whether the DWT **12** is releasably coupled to the main body **14**. As shown in FIGS. 8 and 9, the main body **12** includes a microswitch **40** at a distal face of the main body **12**. The DWT **12** includes a protrusion **42** that extends proximally and that is configured to push the microswitch **40** to activate the microswitch **40** with the DWT **12** releasably coupled to the main body **14**. The activation of the microswitch **40** indicates that the DWT **12** is releasably coupled to the main body **14**. Correspondingly, the microswitch **40** not being activated indicates that the DWT **12** is not releasably coupled to the main body **14**. Instead of or in addition to the protrusion **42**, the DWT **12** can include another element configured to activate the microswitch, such as a pivotable latch that moves into position to engage the microswitch **40** in response to the DWT **12** being coupled to the main body **14**.

[0334] In this illustrated implementation the microswitch **40** is configured to extend distally through a motor bracket **44** of the main body **12**. The motor bracket **44** is configured to help protect the microswitch **40** from being damaged during coupling of the DWT **12** and the main body **14**, during de-coupling of the DWT **12** and the main body **14**, and when the main body **14** is de-coupled from the DWT **12**. The motor bracket **44** has an opening configured to receive the protrusion **42** to allow the protrusion **42** to contact and activate the microswitch **40**. As discussed

further below, the motor bracket **44** is also configured to support a separator motor **46** of the handheld surface cleaner **10** (see FIGS. 2, 3, and 8).

[0335] In this illustrated implementation the protrusion **42** of the DWT **12** extends proximally from a separator bracket **48** of a separator assembly **50** of the DWT **12**, shown in FIGS. 10-13, but can be located elsewhere. As discussed further below, the separator assembly **50** also includes a separator **52** (also see FIG. 14) supported by the separator bracket **48**.

[0336] Instead of or in addition to including the microswitch **40**, the handheld surface cleaner **10** can include a switch in a circuit also including the separator motor **46** and a power supply of the handheld surface cleaner **10**. The protrusion **42** or another protrusion of the DWT **12** can be configured to push the switch, and thus activate the switch, similar to that discussed herein regarding the microswitch **40**. With the switch activated, the switch is closed and the circuit is closed, which allows the power supply to supply power to the separator motor **46**. With the switch deactivated, the switch is open and the circuit is open, which prevents the power supply from supplying power to the separator motor **46**. The switch can be located, for example, at the main body **12** similar to a location described herein for the microswitch **40**.

[0337] In some implementations, as in this illustrated implementation, the separator assembly **50** is configured to be removably coupled to the DWT **12** to facilitate cleaning of the separator **52** and/or emptying of debris from the DWT's collection and storage cavity **16**. The separator assembly **50** is configured to be de-coupled from the main body **14** with a remainder of the DWT **12**, as shown in FIGS. 5 and 6. The separator assembly **50** is configured as a lid that covers the open proximal end **22** of the DWT **12** through which debris is configured to exit the debris collection and storage cavity **16**. Thus, removing the DWT **12** from the main body **14** will not cause debris to prematurely and messily exit the debris collection and storage cavity **16** because of the lid functionality of the separator assembly **50**.

[0338] With the DWT **12** de-coupled from the main body **14**, the separator assembly **50** is configured to be removed from the DWT **12** to allow for emptying and/or cleaning of the debris collection and storage cavity **16** and/or to allow cleaning of the separator assembly **50**. FIGS. 10-13 show the separator assembly **50** as a standalone element de-coupled from the DWT **12**.

[0339] The separator assembly **50** in this illustrated implementation includes a pair of tabs **54** configured to facilitate manual removal of the separator assembly **50** from the DWT **12**. In another implementation, the separator assembly **50** can include a single tab **54** or can include more than two tabs **54**. The tabs **54** are configured to be held by a user to help pull the separator assembly **50** in a proximal direction to remove the separator assembly **50** from the DWT **12**. Similarly, the tabs **54** are configured to be held by a user to help push the separator assembly **50** in a distal direction to re-attach the separator assembly **50** to the DWT **12**. A user may choose to hold only one of the tabs **54** when removing and/or re-attaching the separator assembly **50**.

[0340] The tabs **54** in this illustrated implementation extend proximally and are located opposite one another around a perimeter of the separator assembly **50**. Free ends of the tabs **54** are configured to be located outside of the DWT **12** (and the main body **14**) with the DWT **12** coupled to the main body **14**, as shown in FIG. 1 in which one of the tabs **54** is visible. At least a portion of the tabs **54** is thus outside of a debris flow path and an air flow path in the handheld surface cleaner **10** so that at least the portion of the tabs **54** may remain clean and dry when a user holds the tabs **54**.

[0341] The tabs **54** are configured as a lock configured to lock the separator assembly **50** to the DWT housing **26** (and the DWT lower cover **30** if not omitted). The tabs **54** are configured to move between a locked configuration, shown in FIGS. 1 and 4-7, and an unlocked configuration. In the locked configuration, the tabs **54** are configured to lock the separator assembly **50** to the DWT housing **26** (and the DWT lower cover **30** if not omitted). In the unlocked configuration, the tabs **54** are configured to allow the separator assembly **50** to be removed from the DWT housing **26** (and the DWT lower cover **30** if not omitted).

[0342] With the tabs **54** in the locked configuration and locking the separator assembly **50** to the DWT housing **26** (and the DWT lower cover **30** if not omitted), the tabs **54** are configured to be moved manually by a user from the locked configuration to the unlocked configuration, which allows a user to decide whether and when to unlock the separator assembly **50**. The tabs **54** are configured to be pushed radially inward relative to the DWT housing **26** (and the DWT lower cover **30** if not omitted) to move from the locked configuration to the unlocked configuration. The tabs **54** are biased to the locked configuration and are resilient. Thus, when a user stops pushing the tabs **54** radially inward, the tabs **54** are configured to move automatically from the unlocked configuration to the locked configuration.

[0343] With the tabs **54** in the locked configuration and without the separator assembly **50** being coupled to the DWT housing **26** (and the DWT lower cover **30** if not omitted), the tabs **54** are configured to move automatically from the locked configuration to the unlocked configuration in response to the separator assembly **50** being coupled to the DWT housing **26** (and the DWT lower cover **30** if not omitted). The separator assembly **50** is configured to be inserted into the DWT housing **26**, e.g., into the debris collection and storage cavity **16** of the DWT **12**, in a distal direction through the open proximal end **22** of the DWT **12** with the separator **52** leading. The tabs **54** being resilient allows the tabs **54** to move automatically from the locked configuration, to the unlocked configuration, and back to the locked configuration as the separator assembly **50** is configured to be inserted into the DWT housing **26**.

[0344] The main body **14** includes various electronic components configured to facilitate electronic control of various aspects of the handheld surface cleaner **10**. The electronic components can be part of a circuit board. As shown in FIGS. 2 and 3, the main body **14** includes a printed circuit board (PCB) **56** that includes electronic components including at least a controller (e.g., processor, microcontroller, etc.) and a memory. The PCB **56** is located at a top of the main body **14** adjacent a handle **58** of the handheld surface cleaner **10** at the main body **14** in this illustrated implementation but can be located elsewhere.

[0345] The PCB **56** is operatively coupled to the microswitch **40**. The PCB **56** is configured to determine whether or not the DWT **12** is releasably coupled to the main body **14** based on an activation state of the microswitch **40**. This is a safety feature to prevent suctioning of debris without the DWT **12** being attached properly to the main body **14**. If the microswitch **40** is activated, the PCB **56**, e.g., the controller at the PCB **56**, is configured to determine that the DWT **12** is releasably coupled to the main body **14** and consequently that the handheld surface cleaner **10** can allow use of the handheld surface cleaner **10** to clean a surface by allowing suction to be activated, e.g., by allowing a suction motor **60** of the handheld surface cleaner **10** to be on and therefore provide a suction force. If the microswitch **40** is not activated, the PCB **56**, e.g., the controller at the PCB **56**, is configured to determine that the DWT **12** is not releasably coupled to the main body **14** and consequently is configured to prevent use of the handheld surface cleaner **10** to clean a surface by disallowing activation of suction, e.g., by not allowing the suction motor **60** to be on and therefore not provide a suction force.

[0346] The handheld surface cleaner **10** includes a power supply configured to supply power to the PCB **56**. The power supply includes a plurality of rechargeable batteries in this illustrated implementation but can be another type of power supply, such as a plurality of non-rechargeable batteries, a single rechargeable battery, etc. The main body **14** includes a battery holder **62** (see FIG. 2) in the handle **58** configured to hold the batteries therein. The power supply being on board the handheld surface cleaner **10** may help allow for easy portability and maneuverability of the handheld surface cleaner **10**. The handle **58** in this illustrated implementation includes a charging dock to allow for recharging of the rechargeable batteries.

[0347] As in this illustrated implementation, the handle **58** can define a pistol grip. A pistol grip may be more securely held by a user as the handheld surface cleaner **10** is moved over a surface than a suitcase-type handle where a user positions their fingers through an opening and holds on to

a U-shaped or C-shaped grip. The handle **58** has a substantially cylindrical shape in this illustrated implementation, which may facilitate comfortable holding of the handle **58** around the handle's substantially circular perimeter.

[0348] The handle **58** includes a finger grip **58a** configured to facilitate a user's holding of the handle **58** by providing a surface against which a user's finger can be positioned to provide leverage and/or stability. The handle **58** includes a single finger grip **58a** in this illustrated implementation but can include another number, e.g., two, three, etc. Including a plurality of finger grips may allow for each of a user's fingers to be positioned against a finger grip.

[0349] As shown in FIGS. **1** and **2**, the handle **58** defines a proximal portion of the handheld surface cleaner **10**, and a longitudinal axis defined by an inlet **64** of the handheld surface cleaner **10** intersects the handle **58**. Compared to other surface cleaners that have a handle on a side of the surface cleaner, such as on a top side, the handle **58** may make the handheld surface cleaner **10** easier for a user to hold and use in cleaning a surface since the handheld surface cleaner **10** will typically be used with the inlet **64** facing generally downward. Thus, most of a weight of the handheld surface cleaner **10**, and all of a weight of debris in the debris collection and storage cavity **16**, will be aligned with or below the handle **58** so a user does not have to counteract gravity as much as in using other surface cleaners that have a handle on a side of the surface cleaner, such as on a top side. Additionally, the handle **58** being so located is configured to leverage weight toward a front of the handheld surface cleaner **10**, which may help the inlet **64** be positioned against a surface being cleaned and thus be more likely to draw in more debris.

[0350] The power supply of the handheld surface cleaner **10** is also configured to power the separator motor **46** of the handheld surface cleaner **10**, which is configured to drive rotation of the separator **52**, and to power the suction motor **60** of the handheld surface cleaner **10**, which is configured to provide a suction force for suctioning debris into the inlet **64** (see FIGS. **1-4**) at a distal end of the handheld surface cleaner **10**.

[0351] As shown in FIG. **9**, the suction motor **60** is located at the main body **14** in this illustrated implementation. The suction motor **60** can include any of a variety of motors configured to provide suction, as will be appreciated by a person skilled in the art. Thus, the suction motor **60** is separated from the DWT **12** when the DWT **12** is de-coupled from the main body **14**.

[0352] The separator motor **46** is a direct current (DC) motor in this illustrated implementation but other types of motors are possible. As shown in FIG. **9**, the separator motor **46** is located at the main body **14** in this illustrated implementation. Thus, the separator motor **46** is separated from the DWT **12** when the DWT **12** is de-coupled from the main body **14**.

[0353] The separator motor **46** is operatively coupled to the separator **52** via a shaft **66**. The shaft **66** is configured to be driven by the separator motor **46** to rotate to cause rotation of the separator **52**.

[0354] The separator **52** is configured as a rotor configured to separate air from debris. The separator **52** is configured to rotate relative to the housing **26** of the DWT **12** and, if not omitted, the lower cover **30** of the DWT **12**. Rotation of the separator **52**, e.g., as driven by the separator motor **46**, is configured to cause debris entering the separator **52** to be expelled from the separator **52** into the debris collection and storage cavity **16** of the DWT **12** while allowing air entering the separator **52** to flow distally out of the separator **52** for exit from the handheld surface cleaner **10** through air exit holes **68** (see FIGS. **1**, **8**, and **9**). The air exit holes **68** are formed in the main body **14** adjacent to and laterally outward from the suction motor **60** in this illustrated implementation.

[0355] As shown in FIGS. **10**, **11**, and **14-16**, the separator **52** in this illustrated implementation includes an inner separator **52a** and an outer separator **52b** that is secured to the inner separator **52a**. The inner and outer separators **52a**, **52b** are configured to rotate together as a unit. In other implementations, the separator **52** is a singular member.

[0356] The inner separator **52a** includes a plurality of blades **52c** (also referred to herein as “vanes”) that extend radially outward from a center of the inner separator **52a**, which is also a

center of the separator **52**. Each of the blades **52c** is the same as the other blades **52c**. Each of the blades **52c** has a curvature to direct debris radially outward. A proximal end of the inner separator **52a** is open.

[0357] The outer separator **52b** includes a plurality of holes **52d** in a sidewall of the outer separator **52b**. Proximal and distal ends of the outer separator **52b** are open. Debris flowing into the separator **52** through the open distal end of the outer separator **52b** encounters the blades **52c** of the inner separator **52a**. The blades **52c** direct debris radially outward to fling the debris through the holes **52d** in the outer separator **52b**. Air flows into the separator **52** through the open distal end of the outer separator **52b** and passes through the separator **52** out of the open proximal end of the separator **52** defined by the open proximal ends of the inner and outer separators **52a**, **52b** to allow the air to continue flowing through the handheld surface cleaner **10** for exit from the handheld surface cleaner **10** through the air exit holes **68**.

[0358] As shown in FIG. **14**, the separator **52** includes an inlet opening **52e** and an outlet opening **52f**. Debris and air is configured to enter the separator **52** through the inlet opening **52e**. The debris is configured to exit the separator **52** through the plurality of holes **52d**. The air is configured to exit the separator **52** through the outlet opening **52f**. The separator **52** has a single inlet opening **52e** and a single outlet opening **52f** in this illustrated implementation but can include a plurality of inlet openings and/or a plurality of outlet openings. The inlet opening **52e** and the outlet opening **52f** in this illustrated implementation are each centered along a first longitudinal axis **A1** but the inlet opening **52e** and/or the outlet opening **52f** can be offset from the first longitudinal axis **A1**. The separator **52** is configured to rotate about the first longitudinal axis **A1**.

[0359] As shown in FIGS. **2** and **3**, the debris collection and storage cavity **16** surrounds an entire perimeter of the separator **52**. Thus, debris that the separator **52** flings radially outward through the plurality of holes **52d** will enter the debris collection and storage cavity **16** regardless of a particular rotational position of the separator **52** and regardless of an orientation in which the handheld surface cleaner **10** is being held. The plurality of holes **52d** in this illustrated implementation each have a substantially rectangular shape that is bent to be in two planes, but the plurality of holes **52d** can have another shape, e.g., circular, triangular, teardrop, irregular, or other shape.

[0360] As shown in FIGS. **2**, **3**, and **11**, the separator assembly **50** includes a seal **70** configured to help prevent debris from exiting out of the open proximal end of the separator **52**, e.g., out of the open proximal ends of the inner and outer separators **52a**, **52b**. The seal **70** at the separator **52** in this illustrated implementation is an oil seal but other types of seals may be used.

[0361] The handheld surface cleaner **10** includes two filters configured to allow the handheld surface cleaner collect debris within the handheld surface cleaner **10** while allowing air to flow out of the handheld surface cleaner **10**. In other implementations, a different number and/or different types of filters may be used.

[0362] As shown in FIGS. **2-4**, a first filter **72** of the handheld surface cleaner **10**, in the form of a hair cage in this illustrated implementation, is located at the DWT **12** upstream of (distal to) the separator **52**. The first filter **72** is configured to collect hair and/or other large debris that enters the handheld surface cleaner **10** through the inlet **64** to prevent the hair and/or other large debris from reaching the separator **52**, where the hair and/or other large debris may jam the separator **52**, clog the separator's holes **52d**, clog the separator's open proximal end, and/or and prevent rotation of the separator **52**.

[0363] The first filter **72** is removable from the DWT **12** to facilitate cleaning of the first filter **72**. FIG. **17** shows the first filter **72** as a standalone element. In other implementations, the first filter **72** is not removable from the DWT **12**.

[0364] In some implementations, a seal (not shown) is located upstream of or in the first filter **72** to help prevent backflow of hair and/or other matter trapped by the first filter **72** so the hair and/or other matter does not exit the handheld surface cleaner **10** through the inlet **64** of the handheld surface cleaner **10**. In some implementations, an attachment releasably coupled to the inlet **64** of

the handheld surface cleaner **10** includes such a seal. Attachments are discussed further below.

[0365] As shown in FIGS. **2**, **3**, **8**, and **9**, a second filter **74** of the handheld surface cleaner **10**, in the form of foam in this illustrated implementation, is located at the main body **14** downstream of (proximal to) the separator **52**. The second filter **74** is configured to collect the very small amounts of liquid and/or other debris that may inadvertently exit the separator **52** out of the open proximal end of the separator **52**. The second filter **74** may therefore help prevent liquid and/or other debris from reaching the air exit holes **68** and from exiting the handheld surface cleaner **10** through the air exit holes **68**, which would be messy and/or could clog one or more of the air exit holes **68**.

[0366] The second filter **74** is removable from the main body **14** to facilitate cleaning of the second filter **74**. FIG. **18** shows the second filter **74** as a standalone element. In other implementations, the second filter **74** is not removable from the main body **14**.

[0367] A debris flow path is defined through the handheld surface cleaner **10** by, in order, the inlet **64** at the distal end of the handheld surface cleaner **10**, the first filter **72**, a duckbill valve **76**, and the separator **52**, with the debris being collected in the debris collection and storage cavity **16** of the DWT **12**. The duckbill valve **76**, shown in FIGS. **2**, **3**, and **6**, is located upstream of (distal to) the separator **52** and is a seal configured to allow fluid flow in only one direction toward the separator **52**. The duckbill valve **76** is thus configured to prevent fluid from flowing out of the debris collection and storage cavity **16** of the DWT **12** and into the first filter **72** or out of the handheld surface cleaner **10** through the inlet **64**.

[0368] An air flow path is defined through the handheld surface cleaner **10** by, in order, the inlet **64** at the distal end of the handheld surface cleaner **10**, the first filter **72**, the duckbill valve **76**, the separator **52**, the separator bracket **48**, the second filter **74**, and a gap of space defined between a lower cover **78** (see FIGS. **2**, **3**, and **8**) of the main body **14** and a suction motor housing **80** (see FIG. **2**) of the main body **14**, with the air exiting the handheld surface cleaner **10** through the air exit holes **68**.

[0369] The debris and air flow paths are the same from the inlet **64** to the separator **52**. As shown in FIG. **2**, a flow path from the inlet **64** to the separator **52** defines the first longitudinal axis **A1**. As also shown in FIG. **2**, the handle **58** defines a second longitudinal axis **A2** that is at a transverse angle β relative to the first longitudinal axis **A1**. In an exemplary implementation, the transverse angle β is greater than about 90 degrees and less than about 180 degrees. For example, the angle β can in a range of about 135 degrees to about 175 degrees, e.g., about 140 degrees, about 145 degrees, about 150 degrees, about 155 degrees, about 160 degrees, about 165 degrees, etc. The transverse angle β being at least about 135 degrees and less than about 180 degrees may help provide a center of gravity so a user does not have to counteract gravity as much as with a smaller angle.

[0370] The debris that enters the separator **52** through the separator's inlet opening **52e** along the flow path is configured to travel from the flow path to the debris flow path in the handheld surface cleaner **10** and then flow along the debris flow path. As discussed above, the separator **52** is configured to direct the debris radially outward and into the debris collection and storage cavity **16**. The radial outward flow of the debris is in a direction radially outward from the first longitudinal axis **A1**.

[0371] The air that enters the separator **52** through the separator's inlet opening **52e** along the flow path is configured to travel from the flow path to the air flow path in the handheld surface cleaner **10** and then flow along the air flow path in the handheld surface cleaner **10**. The air flow path is different from the debris flow path. As also discussed above, the air is configured to flow out of the separator **52** through the separator's outlet opening **52f**. This portion of the air flow path is along the first longitudinal axis **A1**. A remainder of the air flow path is not along the first longitudinal axis **A1**. As discussed above, downstream of the separator **52**, the air flow path is through the separator bracket **48**, the second filter **74**, and the gap of space defined between the lower cover **78** and the suction motor housing **80**.

[0372] The handheld surface cleaner **10** includes a plurality of actuators **82**, in the form of buttons in this illustrated implementation, configured to be actuated by a user to control various functions of the handheld surface cleaner **10**. Each of the actuators **82** is operatively connected to the PCB **56**. The handheld surface cleaner **10** includes two actuators **82** in this illustrated implementation but can include another number of actuators **82**, e.g., one, three, four, etc.

[0373] In this illustrated implementation, a first one of the actuators **82** is a power button configured to control on/off of the handheld surface cleaner **10**. If the microswitch **40** is not activated, actuation of the first actuator **82** does not cause suctioning functionality of the handheld surface cleaner **10**. In other words, the controller is configured to allow activation of the suction motor **60** in response to actuation of the first actuator **82** if the microswitch **40** is activated, and the controller is configured to prevent activation of the suction motor **60** in response to actuation of the first actuator **82** if the microswitch **40** is not activated.

[0374] In this illustrated implementation, a second one of the actuators **82** is a boost button configured to control suction boost functionality. In response to actuation of the second actuator **82**, the controller is configured to increase suction provided by the suction motor **60**. In some implementations, the controller is configured to provide suction boost functionality as long as the second actuator **82** is being actuated, e.g., as long as a user holds down the second actuator **82**. Such a configuration gives the user greater suction control. In other implementations, the controller is configured to provide suction boost functionality for a predetermined amount of time in response to actuation of the second actuator **82**. If a user desires additional suction boost, the user can actuate the second actuator **82** again. Such a configuration helps prevent overheating of the suction motor **82**.

[0375] In some implementations, the handheld surface cleaner **10** includes a light emitting diode (LED) or other type of light. For example, each of the actuator(s) **82** can have an associated LED or other type of light. The associated light is configured to provide information to a user regarding the functionality of its associated actuator **82**, such as a light associated with the first actuator **82** being configured to indicate a power status of the handheld surface cleaner, e.g., illuminated for “on” and not illuminated for “off.” For another example, a light can be configured to illuminate to signal to a user that the DWT **12** is full, thereby signaling to the user that the DWT **12** should be emptied to allow for further suctioning of debris. For yet another example, a light associated with the first actuator **82** can be configured to indicate that the first actuator **82** was actuated but the handheld surface cleaner **10** was not powered on due to the DWT **12** not being properly attached to the main body **14**, e.g., as indicated by the activation state of the microswitch **40**.

[0376] In this illustrated implementation, the handheld surface cleaner **10**, e.g., the controller at the PCB **56**, is configured to determine automatically whether the DWT **12** is full. The fullness detection can be achieved in any of a number of ways. For example, the fullness of the DWT **12** can be determined using a current of a motor. If the current is above a predetermined threshold value, the DWT **12** can be considered to be full. For another example, the fullness of the DWT **12** can be determined using a pair of probes configured to extend into the DWT **12** to indicate a liquid level within the DWT **12**. No current will be conducted between the probes until each of the probes is in contact with liquid, e.g., an electrode of each probe is in contact with liquid. The current will thus indicate that a certain fill level in the DWT **12** has been reached.

[0377] The handheld surface cleaner **10**, e.g., the controller, in this illustrated implementation is configured to determine whether the DWT **12** is full based on a current of the separator motor **46**. A current sensor operatively coupled to the controller is configured to measure a current of the separator motor **46**, although current may be measured in another way. If the measured current is not above a predetermined threshold value stored in the memory at the PCB **56**, the handheld surface cleaner **10**, e.g., the controller, is configured to determine that the DWT **12** is not full. If the measured current is above the predetermined threshold value, the handheld surface cleaner **10**, e.g., the controller, is configured to determine that the DWT **12** is full. As debris enters the DWT **12** and

fills the DWT **12** enough to reach the separator **52**, the separator motor **46** pulls more and more current as the DWT's fill level increases and more and more debris is around the separator **52**. If the current of the separator motor **46** rises above the predetermined threshold value, the DWT **12** is considered to be full because of the amount of current being pulled by the separator motor **46**. [0378] In an exemplary implementation, the current of the separator motor **46** must be above the predetermined threshold value for a predetermined period of time before the current of the separator motor **46** is considered to be above the predetermined threshold value. Waiting the predetermined period of time may account for temporary position of the handheld surface cleaner **10** in which the separator motor **46** may be only temporarily pulling current above the predetermined threshold value because, given the orientation that a user is holding the handheld surface cleaner **10**, gravity is causing the debris to settle temporarily around the separator **52** to give a false indication of the DWT **12** being full.

[0379] In response to determining that the DWT **12** is full, the handheld surface cleaner **10**, e.g., the controller, is configured to cause at least one action to be performed. For example, the handheld surface cleaner **10**, e.g., the controller, can be configured to cause a light of the handheld surface cleaner **10** to be illuminated to signal a user that the DWT **12** should be emptied to allow for further suctioning of debris. For another example, the handheld surface cleaner **10**, e.g., the controller, can be configured to cause each of the separator motor **46** and the suction motor **60** to turn off in response to determining that the DWT **12** is full, thereby preventing more debris from being suctioned into the handheld surface cleaner **10**. In some implementations, the handheld surface cleaner **10**, e.g., the controller, can cause the suction motor **60** to turn off before the separator motor **46** is turned off to allow the separator motor **46** to continue running to separate any debris and air that has already entered the handheld suction cleaner **10** and to help clean the separator **52** before a next rotation of the separator **52**. In some implementations, the handheld surface cleaner **10**, e.g., the controller, can cause the suction motor **60** and the separator motor **46** to turn off simultaneously. For yet another example, the handheld surface cleaner **10**, e.g., the controller, can be configured to cause the separator motor **46** to be turned off after the suction motor **60** is turned off in response to determining that the measured current is above a first predetermined threshold value and to cause the suction motor **60** and the separator motor **46** to turn off simultaneously in response to determining that the measured current is above a second predetermined threshold value that is greater than the first predetermined threshold. Using the second predetermined threshold value is a safety feature for short protection.

[0380] The inlet **64** at the distal end of the handheld surface cleaner **10** is configured to releasably couple to an attachment to help facilitate surface cleaning. In an exemplary implementation, a plurality of attachments are provided as a kit with the handheld surface cleaner **10**. The plurality of attachments are configured to be selectively coupled to the handheld surface cleaner **10**, one at a time, based on the particular type of cleaning desired by a user. The attachments described herein are described with respect to a handheld surface cleaner but can similarly be used with another type of cleaner, such as an upright vacuum cleaner including a hose configured to removably attach to an attachment. Additionally, the attachments described herein are described as being removably attachable to a cleaner but can instead be non-removably attached to a cleaner.

[0381] FIGS. **19-22** illustrate one implementation of an attachment in the form of a stain tool attachment **100** configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner **10** of FIG. **1**. FIG. **19** shows the stain tool attachment **100** releasably coupled to the handheld surface cleaner **10** of FIG. **1**. FIGS. **20-22** show the stain tool attachment **100** as a standalone element.

[0382] As shown in FIGS. **20-22**, the stain tool attachment **100** includes an inlet opening **102** and an outlet opening **104**. Debris is configured to enter the stain tool attachment **100** through the inlet opening **102**, e.g., under a suction force provided by a handheld surface cleaner to which the stain tool attachment **100** is attached, and to exit the stain tool attachment **100** through the outlet opening

104, e.g., to pass into a handheld surface cleaner to which the stain tool attachment **100** is attached. The inlet opening **102** is a single opening and has a rectangular shape in this illustrated implementation but can instead include a plurality of openings and/or have a different shape. The outlet opening **104** is a single opening and has a circular shape in this illustrated implementation but can instead include a plurality of openings and/or have a different shape.

[0383] As shown in FIG. **21**, the stain tool attachment **100** includes a seal **106**. The seal **106** is located between the inlet opening **102** and the outlet opening **104** of the stain tool attachment **100**. The seal **106** is configured to move between a closed position (shown in FIG. **21**) and an open position.

[0384] The seal **106** is a dynamic seal configured to move automatically between the open and closed positions in response to application and removal of a suction force, e.g., a suction force provided by a handheld surface cleaner to which the stain tool attachment **100**. The application of the suction force is configured to cause the seal **106** to move from the closed position to the open position, e.g., by pulling back a plurality of flexible flaps **106a** of the seal **106** attached to the stain tool attachment **100** along their radially outward edges. In the open position the seal **106** is configured to allow flow from the inlet opening **102** to the outlet opening **104**. The removal of the suction force is configured to cause the seal **106** to move from the open position to the closed position, e.g., by allowing the plurality of flexible flaps **106a** of the seal **106** to dynamically flex to their default positions (shown in FIG. **21**). In the closed position, the seal **106** is configured to prevent flow from the inlet opening **102** to the outlet opening **104** and to prevent flow from the outlet opening **104** to the inlet opening **102**. Preventing flow from the inlet opening **102** to the outlet opening **104** stops any debris that inadvertently enters the inlet opening **102** from passing out of the outlet opening **104**, whether or not the stain tool attachment **100** is attached to a handheld surface cleaner. Preventing flow from the outlet opening **104** to the inlet opening **102** stops any debris that inadvertently enters the outlet opening **104** from passing out of the inlet opening **102**, whether or not the stain tool attachment **100** is attached to a handheld surface cleaner.

[0385] FIGS. **23A-23C** illustrate another implementation of an attachment in the form of a stain tool attachment **200** configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner **10** of FIG. **1**. FIGS. **23A-23C** show the stain tool attachment **200** as a standalone element. The stain tool attachment **200** of FIGS. **23A-23C** is generally configured and used similar to the stain tool attachment **100** of FIGS. **18-21**, e.g., includes an inlet opening **202**, an outlet opening **204**, and a seal **206**, but has a larger size than the stain tool attachment **100** of FIGS. **19-22**. The inlet opening **202** in this illustrated implementation includes a plurality of openings (3) but can instead have another plural number of openings or, as mentioned above, can instead be a single opening.

[0386] The stain tool attachment **200** includes an agitator **208** configured to agitate debris on a surface being cleaned to help the debris be drawn into the inlet opening **202** of the stain tool attachment **200**. The agitator **208** is located behind (proximal to) the inlet opening **202** in this illustrated implementation.

[0387] The agitator **208** can have a variety of configurations. As shown in FIG. **23B**, the agitator **208** includes a plurality of bristles. Bristles are typically more effective in agitating certain types of debris than other types of debris, such as being generally more effective in agitating solid debris than liquid debris. Rather than require a user to detach the stain tool attachment **200** from a cleaner to which the stain tool attachment **200** is attached, or to retrieve another attachment even if the stain tool attachment **200** is not currently attached to a cleaner, the agitator **208** is configured to removably attach to the stain tool attachment **200** to allow for a different type of agitation mechanism to be coupled to the stain tool attachment **200**. FIGS. **23A-23C** show the agitator **208** removably coupled to the stain tool attachment **200**. FIGS. **24A** and **24B** show the agitator **208** as a standalone element. In some implementations, a different agitator can be coupled to the stain tool attachment **200** to allow for the different type of agitation mechanism. In other implementations, as

in this illustrated implementation, the same agitator **208** is configured to be coupled to the stain tool attachment **200** in different orientations, with each orientation providing for a different agitation mechanism.

[0388] An attachment, e.g., any of the attachments described herein, can include a grip feature. The grip feature can have a variety of configurations. For example, the grip feature can include a textured surface. For another example, the grip feature can include a plurality of raised dimples. For another example, the grip feature can include a plurality of recessed dimples. For yet another example, the grip feature can include a plurality of raised ribs. For still another example, the grip feature can include a plurality of grooves.

[0389] FIG. **23D** illustrates another implementation of an attachment in the form of a stain tool attachment **200'** configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner **10** of FIG. **1**. The stain tool attachment **200'** of FIG. **23D** is the same as the stain tool attachment **200** of **23A-23C** except that the stain tool attachment **200'** of FIG. **23D** includes a grip feature **200''**. The grip feature **200''** is configured to improve a user's hold of the stain tool attachment **200'** and/or to indicate to a user, e.g., visually and/or tactilely, where the attachment **200'** should be held for efficient attachment of the attachment **200'** to and removal of the attachment **200'** from a handheld surface cleaner. The grip feature **200''** can have a variety of configurations. The grip feature **200''** in this illustrated implementation includes a plurality of raised ribs on opposed sides of the attachment **200'** (only one side is visible in the view of FIG. **23D**).

[0390] As shown in FIGS. **24A** and **24B**, the agitator **208** has two faces **208a**, **208b** each with a different type of agitation mechanism. A first face **208a**, shown in FIGS. **23A-24A**, of the agitator **208** includes the plurality of bristles. With the agitator **208** coupled to the stain tool attachment **200** in a first orientation, as shown in FIGS. **23A-23C**, the plurality of bristles are provided as the agitation mechanism. A second face **208b**, shown in FIG. **24B**, of the agitator **208** includes one or more ribs. With the agitator **208** coupled to the stain tool attachment **200** in a second orientation, the rib(s) are provided as the agitation mechanism. Ribs are typically more effective in agitating certain types of debris than other types of debris, such as being generally more effective in agitating liquid debris than solid debris.

[0391] The agitator **208** is configured to removably attach to the stain tool attachment **200** with a snap lock mechanism but another attachment mechanism may be used. The agitator **208** includes a tab **208c** configured to be held to facilitate manual coupling of the agitator **208** to the stain tool attachment **200** and manual de-coupling of the agitator **208** from the stain tool attachment **200**. The tab **208c** includes an opening **208d** configured to releasably receive therein a protrusion **200a** of the stain tool attachment **200** to lock together the agitator **208** and the stain tool attachment **200**. In other implementations, the tab **208** can include the protrusion and the stain tool attachment **200** can include the opening.

[0392] The stain tool attachment **100** of FIGS. **19-22** includes an agitator **102** that is generally configured and used similar to the agitator **208** of FIGS. **23A-24B**, although as mentioned above the agitator **108** can instead be non-removably attached to the stain tool attachment **100**.

[0393] FIGS. **25A-25C** illustrate another implementation of an attachment in the form of a crevice tool attachment **300** configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner **10** of FIG. **1**. The stain tool attachment **300** of FIGS. **25A-25C** is generally configured and used similar to the stain tool attachment **100** of FIGS. **18-21**, e.g., includes an inlet opening **302**, an outlet opening **304**, a seal (obscured in the figures), and an agitator **308**. The agitator **308** is configured to removably attach to the crevice tool attachment **300** but can instead be non-removably attached to the crevice tool attachment **300**. FIG. **25A** shows the agitator **308** coupled to the crevice tool attachment **300**. FIGS. **25B** and **25C** show the agitator **308** de-coupled from the crevice tool attachment **300**. In this illustrated implementation, the agitator **308** is configured to surround the inlet opening **302**, includes an agitation mechanism in the form of

bristles, and includes a pair of prongs **308a** configured to snap lock in a corresponding pair of openings **300a** of the stain tool attachment **300**.

[0394] FIGS. **26-28** illustrate another implementation of an attachment in the form of a crevice tool attachment **310** configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner **10** of FIG. **1**. FIGS. **26-28** show the crevice tool attachment **310** as a standalone element. The crevice tool attachment **310** of FIGS. **26-28** is generally configured and used similar to the stain tool attachment **100** of FIGS. **18-21**, e.g., includes an inlet opening **312**, an outlet opening **314**, a seal **316**, and an agitator **318**. The agitator **318** is non-removably attached to the crevice tool attachment **310** but can instead be configured to removably attach to the crevice tool attachment **310**. In this illustrated implementation, the agitator **318** includes an agitation mechanism in the form of bristles.

[0395] FIGS. **29A-29C** illustrate another implementation of an attachment in the form of a pet hair tool attachment **400** configured to releasably couple to an inlet of a handheld surface cleaner such as the handheld surface cleaner **10** of FIG. **1**. FIGS. **29A-29C** show the pet hair tool attachment **400** as a standalone element. The pet hair tool attachment **400** of FIGS. **29A-29C** is generally configured and used similar to the stain tool attachment **100** of FIGS. **18-21**, e.g., includes an inlet opening **402**, an outlet opening **404**, a seal (obscured in the figures), and an agitator **408**. The agitator **408** is non-removably attached to the pet hair tool attachment **400** but can instead be configured to removably attach to the pet hair tool attachment **400**. In this illustrated implementation, the agitator **408** includes an agitation mechanism in the form of ribs.

[0396] Debris entering the attachments **100**, **200**, **300**, **310** of FIGS. **19-23C** and **25A-28** is configured to pass through the attachment (with the seal in the open position) without the debris being collected on board the attachment **100**, **200**, **300**, **310**. The pet hair tool attachment **400** of **29A-29C** includes a debris storage container **410** configured to collect at least some debris entering the pet hair tool attachment **400**.

[0397] The debris storage container **410** is configured to removably attach to the pet hair tool attachment **400** to facilitate emptying and cleaning of the storage container **410**. FIGS. **29A-29C** show the debris storage container **410** coupled to the pet hair tool attachment **400**. FIGS. **30** and **31** show the debris storage container **410** detached from the pet hair tool attachment **400**.

[0398] In this illustrated implementation, with the debris storage container **410** attached to the pet hair tool attachment **400**, the debris storage container **410** is configured to be rotated in a first direction, e.g., clockwise, to move the debris storage container **410** from a locked configuration, in which the debris storage container **410** is locked to the pet hair attachment tool **400**, to an unlocked configuration, in which the debris storage container **410** is unlocked from the pet hair attachment tool **400**. With the debris storage container **410** detached from the pet hair tool attachment **400**, the debris storage container **410** is configured to be rotated in a second direction, e.g., counterclockwise, to move the debris storage container **410** from the unlocked configuration to the unlocked configuration.

[0399] The debris storage container **410** and the pet hair tool attachment **400** in this illustrated implementation are attachable and detachable via a bayonet mount. The debris storage container **410** includes a plurality of male members and the pet hair tool attachment **400** includes a plurality of female members, although in other implementations the debris storage container **410** can include a plurality of female members and the pet hair tool attachment **400** can include a plurality of male members. With the debris storage container **410** in the locked configuration, the female and male members are misaligned. Moving the debris storage container **410** from the locked configuration to the unlocked configuration is configured to align the female and male members to allow removal of the debris storage container **410** from the pet hair tool attachment **400**. Similarly, moving the debris storage container **410** from the unlocked configuration to the locked configuration is configured to misalign the female and male members.

[0400] As shown in FIG. **30**, the pet hair attachment tool **400** includes a filter **412** located in a flow

path between the inlet opening **402** and the outlet opening **404**. The filter **412** is non-removably attached to the pet hair attachment tool **400** in this illustrated implementation but can instead be removably attached to the pet hair attachment tool **400**.

[0401] Debris small enough to pass through the filter **412** is configured to flow from the inlet opening **402** to the outlet opening **404** (with the seal **406** in the open position). Debris too large to pass through the filter **412** is configured to flow from the inlet opening **402** to a debris collection and storage cavity **410a** of the debris storage container **410** because the filter **412** is configured to flow prevent the too-large debris from continuing to the outlet opening **404**. With the debris storage container **410** attached to the pet hair tool attachment **400**, the filter **412** is at least partially located in the debris collection and storage cavity **410a**. As shown in FIG. **31**, the debris storage container **410** includes a side opening **410b** in fluid communication with the inlet opening **402** to allow debris entering the inlet opening **402** to flow into the debris storage container **410**, e.g., into the debris collection and storage cavity **410a**, with debris less than a certain size defined by the filter **412** passing through the filter **412** and out a top opening **410c** of the debris storage container **410** to continue along the flow path and with debris less than the certain size remaining in the debris storage container **410**, e.g., in the debris collection and storage cavity **410a**.

[0402] FIG. **32** illustrates another implementation of a separator assembly **500** of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The separator assembly **500** of FIG. **32** is generally configured and used similar to the separator assembly **50** of FIGS. **10-13** except that, unlike the side holes **52d** in the separator **52** of FIGS. **10-13** that are open regardless of whether the separator **52** is rotating, holes (obscured by movable covers **502** in FIG. **32**) formed in a sidewall of a separator **504** of the separator assembly **500** of FIG. **32** are not open regardless of whether the separator **504** is rotating. With the separator **504** in a static, non-rotating position, the holes of the separator **504** are closed, as shown in FIG. **32**. With the separator **504** rotating, the holes of the separator **504** are open. The holes being open or closed depending on whether the separator **504** is rotating may help prevent any debris from passing into a static, non-rotating separator **504** from the handheld surface cleaner's collection and storage cavity, as such passage may jam the separator **504** and/or allow for debris to exit out of the handheld surface cleaner's inlet.

[0403] As discussed further below, in an exemplary implementation, each of the covers **502** is configured as a centrifugal seal configured to respond to a centrifugal force to move automatically between sealing the separator's holes and not sealing the separator's holes. Without the covers **502** sealing the separator's holes, debris is free to pass through the holes from inside the separator to outside the separator, e.g., to a DWT of the cleaner that includes the separator. With the covers **502** sealing the separator's holes, debris in the DWT cannot enter the separator through the holes.

[0404] The holes formed in the separator of the separator assembly of FIG. **32**, e.g., formed in an outer separator **506** of the separator **504**, each have an associated movable cover **502** in the form of a flap. The movable cover **502** is attached to the separator **504**, e.g., to the outer separator **506** and/or to the separator's inner separator **508**, along one edge of the movable cover **502**. The edges of the movable covers **502** that are attached to the separator **504** are the trailing edges of the movable covers **502** in a direction of the separator's rotation. In a resting configuration, shown in FIG. **32** and corresponding to the separator **504** being static and non-rotating, the movable covers **502** cover the separator's holes so that the side holes are closed and are sealed. In an active configuration, corresponding to the separator **504** rotating, the movable covers **502** do not cover the separator's holes so that the holes are open and are not sealed. Because the edges of the movable covers **502** that are attached to the separator **504** are the trailing edges of the movable covers **502**, the separator's rotation will urge the movable covers **502** to open automatically in response to centrifugal force. When the separator **504** stops rotating, the movable covers **502** will automatically move from the active configuration to the resting configuration since the centrifugal force is no longer present.

[0405] As in this illustrated implementation, each of the movable covers **502** can be configured to selectively cover one of the separator's holes. In another implementation, one or more of the movable covers **502** can be configured to selectively cover more than one of the separator's holes, e.g., cover two of the separator's holes, etc.

[0406] In an exemplary implementation, the movable covers **502** are formed of rubber and/or other flexible material configured to help "blow" the movable covers **502** off their respective holes and to help the movable covers **502** seal the holes with the movable covers **502** in the resting configuration.

[0407] FIG. **33** illustrates another implementation of a handheld surface cleaner **600** configured to use suction to draw debris into the handheld surface cleaner **600** through an inlet **602** and to separate the air and the debris on board the handheld surface cleaner **600**. The handheld surface cleaner **600** of FIG. **33** is generally configured and used similar to the handheld surface cleaner **10** of FIG. **1** except that instead of a DWT **604** of the handheld surface cleaner **600** being pivotally attached to a main body **606** of the handheld surface cleaner **600**, the DWT **604** has a straight, non-pivoting connection with the main body **604**. The DWT **604** includes a lock **608** configured to be pressed by a user to release the DWT **604** from the main body **606**. FIG. **34** shows the DWT **604** of the handheld surface cleaner **600** of FIG. **33** as a standalone element detached from the main body **606**.

[0408] The lock **608** is spring-loaded to allow a user to push the main body **606** and the dirty water tank **604** together, e.g., moving one or both of the DWT **604** and the main body **606** toward the other in a straight, longitudinal motion, to automatically lock the DWT **604** and the main body **606** together until a user actuates the lock **608**.

[0409] The lock **608** in this illustrated implementation includes a pair of buttons on opposite sides of the DWT **604** around a perimeter of the DWT **604**. The lock **608** is configured to move between a locked configuration, shown in FIGS. **33** and **34**, and an unlocked configuration. In the locked configuration, the lock **608** is configured to lock the DWT **604** and the main body **606** together. In the unlocked configuration, the lock **608** is configured to allow the DWT **604** and the main body **606** to be de-coupled.

[0410] With the lock **608** in the locked configuration and locking the DWT **604** and the main body **606** together, the lock **608** is configured to be moved manually by a user from the locked configuration to the unlocked configuration, which allows a user to decide whether and when to de-couple the DWT **604** and the main body **606**. The buttons of the lock **608** are configured to be pushed radially inward relative to a housing **610** of the DWT **604** (and a lower cover **612** of the DWT **604** if not omitted) to move from the locked configuration to the unlocked configuration. The lock **608** is biased to the locked configuration, and the buttons of the lock **608** are spring-loaded. Thus, when a user stops pushing the buttons radially inward, the lock **608** is configured to move automatically from the unlocked configuration to the locked configuration.

[0411] With the lock **608** in the locked configuration and without DWT **604** being coupled to the main body **606**, the lock **608** is configured to move automatically from the locked configuration to the unlocked configuration and back to the locked configuration in response to the DWT **604** being coupled to the main body **606**. Pushing the DWT **604** and the main body **606** together longitudinally is configured to cause the lock **608** to move automatically from the locked configuration to the unlocked configuration and back to the locked configuration. The spring-loading of the lock **608** is configured to allow the automatic movement of the lock **608** between the unlocked and locked configurations. A user thus does not need to take any specific action to lock the DWT **604** and the main body **606** together because merely attaching the DWT **604** and the main body **606** together automatically locks the DWT **604** and the main body **606** together until the user decides to unlock the DWT **604** from the main **606**.

[0412] FIG. **35** illustrates another implementation of a handheld surface cleaner **700** configured to use suction to draw debris into the handheld surface cleaner **700** and to separate the air and the

debris on board the handheld surface cleaner **700**. The handheld surface cleaner **700** of FIG. **35** is generally configured and used similar to the handheld surface cleaner **10** of FIG. **1** except that, instead of the handheld surface cleaner's separator motor **46** being located at the handheld surface cleaner's main body **14**, a separator assembly **702** of the handheld surface cleaner's DWT **704** includes a separator motor **706**, as shown in FIGS. **35-37**.

[0413] To allow the separator motor **706** to be powered by a power supply located at the main body **708**, e.g., by one or more batteries in a battery holder at a handle **710** of the main body **708**, the DWT **704** includes at least one electrical connection **712** (two electrical connections are shown in the illustrated implementation of FIG. **37**), e.g., a conductive plate or other electrical connector, operatively connected to the separator motor **706** and configured to contact at least one corresponding electrical connection **714** (two electrical connections are shown in the illustrated implementation of FIG. **38**), e.g., a conductive pin or other electrical connector, of the main body **708** that is operatively connected to the power supply. The attachment of the DWT **704** and the main body **708** is configured to cause each of the DWT's one or more electrical connections **712** to conductively engage automatically with the main body's one or more electrical connections **714**. Detaching the DWT **704** and the main body **708** is configured to cause the one or more electrical connections **712** to automatically no longer conductively engage with the main body's one or more electrical connections **714**.

[0414] The electrical connection between the DWT **704** and the main body **708** is configured to be isolated from any air flowing the handheld surface cleaner **700** and from any debris in the handheld surface cleaner **700**, whether the debris is flowing in the handheld surface cleaner **700** or is collected in the DWT's debris collection and storage cavity **716**. Such isolation may help prevent a short circuit and/or may help prevent rusting and/or other damage to the DWT's one or more electrical connections **712** and/or the main body's one or more electrical connections **714**. With the DWT **704** and the main body **708** attached together, the DWT's one or more electrical connections **712** and the main body's one or more electrical connections **714** are configured to be isolated from any debris flowing in the handheld surface cleaner **700**, from any air flowing in the handheld surface cleaner **700**, and from any debris collected in the DWT's debris collection and storage cavity **716**. The electrical connection between the DWT **704** and the main body **708** is thus isolated from any debris flowing in the handheld surface cleaner **700**, from any air flowing in the handheld surface cleaner **700**, and from any debris collected in the DWT's debris collection and storage cavity **716**.

[0415] In this illustrated implementation, the at least one electrical connection **712** of the DWT **704** includes a pair of conductive plates, and the at least one corresponding electrical connection **714** of the main body **708** includes a pair of pogo pins. The attachment of the DWT **704** and the main body **708** is configured to cause each of the pogo pins **714** to compress automatically in response to pressing against a respective one of the conductive plates **712**. Detaching the DWT **704** and the main body **708** is configured to cause the pogo pins **714** to decompress automatically in response to no longer pressing against either of the conductive plates **712**. In another implementation, the DWT **704** includes the one or more pogo pins, and the main body **708** includes the one or more conductive plates.

[0416] FIG. **39** illustrates another implementation of a handheld surface cleaner **800** configured to use suction to draw debris into the handheld surface cleaner **800** and to separate the air and the debris on board the handheld surface cleaner **800**. The handheld surface cleaner **800** of FIG. **39** is generally configured and used similar to the handheld surface cleaner **10** of FIG. **1** except that a separator assembly **802** of the handheld surface cleaner's DWT **804** includes a separator motor **806**, in addition to a separator **808**, a separator bracket **810**, a shaft **812**, and a seal **814** (an oil seal in this illustrated implementation), similar to the implementation shown in FIG. **35**, and except that, as shown in FIGS. **39** and **40**, no drive dog is used in driving rotation of the handheld surface cleaner's separator motor **806** (see FIG. **5** showing a drive dog **84** of the handheld surface cleaner

10).

[0417] FIG. 41 illustrates another implementation of a handheld surface cleaner 900 configured to use suction to draw debris into the handheld surface cleaner 900 and to separate the air and the debris on board the handheld surface cleaner 900. The handheld surface cleaner 900 of FIG. 41 is generally configured and used similar to the handheld surface cleaner 10 of FIG. 1 except that, as shown in FIGS. 41 and 42, a separator assembly 902 of the handheld surface cleaner's DWT 904 includes a separator motor 906, in addition to a separator 908, a separator bracket 910, and a shaft 912 similar to the implementation shown in FIG. 35; except that, as shown in FIGS. 42 and 43, holes (obscured in FIG. 43 by movable covers 914) formed in the separator 908 of the separator assembly 902 of FIGS. 42 and 43 each have an associated movable cover 914 in the form of a flap similar to the implementation shown in FIG. 32; except that, as shown in FIG. 41, the DWT 904 includes a spring-loaded spout 916 instead of a cap; and except that, as shown in FIGS. 41-44, the handheld surface cleaner 900 includes a centrifugal seal 918 upstream of (distal to) the separator 908 and does not include a duckbill valve upstream of the separator 908. The centrifugal seal 918 is a seal configured to allow fluid flow toward the separator 908. The centrifugal seal 918 is also configured to prevent fluid from flowing out of a debris collection and storage cavity 920 of the DWT 904 and into a hair cage 922 or an inlet 924 of the handheld surface cleaner 900.

[0418] As in this illustrated implementation, each of the movable covers 914 can be configured to selectively cover one of the separator's holes. In another implementation, one or more of the movable covers 914 can be configured to selectively cover more than one of the separator's holes, e.g., cover two of the separator's holes, etc.

[0419] As shown in FIG. 43, the separator assembly 902 includes the centrifugal seal 918. The centrifugal seal 918 is thus removable from the DWT 904 with a remainder of the separator assembly 902.

[0420] The centrifugal seal 918 is configured to move between a first configuration and a second configuration. The centrifugal seal 918 is configured to be in the first configuration with the separator 908 being in a resting configuration in which the separator 908 is not rotating. In the first configuration, the centrifugal seal 918, e.g., a distal end of the centrifugal seal 918, abuts, as shown in FIGS. 41 and 42, an inlet hosing 926, e.g., a proximal end of the inlet hosing 926, in which the hair cage 922 is at least partially disposed. With the centrifugal seal 918 abutting the inlet hosing 926, fluid cannot flow out of the debris collection and storage cavity 920 of the DWT 904 and into the hair cage 922 or the inlet 924. The movable covers 914, e.g., flaps, covering the holes of the separator 908 with the separator 908 in the resting configuration are configured to prevent any debris in the handheld surface cleaner's debris collection and storage cavity 920 from passing into the separator 908 and from the separator 908 into the centrifugal seal 918 where debris may be free to exit through the handheld surface cleaner's inlet 924. The centrifugal seal 918 and the movable covers 914 are thus configured to cooperate to prevent debris in the debris collection and storage cavity 920 from inadvertently exiting the handheld surface cleaner 900 when suction is not being applied and the separator 908 is not rotating.

[0421] The centrifugal seal 918 is configured to be in the second configuration with the separator 908 being in a rotating configuration in which the separator 908 is rotating. In the second configuration, the centrifugal seal 918, e.g., the distal end of the centrifugal seal 918, does not abut the air inlet hosing 926, e.g., the proximal end of the air inlet hosing 926, and a gap of space exists between the centrifugal seal 918 and the air inlet hosing 926. The centrifugal seal 918 is hollow and has a passageway extending therethrough so that with the separator 908 and the centrifugal seal 918 rotating, debris and air can pass from the air inlet hosing 926 and through the centrifugal seal 918 to the separator 908 to allow the separator 908 to separate the debris and the air. Because of the suction force provided via the handheld surface cleaner's suction motor 928 (at the handheld surface cleaner's main body 930, as shown in FIG. 41), debris and air passing out of the air inlet hosing 926 will be urged into the centrifugal seal 918 and thus into the separator 908 instead of

passing through the gap of space into the handheld surface cleaner's debris collection and storage cavity **920**.

[0422] The centrifugal seal **918** is attached to the separator **908**, as shown in FIGS. **41-44**, e.g., by a proximal end of the centrifugal seal **918** being attached to a distal end of the separator **908**. The centrifugal seal **918** is thus configured to rotate with the separator **908**. The centrifugal seal **918** is configured to move automatically from the first configuration to the second configuration in response to the rotation of the centrifugal seal **918** and thus in response to the rotation of the separator **908**. The centrifugal seal **918** is configured to move automatically from the second configuration to the first configuration in response to the rotation of the centrifugal seal **918** stopping and thus in response to the rotation of the separator **908** stopping. In an exemplary implementation, the centrifugal seal **918** is formed of rubber and/or other flexible material configured to help allow flexing of the centrifugal seal **918** as the centrifugal seal **918** moves dynamically between the first and second configurations.

[0423] As shown in FIGS. **41**, **42**, and **44**, the separator assembly **902** also includes a seal **929** configured to help prevent debris from exiting out of the open proximal end of the separator **908**, e.g., out of the open proximal ends of the separator's inner and outer separators **908a**, **908b**. The seal **928** at the separator **908** in this illustrated implementation is an oil seal but other types of seals may be used.

[0424] FIG. **45** illustrates another implementation of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The handheld surface cleaner of FIG. **45** is generally configured and used similar to the handheld surface cleaner of FIG. **41** except that a bearing **1000** is used instead of the oil seal **929**.

[0425] FIG. **46** illustrates another implementation of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The handheld surface cleaner of FIG. **46** is generally configured and used similar to the handheld surface cleaner of FIG. **41** except that the oil seal **929** is replaced with a rotatable V-ring face seal **1100**.

[0426] FIG. **47** illustrates another implementation of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The handheld surface cleaner of FIG. **47** is generally configured and used similar to the handheld surface cleaner of FIG. **41** except that a static (non-rotating) V-ring face seal **1200** is used instead of the oil seal **929**.

[0427] FIG. **48** illustrates another implementation of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The handheld surface cleaner of FIG. **48** is generally configured and used similar to the handheld surface cleaner of FIG. **41** except that a convoluted path **1300** is used instead of the oil seal **929**.

[0428] FIG. **49** illustrates another implementation of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The handheld surface cleaner of FIG. **19** is generally configured and used similar to the handheld surface cleaner of FIG. **41** except that a four lip seal **1400** is used instead of the oil seal **929**.

[0429] FIGS. **50A-50D** illustrate another implementation of a separator **1500** of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The separator **1500** of FIGS. **50A-50D** is generally configured and used similar to the separator **908** of FIGS. **41-44** except that the separator **1500** of FIGS. **50A-50D** is attached to a rim **1502** and the separator **1500** does not include inner and outer separators.

[0430] The separator **1500** in this illustrated implementation is a singular member. Movable covers

1504, which are configured and used similar to the movable covers **502**, **914** discussed above, are attached to the separator **1500**, such as being glued to the separator **1500**, press fit with the separator **1500**, overmolded with the separator **1500**, or otherwise attached, and cover the separator's holes so that the holes are closed with the separator **1500** being static and non-rotating. FIG. **50C** shows a connection area **1506** for the separator **1500** and one of the movable covers **1502** as representative of each of the movable covers **1502**.

[0431] As in this illustrated implementation, each of the movable covers **1504** can be configured to selectively cover one of the separator's holes. In another implementation, one or more of the movable covers **1504** can be configured to selectively cover more than one of the separator's holes, e.g., cover two of the separator's holes, etc.

[0432] The rim **1502** is attached on a first side, e.g., proximal side, to a distal end of the separator **1500** and on a second, opposite side, e.g., distal side, to a centrifugal seal **1508**. The separator **1500** is thus attached indirectly to the centrifugal seal **1504** via the rim **1502**. The centrifugal seal **1504** is generally configured and used similar to the centrifugal seal **918** of FIGS. **41-44**. The rim **1502** is configured to rotate as a unit with the separator **1500** and the centrifugal seal **1504**.

[0433] The rim **1502** extends entirely around a perimeter of the separator's distal end and entirely around a perimeter of the centrifugal seal's proximal end. The rim **1502** is hollow and is configured to allow any debris flowing proximally through the centrifugal seal **1508** to pass through the rim **1502** to the separator **1508**. The rim **1502** is configured to provide a barrier for backflow so any debris in a debris collection and storage cavity of the handheld surface cleaner's dirty water tank cannot enter one or both of the centrifugal seal **1508** and the separator **1500** through any space between the centrifugal seal **1508** and the separator **1500** since the rim **1502** is blocking any such space. The rim **1502** and the movable covers **1504** are thus configured to cooperate to prevent debris in the debris collection and storage cavity from inadvertently exiting the handheld surface cleaner when suction is not being applied and the separator **1500** is not rotating.

[0434] As shown in FIG. **50A**, the separator **1500** includes an inlet opening **1500a** and an outlet opening **1500b**. Debris and air is configured to enter the separator **1500** through the inlet opening **1500a**. The debris is configured to exit the separator **1500** through the separator's plurality of holes. The air is configured to exit the separator **1500** through the outlet opening **1500b**. The separator **1500** has a single inlet opening **1500a** and a single outlet opening **1500b** in this illustrated implementation but can include a plurality of inlet openings and/or a plurality of outlet openings. The inlet opening **1500a** and the outlet opening **1500b** in this illustrated implementation are each centered along a central rotational axis **A3** of the separator **1500** but the inlet opening **1500a** and/or the outlet opening **1500b** can be offset from the central rotational axis **A3**.

[0435] FIG. **50D** shows an implementation of a separator assembly **1510** including the separator **1500**, the rim **1502**, and the centrifugal seal **1508** and also including a separator bracket **1512**, a motor (obscured in FIG. **50D**), and a shaft (obscured in FIG. **50D**). The separator assembly **1510** is generally configured and used similar to the separator assembly **902** of FIG. **43**.

[0436] FIG. **51** illustrates another implementation of a separator **1600** of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The separator **1600** of FIG. **51** is generally configured and used similar to the separator **908** of FIGS. **41-44** except that the separator **1600** of FIG. **51** is not attached to a centrifugal seal (directly or indirectly), does not include movable covers, and does not include inner and outer separators. The separator **1600** in this illustrated implementation is a singular member.

[0437] As shown in FIG. **51**, a separator bracket **1602** is located downstream of the separator **1600**. The separator bracket **1602** is generally configured and used similar to the separator bracket **48** of FIGS. **2** and **3** discussed above. In this illustrated implementation, a seal **1604** is positioned between the separator **1600** and the separator bracket **1602**. The seal **1604** is attached to the separator bracket **1602** and thus does not rotate with the separator **1600**.

[0438] The seal **1604** is hollow with a continuous sidewall. With the separator **1600** rotating, the seal **1604** is configured to allow any debris flowing through the separator **1600** that is not directed by the separator **1600** into a debris collection and storage cavity of the handheld surface cleaner's dirty water tank to pass through the seal **1604** to the separator bracket **1602**.

[0439] The seal **1604** is configured to move between a first configuration and a second configuration. The movement of the seal **1604** is relative to the separator bracket **1602**. The seal **1604** is thus non-removably but movably attached to the separator bracket **1602**.

[0440] The seal **1604** is configured to be in the first configuration with the separator **1600** being in the resting configuration in which the separator **1600** is not rotating. In the first configuration, the seal **1604** is in a radially outward position, represented by reference R1 in FIG. 51. In the first configuration, the seal **1604** is configured to provide a barrier for backflow so any debris in the debris collection and storage cavity cannot enter one or both of the separator bracket **1602** and the separator **1600** through a space between the separator bracket **1602** and the separator **1600** since the seal **1604** is blocking the space. The seal **1604**, the separator **1600**, and the separator bracket **1602** are thus configured to cooperate to prevent debris in the debris collection and storage cavity from inadvertently exiting the handheld surface cleaner when suction is not being applied and the separator **1600** is not rotating.

[0441] The seal **1604** is configured to be in the second configuration with the separator **1600** being in a rotating configuration in which the separator **1600** is rotating. In the second configuration, the seal **1604** is in a radially inward position, represented by reference R2 in FIG. 51. The seal **1604** is configured to move automatically from the first configuration to the second configuration in response to the rotation of the separator **1600** and to move automatically from the second configuration to the first configuration in response to the separator **1600** stopping rotating. In the second configuration, the seal **1604** no longer blocks the space between the separator bracket **1602** and the separator **1600**. Debris flowing in the separator **1600** and directed radially outward by the separator **1600** is configured to pass through the space into the debris collection and storage cavity. Because of the suction force provided via the handheld surface cleaner's suction motor (not shown in FIG. 51), air passing out of the separator **1600** will be urged into the separator bracket **1602** instead of passing through the space into the handheld surface cleaner's debris collection and storage cavity.

[0442] FIG. 52 illustrates another implementation of a separator **1700** of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner and to separate the air and the debris on board the handheld surface cleaner. The separator **1700** of FIG. 52 is simplified for purposes of illustration and is generally configured and used similar to other separators described herein except that the separator **1700** of FIG. 52 includes a plurality of external fins **1702**. The fins **1702** are located equidistantly around a perimeter of the separator **1700** in this illustrated implementation, although other spacing of the fins **1702** is possible. The fins **1702** each have a partial elliptical shape in this illustrated implementation, but other shapes are possible.

[0443] The fins **1702** extend radially outward and are thus configured to contact debris collected in a debris collection and storage cavity of the handheld surface cleaner's DWT before any other portion of the separator **1700**. One or more of fins' contact with the debris in the debris collection and storage cavity is configured to add resistance to the separator's rotation. Thus, in implementations in which the handheld surface cleaner, e.g., the handheld surface cleaner's controller, is configured to determine whether the DWT is full based on a current of the handheld surface cleaner's separator motor, the current will rise above the predetermined threshold value sooner than in implementations in which the fins **1702** are not present. The DWT will thus be considered to be full sooner, which may help prevent DWT overflow and/or facilitate user emptying of the DWT since the DWT will be less likely to be completely full when emptied.

[0444] In addition to or instead of a plurality of fins extending radially outward from the separator, a plurality of fins can extend radially outward from another element that rotates with the separator.

For example, in implementations including a rim attached to a separator, a plurality of fins can extend radially outward from the rim in addition to or instead of a plurality of fins extending radially outward from the separator.

[0445] FIG. 53 illustrates another implementation of a handheld surface cleaner **1800** configured to use suction to draw debris into the handheld surface cleaner **1800** and to separate the air and the debris on board the handheld surface cleaner **1800**. The handheld surface cleaner **1800** of FIG. 53 is generally configured and used similar to the handheld surface cleaner **900** of FIG. 41 except that the handheld surface cleaner's air inlet hosing **1802** includes a proximal overhang **1802a** and the handheld surface cleaner's separator bracket **1804** includes a distal lip **1804a**.

[0446] The proximal overhang **1802a** of the air inlet hosing **1802** extends proximally along a length of the handheld surface cleaner's centrifugal seal **1806**, at least when the centrifugal seal **1806** is not rotating, to cover and extend radially around that length of the centrifugal seal **1806**. For example, the proximal overhang **1802a** can extend along about 30% to about 60% of the length of the centrifugal seal **1806**, e.g., about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, or about 60%. The proximal overhang **1802a** terminates distal to the handheld surface cleaner's separator **1808** and is spaced a distance away from the separator **1808** to help prevent wear on the separator **1808** and/or the air inlet hosing **1802** caused by the separator's rotation relative to the air inlet hosing **1802**. In implementations in which a proximal rim **1810** is attached to the separator **1808**, as in this illustrated implementation, the proximal overhang **1802a** terminates distal to the rim **1810** with a gap of space between the proximal overhang **1802a** and the rim **1810** to help prevent wear on the rim **1810** and/or the air inlet hosing **1802** caused by the separator's and rim's rotation relative to the air inlet hosing **1802**.

[0447] As discussed above, the centrifugal seal **1806** is configured to move between first and second configurations where, in the second configuration, the centrifugal seal **1808**, e.g., a distal end of the centrifugal seal **1808**, does not abut the air inlet hosing **1802** and a gap of space exists between the centrifugal seal **1808** and the air inlet hosing **1802**. The proximal overhang **1802a** is configured to fully cover the gap of space to help prevent any debris flowing distally through the air inlet hosing **1802** from passing into the handheld surface cleaner's debris collection and storage cavity before reaching the separator **1808** and to help prevent backflow by helping to prevent any debris in the debris collection and storage cavity from entering the air inlet hosing **1802**.

[0448] The distal lip **1804a** of the separator bracket **1804** extends distally along a length of the separator **1808** at least enough to cover and extend radially around the handheld surface cleaner's oil seal **1812**. Over time, the oil seal **1812** may experience wear. The distal lip **1804a** of the separator bracket **1804** is configured to reduce wear on the handheld surface cleaner's oil seal **1812** by providing a tortuous air path.

[0449] FIG. 54A illustrates another implementation of a handheld surface cleaner configured to use suction to draw debris into the handheld surface cleaner **1900** and to separate the air and the debris on board the handheld surface cleaner **1900**. The handheld surface cleaner **1900** of FIG. 54A is generally configured and used similar to the handheld surface cleaner **900** of FIG. 41 except that the handheld surface cleaner includes a rim **1902** that is generally configured and used similar to the rim **1502** of FIGS. 50A-50B. In this illustrated implementation, the rim **1902** includes a distal overhang **1902a**. The distal overhang **1902a** can be a single circumferential overhang or can include a plurality of fins similar to the fins **1702** of FIG. 52 discussed above.

[0450] The distal overhang **1902a** extends distally past a centrifugal seal **1904** attached to the rim **1902**. The distal overhang **1902a** is thus configured to contact debris collected in a debris collection and storage cavity **1906** of the handheld surface cleaner's DWT **1908** before any the centrifugal seal **1904** with the handheld surface cleaner in at least some orientations relative to ground. Debris in the debris collection and storage cavity **1906** will tend to settle toward ground due to gravity. Thus, in implementations in which the handheld surface cleaner, e.g., the handheld surface cleaner's controller, is configured to determine whether the DWT **1908** is full based on a current of

the handheld surface cleaner's separator motor, the current will rise above the predetermined threshold value sooner than in implementations in which the distal overhang **1902a** is not present because the distal overhang **1902a** will contact the debris. The DWT **1908** will thus be considered to be full sooner, which may help prevent DWT **1908** overflow and/or facilitate user emptying of the DWT **1908** since the DWT **1908** will be less likely to be completely full when emptied.

[0451] FIG. **54A** shows an amount of debris **1910** in the debris collection and storage cavity **1906**. The debris is water in this illustrated implementation but can include solid(s) and/or other types of liquid. The amount of debris **1910** is 150 ml in FIG. **54A**, which in this illustrated implementation corresponds to the DWT **1908** being considered full. As shown in FIG. **54A**, with the handheld surface cleaner **1900** in a substantially vertical orientation with the handheld surface cleaner's handle being located upwardly (to the top in the view of FIG. **54A**), the debris **1910** settles downward due to gravity with a length **1902b** of the distal overhang **1902a** being located in the debris **1910**. The debris **1910** will thus provide resistance to the rotating rim **902**.

[0452] By way of comparison, FIG. **54B** shows the handheld surface cleaner **1900** of FIG. **54A** without the rim **1902** including the distal overhang **1902a**. As shown in FIG. **54B**, with the same amount of debris **1910** (150 ml) in the debris collection and storage cavity **1906**, none of the rotating rim **1902**, centrifugal seal **1904**, and separator **1912** contact the debris **1910**. The DWT **1908** is thus full, but the handheld surface cleaner, e.g., the handheld surface cleaner's controller, will not determine the DWT **1908** to be full, e.g., using a pair of probes, based on a current of the handheld surface cleaner's separator motor, or detecting fullness in another way, at least not until more debris **1910** collects in the debris collection and storage cavity **1906**.

[0453] In implementations that do not include a rim, the separator can include a distal overhang that is generally configured and used similar to the distal overhang **1902a** of FIG. **54A**.

[0454] FIG. **55** illustrates another implementation of a handheld surface cleaner **2000** configured to use suction to draw debris into the handheld surface cleaner **2000** and to separate the air and the debris on board the handheld surface cleaner **2000**. The handheld surface cleaner **2000** of FIG. **55** is generally configured and used similar to the handheld surface cleaner **1800** of FIG. **53** except that the handheld surface cleaner **2000** includes a duckbill valve **2002**. The duckbill valve **2002** is generally configured and used similar to the duckbill valve **76** of FIGS. **2**, **3**, and **6** and is located upstream of (distal to) the handheld surface cleaner's separator **2004**, downstream of (proximal to) the handheld surface cleaner's hair cage **2006**, and is a seal configured to allow fluid flow in only one direction toward the separator **2004**. The duckbill valve **2002** in this illustrated implementation is attached to the handheld surface cleaner's inlet hosing **2008** and extends proximally into the handheld surface cleaner's centrifugal seal **2010**, terminating upstream of the handheld surface cleaner's rim **2012**.

[0455] FIG. **56A** illustrates another implementation of a handheld surface cleaner **2100** configured to use suction to draw debris into the handheld surface cleaner **2100** and to separate the air and the debris on board the handheld surface cleaner **2100**. The handheld surface cleaner **2100** of FIG. **56A** is generally configured and used similar to the handheld surface cleaner **1800** of FIG. **53** except for some differences discussed below.

[0456] The handheld surface cleaner **2100** of FIG. **56A** includes a seal **2102** at an inlet **2104** at a distal end of the handheld surface cleaner **2100**. For comparison, for example, see the inlet **64** shown in FIG. **1** that does not include a seal. The seal **2102** in this illustrated implementation is a dynamic seal considered and used similar to the seal **106** of FIG. **21** discussed above, e.g., configured to move automatically between open and closed positions in response to application and removal of a suction force provided using a suction motor **2106** of the handheld surface cleaner **2100**. The seal **2102** is located upstream (distal to) a hair cage **2108** of the handheld surface cleaner **2100**. In the open position the seal **2102** is configured to allow flow from the inlet **2104** to the hair cage **2108**. The removal of the suction force is configured to cause the seal **2102** to move from the open position to the closed position, e.g., by allowing the seal's plurality of flexible flaps **2102a** of

the seal **2102** to dynamically flex to their default positions (shown in FIG. 56A). In the closed position, the seal **2102** is configured to prevent flow from the inlet **2104** to the hair cage **2108**. [0457] The handheld surface cleaner **2100** of FIG. 56A includes a separator bracket **2110** that includes a distal lip **2110a** similar to the distal lip **1804a** of FIG. 53 discussed above. However, in this illustrated implementation, the handheld surface cleaner **2100** includes a bearing **2112** instead of an oil seal. The distal lip **2110a** extends distally along a length of the handheld surface cleaner's separator **2114** at least enough to cover and extend radially around the bearing **2112**. The bearing **2112** is configured and used similar to the bearing **1000** of FIG. 45.

[0458] The handheld surface cleaner **2100** of FIG. 56A is configured to automatically detect whether the handheld surface cleaner's DWT **2116** is releasably coupled to the handheld surface cleaner's main body **2118**. However, unlike the main body **14** and DWT **12** of FIGS. 8 and 9 that include a cooperating microswitch **40** and protrusion **42**, respectively, for detecting DWT coupling, the main body **2118** and DWT **2116** of FIG. 56A include a cooperating magnet **2120** and hall effect sensor **2122**, respectively. The hall effect sensor **2122** interacting with the magnetic field of the magnet **2120** indicates, e.g., to the handheld surface cleaner's controller configured to be operatively coupled with the hall effect sensor **2122**, that the DWT **2116** is releasably coupled to the main body **2118**. Correspondingly, the hall effect Hall Effect sensor **2122** not interacting with the magnetic field of the magnet **2120** indicates that the DWT **2116** is not releasably coupled to the main body **2118**.

[0459] The shaft of a handheld surface cleaner can be a single shaft or a plurality of shafts. In this illustrated implementation, the shaft includes a first shaft associated with the handheld surface cleaner's separator motor **2121** and a second shaft associated with the separator **2114**.

[0460] As in this illustrated implementation, a drive dog can be used in driving rotation of the separator **2114**. As shown in FIGS. 56A, 56B, and 56C, the drive dog includes an outer bearing **2124** and an inner plug **2126** secured to the outer bearing **2124**. The outer bearing **2124** includes an inner passageway **2124a** configured to seat the inner plug **2126** therein in a fixed rotational relationship. For example, as in this illustrated implementation, the inner plug **2126** and the outer bearing **2124** can be attached together in a friction fit with the inner plug **2126** being plastic and the outer bearing **2124** being rubber. The outer bearing **2124** being rubber may help reduce vibration caused by running of the separator motor **2121** as compared to the outer bearing **2124** being plastic or metal.

[0461] As in this illustrated implementation, the inner plug **2126** and the outer bearing **2124** can be keyed to help prevent relative rotation between the inner plug **2126** and outer bearing **2124**. The inner plug **2126** has a hexagonal cross-sectional shape and the outer bearing's inner passageway **2124a** has a corresponding hexagonal cross-sectional shape in this illustrated implementation.

[0462] The handheld surface cleaner's second shaft is configured to extend into the outer bearing's inner passageway **2124a** and be secured to the outer bearing **2124** in a fixed rotational relationship. The handheld surface cleaner's first shaft is configured to extend into an inner passageway **2126a** of the inner plug **2126** and be secured to the inner plug **2126** in a fixed rotational relationship. As discussed herein, the separator motor **2121** is configured to drive rotation of the first shaft. The rotation of the first shaft is configured to cause the inner plug **2126** to rotate and thus cause the outer bearing **2124** and the second shaft to rotate, thereby causing the separator **2114** to rotate.

[0463] FIG. 57A illustrates another implementation of a handheld surface cleaner (also referred to herein as a "handheld vacuum cleaner") **2200** configured to use suction to draw debris into the handheld surface cleaner **2200** and to separate the air and the debris on board the handheld surface cleaner **2200**. The handheld surface cleaner **2200** of FIG. 57A is generally configured and used similar to the handheld surface cleaner **10** of FIG. 1 except for some differences discussed below.

[0464] As shown, the handheld vacuum cleaner **2200** includes a main body **2202** having a handle **2204** and a separation assembly **2206** removably coupled to the main body **2202**. The separation assembly **2206** includes a separation body **2208** configured to removably couple to the main body

2202, a separator assembly **2210** (shown schematically for illustration purposes in hidden lines) disposed within the separation body **2208**, a dust cup **2212** pivotally coupled to the separation body **2208**, and an inlet **2214** defined in the separation body **2208**. The separation body **2208** defines a DWT. The separation body **2208** defines a DWT of the handheld vacuum cleaner **2200**. The separation body **2208** and the main body **2202** are configured to be removably coupled together using one or more latches **2216**. For example, as in this illustrated implementation, a latch body **2218** is pivotally coupled to the main body **2202** and the separation body **2208** includes a latch receptacle **2220** for selectively receiving at least a portion of the latch body **2218**.

[0465] In some implementations, and as shown in this illustrated implementation, the separation assembly **2206** further includes a dust cup release **2222**. The dust cup release **2222** is configured to selectively retain the dust cup **2212** in a closed (or use) position. Actuation of the dust cup release **2222** is configured to allow the dust cup **2212** to transition to (e.g., pivot to) an open (or emptying) position.

[0466] In some implementations, and as shown in this illustrated implementation, the dust cup release **2222** is slidably coupled to the separation body **2208** and is configured to selectively engage a catch **2224** on the dust cup **2212**. Such a configuration may allow debris to be emptied from the dust cup **2212** without removal of the separation assembly **2206** from the main body **2202** and/or may allow debris to be emptied from the dust cup **2212** after removal of the separation assembly **2206** from the main body **2202**. Additionally, or alternatively, in some instances, the main body **2202** includes a dust cup release configured to selectively retain the dust cup **2212** in the closed position. Positioning a dust cup release on the main body **2202** may allow a user to empty the dust cup **2212** with the same hand that is grasping the handle **2204**. For example, and with reference to FIG. 57B, a handheld vacuum cleaner **2300**, which is an example of the handheld vacuum cleaner **2200**, is shown which includes a forward dust cup release **2302** disposed on a separation assembly **2304** and a rearward dust cup release **2306** disposed on a main body **2308** proximate a handle **2300** of the main body **2308**.

[0467] Referring again to FIG. 57A, in some implementations, and as shown in this illustrated implementation, the dust cup **2212** includes a debris quantity sensor to detect a quantity of debris within the dust cup **2212**. For example, the debris quantity sensor can include a float configured to indicate a liquid level within the dust cup **2212** (e.g., to alert a user when to empty the dust cup **2212** and/or to disable cleaning functions of the handheld vacuum cleaner **2200** in response to the liquid level reaching a predetermined threshold). For another example, the debris quantity sensor can include a pair of probes configured to extend into the dust cup **2212** to indicate a liquid level within the dust cup **2212**. No current will be conducted between the probes until each of the probes is in contact with liquid, e.g., an electrode of each probe is in contact with liquid. The current will thus indicate that a certain fill level has been reached.

[0468] FIG. 58 shows a cross-sectional view of the handheld vacuum cleaner **2200**. As shown, the main body **2202** defines a cavity **2300** for receiving a suction motor **2302** and a separator motor **2304**. The suction motor **2302** and the separator motor **2304** are arranged in a stacked configuration such that output drive shafts of each of the suction motor **2302** and the separator motor **2304** are substantially parallel to each other and spaced apart from each other. The suction motor **2302** and/or the separator motor **2304** can be, for example, a permanent magnet direct current (PMDC) motor.

[0469] The separator motor **2304** is configured to cooperate with a drivetrain **2306**. The drivetrain **2306** is configured transfer rotational motion from the separator motor **2304** to a separator **2326** of the separator assembly **2210**. The drivetrain **2306** includes a drive pulley **2312** connected to a driven pulley **2314**. The drive pulley **2312** and the driven pulley **2314** are connected via a belt. The drivetrain **2306** further includes a drive-side drive coupling **2316** connected to the driven pulley **2314** and a driven-side drive coupling **2318** connected to the separator assembly **2210**. The drive-side drive coupling **2316** is configured to be selectively coupled to the driven-side drive coupling

2318. For example, when the separation body **2308** is decoupled from the main body **2202** (e.g., for emptying or cleaning), the driven-side drive coupling **2318** is decoupled from the drive-side drive coupling **2316**. In this example, the driven-side drive coupling **2318** is coupled to the separation body **2208** (e.g., coupled to a portion of the separator assembly **2210**). The drive-side and driven-side couplings **2316** and **2318** include mating gears (e.g., mating gears **2301** and **2303** of FIG. 57B), clutch plates (e.g., high friction pads, such as rubber or foam pads, configured to cooperate to transfer rotational motion as shown in FIG. 58), and/or the like. Use of the drivetrain **2306** to transfer rotational motion from the separator motor **2304** to the separator **2326** may allow one or more electronic components (e.g., the separator motor **2304**) to be separate from those components exposed to debris (e.g., the separation assembly **2206**) which may encourage easier cleaning of the components exposed to debris (e.g., by allowing a user to submerge the components exposed to debris within water).

[0470] The driven side coupling **2318** includes a drive shaft **2320** extending within an air channel **2322** of the separator **2326**. The air channel **2322** includes opposing open ends **2323** and **2325** such that air can pass through the air channel **2322**. The first open end **2323** is fluidly coupled to the suction motor **2302** downstream of the second open end **2325**. A pre-motor filter **2327** (e.g., a foam filter, a pleated filter, a cyclone filter, and/or any other type of filter) is disposed such that air that exits through the first open end **2323** passes through the pre-motor filter **2327** before entering the suction motor **2302**. Such a configuration may mitigate a risk of detrimental quantities of residual liquid (e.g., liquid not separated out of the air flow by the separator **2326**) and/or solid debris from reaching the suction motor **2302**.

[0471] The drive shaft **2320** is rotationally supported within the air channel **2322** by one or more bearings **2324**. The drive shaft **2320** is coupled to the separator **2326** such that the separator **2326** rotates with the drive shaft **2320** about a rotation axis **2331**. The rotation axis **2331** can be aligned with a longitudinal axis of the handle **2204**, as in this illustrated implementation. The rotation axis **2331** is offset from a separation assembly central axis **2333** of the separation assembly **2206**, and the rotation axis **2331** and the separation assembly central axis **2333** are substantially parallel. The longitudinal axis of the handle **2204** is thus also offset from and parallel to the separation assembly central axis **2333**. The separation assembly central axis **2333** can intersect one or more of the handle **2204** and/or the dust cup **2212**, with the separation assembly central axis **2333** intersecting both in this illustrated implementation.

[0472] The separator **2326** is configured such that liquid entrained within an airflow that is incident on the separator **2326** is urged outwardly from the separator **2326** and into one or more internal sidewalls **2328** of the separation body **2208**. The liquid may in some instances not be urged into contact with any of the internal sidewalls **2328**, e.g., depending on an orientation in which the cleaner **2200** is being held during and/or after the liquid is urged outwardly. As shown, the separator **2326** partially obstructs the second open end **2325** and extends at least partially around the second open end **2325**. In some instances, a portion of the separator **2326** overlaps with and extends around a portion of the air channel **2322** at the second open end **2325**.

[0473] With additional reference to FIG. 59A, the separator **2326** includes a separator body **2400** and a plurality of vanes **2402** extending from the separator body **2400**. The plurality of vanes **2402** in this illustrated implementation extend from a center **2404** of the separator body **2400**. For example, as shown in this illustrated implementation, the plurality of vanes **2402** extend radially outwardly from the center **2404** to a peripheral edge **2405** of the separator body **2400**. In some instances, and as shown in FIG. 58, the separator body **2400** may also include an at least partially conical shape (e.g., a frustoconical shape). The plurality of vanes **2402** and/or the separator body **2400** may generally be described as being configured to encourage liquid incident thereon to move radially outwardly. As in this illustrated implementation, at least one of the plurality of vanes **2402** may include an arcuate shape.

[0474] As also shown in this illustrated implementation, the separator **2326** includes a shroud **2406**

that extends around (e.g., entirely around a perimeter of) the separator body **2400**. The shroud **2406** is coupled to the separator body **2400** using one or more supports **2408** such that the shroud **2406** is spaced apart from the separator body **2400** by a shroud separation distance **2410** to form at least one air passthrough **2412**. The shroud **2406** further includes a plurality of liquid passthroughs **2414** through which liquid urged radially outward may pass. In some instances, and as shown in this illustrated implementation, the at least one air pass through **2402** and the plurality of liquid passthroughs **2414** may extend within different (e.g., perpendicular) planes.

[0475] With additional reference to FIG. **59B**, the shroud **2406** further extends around (e.g., entirely around a perimeter of) and along a portion of an outer surface **2330** of the air channel **2322** at the second open end **2325**. The shroud **2406** is spaced apart from the outer surface **2330** to form a seal gap **2332**. The seal gap **2332** is configured to receive a seal configured to at least partially restrict air flow through the seal gap **2332**. The seal may be, for example, an oil seal, a bushing, a bearing, a bristle strip, a felt strip, and/or any other type of seal capable of being disposed within the seal gap **2332** while permitting rotational movement between the shroud **2406** and the air channel **2322**.

[0476] The shroud **2406** includes an upstream end **2334** having an upstream end width **2336** (e.g., a diameter) and a downstream end **2338** having a downstream end width **2340** (e.g., a diameter). The upstream end width **2336** is less than the downstream end width **2340** in this illustrated implementation. In some instances, the upstream end width **2336** may be less than or equal to a maximum width (e.g., a diameter of) the separator body **2400**.

[0477] While the handheld vacuum cleaner **200** is shown as including the separator motor **2304**, other configurations for rotating the separator **2326** are possible. For example, the suction motor **2302** can be configured to rotate the separator **2326**. In such an example, the suction motor **2302** is configured to generate suction while causing the separator **2326** to rotate (e.g., using a drive train coupled to a driveshaft of the suction motor **2302**, using a turbine configured to generate rotational motion through airflow generated by the suction motor **2302**, or the like).

[0478] With reference again to FIG. **58**, the separator assembly **2210** includes one or more guide ribs **2342** extending along the one or more sidewalls **2328** of the separation body **2208**. The one or more guide ribs **2342** are configured to encourage (e.g., guide) liquid incident on the one or more sidewalls **2328** to flow along the one or more sidewalls **2328** and into the dust cup **2212**. The one or more guide ribs **2342** extend from a platform **2344** and are coupled to the air channel **2322**. At least one of the one or more guide ribs **2342** include a first guide rib section **2346** and a second guide rib section **2348**. The first and second guide rib sections **2346** and **2348** intersect to form a rib α .

[0479] With additional reference to FIG. **60**, the platform **2344**, the one or more guide ribs **2342**, the air channel **2322**, and the separator **2326** is configured to be collectively removed as a single assembly from the separation body **2208**. In other words, the separator assembly **2210** is configured to be removable from the separation body **2208** as a single assembly. Such a configuration may encourage easier cleaning of the separator assembly **2210**. In some instances, one or more of the one or more guide ribs **2342** and/or the platform **2344** may not be removed with the air channel **2322** and the separator **2326**. In these instances, the separator assembly **2210** may be removable as two or more assemblies. Additionally, or alternatively, one or more of the one or more guide ribs **2342** and/or the platform **2344** may be non-removable from the separation body **2208**.

[0480] The platform **2344** includes a platform filter **2500** (e.g., a screen, a cyclone separator, and/or any other type of filter) that extends between the inlet **2214** and the separator **2326**. The platform filter **2500** is configured to reduce, which may in at least some instances be to eliminate, a quantity of debris greater than a predetermined size (e.g., fibrous debris such as hair) from reaching the separator **2326**. Such a configuration may reduce a risk of the rotation of the separator **2326** being impeded by debris (e.g., fibrous debris wrapped around the separator **2326**). Debris that does not

pass through the platform filter **2500** collects within the dust cup **2212** for later disposal. The platform **2344** further includes a plurality of liquid cutouts **2501** disposed on opposing sides of a respective guide rib **2342** such that liquid flowing along the respective guide rib **2342** passes through a corresponding liquid cutout **2501**.

[0481] As shown, the separator assembly **2210** further includes a pour spout **2502**. The pour spout **2502** includes a spout outlet **2504** at the first open end of the **2323** of the air channel **2322**. The pour spout **2502** defines a portion of a liquid flow path **2506** that extends from the dust cup **2212** to the spout outlet **2504**. The liquid flow path **2506** does not flow through the air channel **2322**. In operation, a user may empty dirty liquid from the dust cup **2212** while the separator assembly **2210** is received within the separation body **2208**. In this configuration, the platform filter **2500** reduces (e.g., prevents) a quantity of debris greater than a predetermined size (e.g., fibrous debris such as hair) from passing through the pour spout **2502**. As such, the platform filter **2500** may generally be described as being configured to provide a filtering function in both a cleaning operation and an emptying operation.

[0482] With reference again to FIG. **58**, a closure **2350** is movably (e.g., pivotally) coupled to the separation body **2208**. The closure **2350** is configured to transition between an open position when the suction motor **2302** is generating suction and a closed position when the suction motor **2302** is not generating suction. In the closed position, the closure **2350** is configured to substantially fluidically isolate the inlet **2214** from the dust cup **2212**. Such a configuration may mitigate, which may in at least some instances be to prevent, collected debris (solid and/or liquid) from inadvertently passing through the inlet **2214** when the suction motor **2302** is disabled.

[0483] As also shown, the handle **2204** defines a battery cavity **2352**. The battery cavity is configured to receive at least one battery **2354**. The at least one battery **2354** is configured to provide power to at least one of the suction motor **2302** and/or the separator motor **2304**.

[0484] When the suction motor **2302** is powered, air (which may include liquid and solid debris entrained therein at least before encountering the separator **2326**) flows along an airflow path **2356**. The airflow path **2356** enters the handheld vacuum cleaner **2200** via the inlet **2214**. From the inlet **2214**, the airflow path **2356** passes through the platform filter **500** to be incident on the separator **2326**. The separator **2326** imparts a rotational flow to the air incident thereon to propel at least a portion of entrained liquid out of entrainment with the air. The airflow path **2356** flows along the separator body **2400** and passes through the at least one air passthrough **2412** to enter the air channel **2322**. From the air channel **2322** the airflow path **2356** extends through the pre-motor filter **2327** to enter the suction motor **2302**. From the suction motor **2302** the airflow path **2356** exits the handheld vacuum cleaner **2200** and enters a surrounding environment. In some instances, the airflow path **2356** may pass through an exit filter, such as a high efficiency particulate air (HEPA) filter, before entering the surrounding environment and after passing through the suction motor **2302**.

[0485] FIG. **61** shows a side view of the handheld vacuum cleaner **2200** having the separation assembly **2206** in a maintenance position. As shown, the separation assembly **2206** is pivotally coupled to the main body **2202** such that the separation assembly **2206** pivots when transitioning between a use position (shown in FIG. **57A**) and the maintenance position. In an exemplary implementation, the pivoting between the use and maintenance positions is achieved with a rotational motion, which can be rotational movement of the separation assembly **2206** relative to the main body **2202**, rotational movement of the main body **2202** relative to the separation assembly **2206**, or rotational movement of each of the main body **2202** and the separation assembly **2206** relative to one another. The one or more latches **2216** selectively retain the separation assembly **2206** in the use position. In some instances, the main body **2202** may include a sensor (e.g., a mechanical switch, a hall effect sensor, or the like) configured to determine a state of the separation assembly **2206** (e.g., whether or not the separation assembly **2206** is in the use position).

[0486] When in the maintenance position, the separator assembly **2210** is configured to be

removable from the separation body **2208** and/or the separation assembly **2206** is configured to be removable from the main body **2202**. For example, when in the maintenance position, application of a removal force on the separator assembly **2210** in a direction substantially parallel to the separation assembly central axis **2333** may result in the separator assembly **2210** being removed from the separation body **2208**. By way of further example, when in the maintenance position, application of a removal force on the separation body **2208** in a direction substantially parallel to the separation assembly central axis **2333** results in the separation body **2208** being removed from the main body **2202**.

[0487] As shown, the main body **2202** includes a first hinge portion **2600** and the separation assembly **2206** includes a second hinge portion **2602** pivotally coupled to the first hinge portion **2600**. The second hinge portion **2602** includes a catch **2604** configured to cooperate with a release **2606** on the separation assembly **2206**. The catch **2604** and the release **2606** are configured to selectively couple the separation assembly **2206** to the main body **2202**. For example, application of a removal force on the separation body **2208** in a direction substantially parallel to the separation assembly central axis **2333** results in the catch **2604** disengaging the release **2606**.

[0488] While FIG. **61** shows the separation assembly **2206** pivotally coupled to the main body **2202**, other configurations are possible. For example, the separation assembly **2206** may decouple from the main body **2202** in response to actuation of the one or more latches **2216** and the application of a removal force in a direction substantially parallel to the separation assembly central axis **2333**.

[0489] FIG. **62** shows a perspective end view of the main body **2202** when the separation assembly **2206** is removed therefrom. As shown, the pre-motor filter **2327** is removably disposed within a filter receptacle **2201** of the main body **2202**. As shown, the pre-motor filter **2327** includes a drive passthrough **2203** through which the drive-side drive coupling **2316** is exposed and engageable by the driven-side drive coupling **2318** (FIG. **58**). In some instances, and as shown in this illustrated implementation, the drive passthrough **2702** may be positioned such that the pre-motor filter **2327** (and/or a frame retaining the pre-motor filter **2327**) completely surrounds the drive-side drive coupling **2316**. The drive-side drive coupling **2316** is configured to selectively engage and disengage the driven-side drive coupling **2318** in response to the separation assembly **2206** being removed from or coupled to the main body **2202** or being pivoted between the maintenance position and the use position.

[0490] FIGS. **63A-63E** illustrate another implementation of a handheld surface cleaner **2700** configured to use suction to draw debris into the handheld surface cleaner **2700** and to separate the air and the debris on board the handheld surface cleaner **2700**. The handheld surface cleaner **2700** includes a DWT **2702**, a main body **2704**, a debris collection and storage cavity **2706** in the DWT **2702**, a cap **2708**, a lower cover **2710** of the DWT **2702**, a latch **2712** (also shown in FIG. **63S**), a microswitch **2714** (also shown in FIGS. **63T** and **63U**), a motor bracket **2716**, a separator motor **2718**, a separator bracket **2720**, a separator assembly **2722** (see FIGS. **63K** and **63L**), a separator **2724**, a PCB **2726**, a suction motor **2728**, a suction motor housing **2729**, a handle **2730**, a battery holder **2732** (shown in FIG. **63E** holding four rechargeable batteries **2734**), an inlet **2736**, an inlet seal **2738**, a shaft **2740**, air exit holes **2742**, a seal **2744** (a bearing in this illustrated implementation, although other seals are possible as discussed herein) at the separator **2724**, a first filter **2746** (a hair cage in this illustrated implementation, also shown in FIGS. **63P-63R**) upstream of the separator **2724**, a second filter **2748** (filter foam in this illustrated implementation) downstream of the separator **2724**, a duckbill valve **2750** (also shown in FIGS. **63Y-63AA**), a lower cover **2752** of the main body **2704**, an actuator **2754** (a depressible button in this illustrated implementation), a charging port **2756**, and inlet hosing **2758**. The handheld surface cleaner **2700** of FIGS. **63A-63E** is generally configured and used similar to handheld surface cleaners discussed above except for some differences discussed below.

[0491] The main body **2704** is configured to releasably couple to the DWT **2702**. In this illustrated

implementation, the DWT **2702** is configured to be completely released from the main body **2704**. FIGS. **63A-63F** show the DWT **2702** and the main body **2704** releasably attached together. FIGS. **63G** and **63H** show the DWT **2702** as a standalone element. FIGS. **63I** and **63J** show the main body **2704** as a standalone element.

[0492] In this illustrated implementation, the handheld surface cleaner **2700**, e.g., a controller at the PCB **2726**, is configured to determine automatically whether the DWT **2702** is full. As in this illustrated implementation, the fullness of the DWT **2702** can be determined using a current of the separator motor **2718**.

[0493] In this illustrated implementation, the handle **2730** defines a pistol grip and includes a finger grip **2730a**. The handle **2730** thus extends downwardly relative to the flow path from the inlet **2736** to the separator **2724**, where downward and upward are relative directions depending on an orientation of the handheld surface cleaner **2700**. Downward and upward directions are labeled in the views of FIGS. **63D** and **63V**.

[0494] In this illustrated implementation, debris and air flow paths in the handheld surface cleaner **2700** are the same from the inlet **2736** to the separator **2724**. As shown in FIG. **63E**, a flow path from the inlet **2736** to the separator **2724** defines a first longitudinal axis **A4**. As also shown in FIG. **63E**, the handle **2730** defines a second longitudinal axis **A5** that is at a transverse angle θ relative to the first longitudinal axis **A4**. As discussed above, in an exemplary implementation, the transverse angle θ is greater than about 90 degrees and less than about 180 degrees, for example in a range of about 135 degrees to about 175 degrees, e.g., about 140 degrees, about 145 degrees, about 150 degrees, about 155 degrees, about 160 degrees, about 165 degrees, etc.

[0495] FIGS. **63E** and **63V** show a third axis **A6** that is substantially perpendicular to the first longitudinal axis **A4** and is thus substantially perpendicular to the flow path from the inlet **2736** to the separator **2724**. FIG. **63V** shows that a fourth longitudinal axis **A7** is at an acute angle δ relative to the third axis **A6**. The fourth longitudinal axis **A7** is substantially parallel to the second longitudinal axis **A5** and is defined by a bottom of the handle **2730**. The handle **2730** is thus at the acute angle δ relative to the third axis **A6** and extends from a distal portion of the main body **2704** at the acute angle δ . In an exemplary implementation, the acute angle δ is in a range of about 60 degrees to about 80 degrees, e.g., in a range of about 60 degrees to about 75 degrees, in a range of about 70 degrees to about 78 degrees, about 60 degrees, about 65 degrees, about 70 degrees, about 75 degrees, etc. The handle **2730** extending downwardly at the acute angle δ is configured to facilitate stable resting of the handheld surface cleaner **2700** on a support surface, e.g., a floor, a tabletop, a countertop, etc., with a downward-most portion **2730b** of the handle **2730** being configured to rest on the support surface.

[0496] FIG. **63V** also shows that the fourth longitudinal axis **A7** (and thus also the second longitudinal axis **A5**) is at an obtuse angle σ relative to the third axis **A6**. In an exemplary implementation, the obtuse angle σ is in a range of about 95 degrees to about 135 degrees, e.g., about 95 degrees, about 100 degrees, about 105 degrees, about 110 degrees, in a range of about 100 degrees to about 110 degrees, in a range of about 100 degrees to about 105 degrees, in a range of about 105 degrees to about 110 degrees, etc. The obtuse angle σ is also configured to facilitate stable resting of the handheld surface cleaner **2700** on a support surface with the downward-most portion **2730b** of the handle **2730** being configured to rest on the support surface.

[0497] As shown in FIGS. **63K** and **63L**, the separator assembly **2722** in this illustrated implementation includes the separator motor **2718**, the separator **2724** (also shown in FIGS. **63M-63O**), the separator bracket **2720**, the shaft **2740** (obscured in FIGS. **63K** and **63L**), a centrifugal seal **2760** (also shown in FIGS. **63M-63O**), a rim **2762** (also shown in FIGS. **63M-63O**), a separator base **2763** (see FIG. **63M**), and a plurality of movable covers **2764** (also shown in FIGS. **63M-63O**). As discussed herein, the movable covers **2764** are each configured to move between configurations in which the movable covers **2764** seal holes **2766** formed in a body **2768** of the separator **2724** and do not seal the holes **2766**. For purposes of illustration, FIG. **63O** shows one of

the movable covers **2764** removed from the separator body **2768** to show the movable cover's associated hole **2766**. The holes **2766** each have a substantially rectangular shape in this illustrated implementation, but as mentioned above, other hole shapes are possible.

[0498] The movable covers **2764** in this illustrated implementation are each attached to the separator body **2768** along one edge **2764a** of the cover **2764**. The movable covers **2764** can each be attached to the separator body **2768** in a variety of ways. As in this illustrated implementation, each of the movable covers **2764** can be hingedly attached to the separator body **2768**, e.g., by including a hinge pin **2764b** hingedly attached to a corresponding hinge knuckle **2768a** of the separator body **2768**.

[0499] As in this illustrated implementation, each of the movable covers **2764** can be configured to selectively cover one of the separator's holes **2766**. In another implementation, one or more of the movable covers **2764** can be configured to selectively cover more than one of the separator's holes **2766**, e.g., cover two of the separator's holes **2766**, etc.

[0500] The separator **2724** in this illustrated implementation is a singular member. The separator **2724** is configured to rotate, as discussed herein, with the separator body **2768**, the centrifugal seal **2760**, and the rim **2762** being configured to rotate as a unit.

[0501] A separator, e.g., any of the separators described herein, can include a vibration reduction feature configured to reduce vibration of components adjacent to the separator during the separator's rotation. Reducing vibration may help reduce wear experienced by the adjacent components and/or may improve user experience by reducing noise heard by the user during separator rotation.

[0502] The separator **2724** of the handheld surface cleaner **2700** of FIG. **63E** includes one implementation of a vibration reduction feature. As in this illustrated implementation, the vibration reduction feature can include a plurality of proximal arms **2724a** and a plurality of distal arms **2724b**. The separator **2724** includes three proximal arms **2724a** spaced substantially equidistantly around the separator body **2728** and three distal arms **2724b** spaced substantially equidistantly around the separator body in this illustrated implementation, but other spacings and numbers of proximal arms **2724a** and distal arms **2724b** are possible. Including at least two proximal arms **2724a** spaced equidistantly around the body **2728** may help limit reduce vibration at a proximal end of the separator **2724** regardless of the separator's rotational position. Including at least two distal arms **2724b** spaced equidistantly around the body **2728** may help limit reduce vibration at a distal end of the separator **2724** regardless of the separator's rotational position. In some implementations, a separator includes only proximal arms or only distal arms.

[0503] The separator's plurality of proximal arms **2724a** extend proximally from the body **2768** of the separator **2724**. The plurality of proximal arms **2724a** are configured to rotate around the duckbill valve **2750**. With the separator **2724** coupled to the DWT **2702**, e.g., with the separator assembly **2722** removably coupled to the DWT **2702**, and with the duckbill valve **2750** coupled to the DWT **2702**, e.g., with the duckbill valve assembly (discussed further below) removably coupled to the DWT **2702**, the plurality of proximal arms **2724a** are configured to extend proximally past a proximal end of the separator body **2768** and a distal end of the duckbill valve **2750**, as shown in FIG. **63F**.

[0504] The plurality of distal arms **2724b** extend distally from the body **2768** of the separator **2724**. The plurality of proximal arms **2724a** are configured to rotate around the separator bracket **2720**. With the separator **2724** coupled to the DWT **2702**, e.g., with the separator assembly **2722** removably coupled to the DWT **2702**, the plurality of distal arms **2724b** are configured to extend distally past a distal end of the separator body **2768** and a proximal end of the separator bracket **2720**, as shown in FIG. **63F**.

[0505] In this illustrated implementation, the debris that enters the handheld surface cleaner **2700** through the inlet **2736** and flows into the separator **2724** through the separator's inlet opening along the flow path is configured to travel from the flow path to the debris flow path in the handheld

surface cleaner **2700** and then flow along the debris flow path. The separator **2724** is configured to direct the debris radially outward through the holes **2766** and into the debris collection and storage cavity **2706**. The radial outward flow of the debris is in a direction radially outward from the first longitudinal axis **A4**.

[0506] In this illustrated implementation, the air that enters the handheld surface cleaner **2700** through the inlet **2736** and flows into the separator **2724** through the separator's inlet opening along the flow path is configured to travel from the flow path to the air flow path in the handheld surface cleaner **2700** and then flow along the air flow path in the handheld surface cleaner **2700**. The air flow path is different from the debris flow path. The air is configured to flow out of the separator **2724** through the separator's outlet opening. This portion of the air flow path is along the first longitudinal axis **A4**. A remainder of the air flow path is not along the first longitudinal axis **A4**. Downstream of the separator **2724**, the air flow path is through the separator bracket **2720**, the second filter **2748**, and a gap of space defined between the main body lower cover **2752** and the suction motor housing **2729**.

[0507] The separator assembly **2722** in this illustrated implementation includes a pair of tabs **2770** configured to facilitate manual removal of the separator assembly **2722** from the DWT **2702**. In another implementation, the separator assembly **2722** can include a single tab **2770** or can include more than two tabs **2770**. The tabs **2770** are configured to be held by a user to help pull the separator assembly **2722** in a proximal direction to remove the separator assembly **2722** from the DWT **2702**. Similarly, the tabs **2770** are configured to be held by a user to help push the separator assembly **2722** in a distal direction to re-attach the separator assembly **2722** to the DWT **2702**. A user may choose to hold only one of the tabs **2770** when removing and/or re-attaching the separator assembly **2722**.

[0508] The tabs **2770** in this illustrated implementation extend proximally and are located opposite one another around a perimeter of the separator assembly **2722**. Free ends of the tabs **2770** are configured to be located outside of the DWT **2702** (and the main body **14**) with the DWT **2702** coupled to the main body **2704**, as shown in FIGS. **63A-63D** in which one of the tabs **2770** is visible. At least a portion of the tabs **2770** is thus outside of the debris flow path in the handheld surface cleaner **2700** so that at least the portion of the tabs **2770** may remain clean and dry when a user holds the tabs **2770**.

[0509] The tabs **2770** are configured as a lock configured to lock the separator assembly **2722** to the DWT's housing **2772** (and the DWT lower cover **2710** if not omitted), similar to the tabs **54** discussed above with respect to the handheld surface cleaner **10** of FIG. **1**.

[0510] To allow the separator motor **2718** to be powered by a power supply at the main body **2704**, e.g., by the rechargeable batteries **2734** in the handle **2730**, the DWT **2702** includes at least one electrical connector **2774**, e.g., a conductive plate or other electrical connector, operatively connected to the separator motor **2718** and configured to contact at least one corresponding electrical connector **2776**, e.g., a conductive pin or other electrical connector, of the main body **2704** that is operatively connected to the power supply. The attachment of the DWT **2702** and the main body **2704** is configured to cause each of the DWT's one or more electrical connectors **2774** to conductively engage automatically with the main body's one or more electrical connectors **2776**. Detaching the DWT **2702** and the main body **2704** is configured to cause the DWT's one or more electrical connectors **2774** to automatically no longer conductively engage with the main body's one or more electrical connectors **2776**. Two electrical connectors **2774**, positive and negative, are shown in the illustrated implementation of FIGS. **63G**, **63H**, and **63K**. Two electrical connectors **2776**, positive and negative, are shown in the illustrated implementation of FIGS. **63I** and **63J**.

[0511] As in this illustrated implementation, the electrical connectors **2774**, **2776**, and thus the electrical connection between the DWT **2702** and the main body **2704**, can be configured to be isolated from any air flowing the handheld surface cleaner **2700** and from any debris in the handheld surface cleaner **2700**, whether the debris is flowing in the handheld surface cleaner **2700**

or is collected in the DWT's debris collection and storage cavity **2706**. Such isolation from debris and air may help prevent a short circuit, and/or may help prevent rusting and/or other damage to the DWT's one or more electrical connectors **2774** and/or the main body's one or more electrical connectors **2776**.

[0512] In this illustrated implementation, the at least one electrical connector **2774** of the DWT **2702** includes a pair of conductive plates, and the at least one electrical connector **2776** of the main body **2704** includes a pair of pogo pins. The attachment of the DWT **2702** and the main body **2704** is configured to automatically cause each of the pogo pins **2776** to compress in response to pressing against a respective one of the conductive plates **2774**. Detaching the DWT **2702** and the main body **2704** is configured to automatically cause the pogo pins **2776** to decompress in response to no longer pressing against either of the conductive plates **2774**. In another implementation, the DWT **2702** includes the one or more pogo pins, and the main body **2704** includes the one or more conductive plates.

[0513] The shaft **2740** of the handheld surface cleaner **2700** can be a single shaft or a plurality of shafts. In this illustrated implementation, the shaft **2740** includes a first shaft associated with the handheld surface cleaner's separator motor **2121** and a second shaft associated with the separator **2114**.

[0514] As in this illustrated implementation, a drive dog can be used in driving rotation of the handheld surface cleaner's separator motor **2718**. As shown in FIGS. **63F**, **63W**, and **63X**, the drive dog includes an outer bearing **2778** and an inner plug **2780** attached to the outer bearing **2778**. The outer bearing **2778** includes an inner passageway **2778a** configured to seat the inner plug **2780** therein in a fixed rotational relationship. For example, as in this illustrated implementation, the inner plug **2780** and the outer bearing **2778** can be attached together in a friction fit with the inner plug **2780** being plastic and the outer bearing **2778** being rubber. The outer bearing **2778** being rubber may help reduce vibration caused by running of the separator motor **2718** as compared to the outer bearing **2778** being plastic or metal.

[0515] The handheld surface cleaner's second shaft is configured to extend into the outer bearing's inner passageway **2778a** and be secured to the outer bearing **2778** in a fixed rotational relationship. The handheld surface cleaner's first shaft is configured to extend into an inner passageway **2780a** of the inner plug **2780** and be secured to the inner plug **2780** in a fixed rotational relationship. As discussed herein, the separator motor **2718** is configured to drive rotation of the first shaft. The rotation of the first shaft is configured to cause the inner plug **2780** to rotate and thus cause the outer bearing **2778** and the second shaft to rotate, thereby causing the separator **2724** to rotate.

[0516] As in this illustrated implementation, the inner plug **2780** and the outer bearing **2778** can be keyed to help prevent relative rotation between the inner plug **2780** and outer bearing **2778**. In this illustrated implementation, the inner plug **2780** has a generally X-shaped cross-sectional shape including four lobes and the outer bearing's inner passageway **2780a** has a corresponding generally X-shaped cross-sectional shape including four lobes to help prevent relative rotation between the inner plug **2780** and outer bearing **2778**. Although four lobes are used in this illustrated implementation, another number of lobes may be used. As compared to the implementation of FIGS. **56A-56C** in which the inner plug **2126** has a hexagonal cross-sectional shape and the outer bearing's inner passageway **2124a** has a corresponding hexagonal cross-sectional shape, the inner plug **2780** of FIG. **63F** has more contact surface area with the outer bearing **2778** than the inner plug **2126** of FIG. **56A** has with the outer bearing **2124**. Having more contact surface area may reduce current draw of the separator motor **2718** and thus operate more efficiently and better transfer motion to the separator **2724**.

[0517] A handheld surface cleaner can include a separator and a valve that is located upstream of the separator and that is configured to allow flow through the valve in a direction toward the separator. The valve can have a variety of configurations. For example, various illustrated implementations of handheld surface cleaners described herein include the valve as a duckbill

valve, although another type of valve can be used. As mentioned above, the handheld surface cleaner **2700** of FIG. **63E** includes the duckbill valve **2750**, which is configured to allow flow through the duckbill valve **2750** in a direction toward the separator **2724**.

[0518] The duckbill valve **2750** of the handheld surface cleaner **2700** is generally configured and used similar to the duckbill valve **76** of FIGS. **2**, **3**, and **6** and is located upstream of (distal to) the separator **2724**, downstream of (proximal to) the hair cage **2746**, and is a seal configured to allow fluid flow in only one direction toward the separator **2724**. The duckbill valve **2750** in this illustrated implementation is attached to the handheld surface cleaner's inlet hosing **2758**, as shown in FIGS. **63F** and **63AA**, and terminates upstream of the handheld surface cleaner's rim **2762** and centrifugal seal **2760**.

[0519] A handheld surface cleaner's duckbill valve can be non-removable from the handheld surface cleaner's DWT or can be removable from the DWT. A duckbill valve being non-removable from the DWT may help ensure that the duckbill valve is properly positioned in the DWT, is always present to provide sealing functionality, and/or does not get lost. A duckbill valve being removable may facilitate a user cleaning of the duckbill valve and/or may allow for the duckbill valve to be replaced if the duckbill valve gets torn or otherwise damaged, e.g., by debris flowing through the duckbill valve.

[0520] The duckbill valve **76** of FIGS. **2**, **3**, and **6**, a duckbill valve **718** of FIG. **35**, a duckbill valve **817** of FIG. **39**, and the duckbill valve **2002** of FIG. **55** are each non-removable from their respective handheld surface cleaner's DWT. (The duckbill valve **817** is omitted from FIG. **40** for purposes of illustration.)

[0521] The duckbill valve **2750** of FIGS. **63E** and **63F** is configured to be removed from the DWT **2702**. The handheld surface cleaner **2700** includes a duckbill valve housing **2782** that is attached to the duckbill valve **2750**. The duckbill valve **2750** and the duckbill valve housing **2782** define a duckbill valve assembly. The duckbill valve housing **2782** is configured to be removably coupled with the DWT **2702** with the duckbill valve **2750**. The duckbill valve **2750** and the duckbill valve housing **2782** are shown attached to the DWT **2702** in FIGS. **63E**, **63F**, and **63AA**. The duckbill valve **2750** and the duckbill valve housing **2782** are shown in FIGS. **63Y** and **63Z** removed from the DWT **2702**.

[0522] The duckbill valve housing **2782** includes an inner passageway having an open distal end and an open proximal end. With the duckbill valve **2750** open, e.g., under a suction force provided by the suction motor **2728**, the inner passageway of the duckbill valve housing **2782** is configured to allow debris and air to flow through the duckbill valve housing **2782** and the duckbill valve **2750** toward the separator **2724**.

[0523] The duckbill valve housing **2782** includes a bayonet mechanism **2782a** configured to engage a corresponding bayonet feature of the inlet hosing **2758**. With the bayonet mechanism **2782a** and the bayonet feature engaged, the duckbill valve housing **2782** and the duckbill valve **2750** are attached to the DWT **2702** via the inlet hosing **2758**. The bayonet mechanism **2782a** of the duckbill valve housing **2782** is configured to be disengaged from the bayonet feature of the inlet hosing **2758** to detach the duckbill valve housing **2782** from the inlet hosing **2758** to allow the duckbill valve housing **2782**, and the duckbill valve **2750** attached to the duckbill valve housing **2782**, to be removed from the DWT **2702**.

[0524] The bayonet mechanism **2782a** of the duckbill valve housing **2782** can be configured to be disengaged manually from the bayonet feature of the inlet hosing **2758** by a user holding and rotating the duckbill valve housing **2782** relative to the inlet hosing **2758**. However, it may be difficult for some users to comfortably put their hand into the DWT **2702** to access the duckbill valve housing **2782** and/or a user may not want to put their hand in the DWT **2702**.

[0525] The bayonet mechanism **2782a** of the duckbill valve housing **2782** can be configured to be disengaged manually from the bayonet feature of the inlet hosing **2758** using a tool inserted into the DWT **2702** and into a proximal opening **2782b** of the duckbill valve housing **2782**. Using the

tool does not necessarily require a user to put their hand into the DWT 2702.

[0526] FIG. 63AA shows one implementation of a tool 2784 configured to remove the duckbill valve 2750 from the DWT 2702 and to attach the duckbill valve 2750 to the DWT 2702. The tool 2784 is also configured to remove the duckbill valve housing 2782 with the duckbill valve 2750 and to attached the duckbill valve housing 2782 to the DWT 2702 with the duckbill valve 2750.

[0527] As in this illustrated implementation, the separator assembly 2722 can be removable from the DWT 2702 and can be configured as a lid that covers an open proximal end 2786 of the DWT 2702 through which debris is configured to exit the debris collection and storage cavity 2706. FIG. 63AA shows the DWT 2702 with the separator assembly 2722 removed from the DWT 2702.

[0528] In an exemplary implementation, the tool 2784 has a length that is greater than a distance between the open proximal end 2786 of the DWT 2702 and the duckbill valve housing 2782 with the duckbill valve housing 2782 attached to the DWT 2702, e.g., attached to the inlet hosing 2758. A user may thus hold a handle 2784a of the tool 2784 outside of the DWT 2702 to access the duckbill valve housing 2782, and thus access the duckbill valve 2780, without the user's hand being inside the DWT 2702.

[0529] With the separator assembly 2722 removed from the DWT 2702, the tool 2784 is configured to be inserted into the DWT 2702, e.g., into the debris collection and storage cavity 2706, through the DWT's open proximal end 2786 and engage the duckbill valve housing 2782. The engagement can be, as shown for example in FIG. 63AA, a distal end of the tool 2784 being inserted through the proximal opening 2782b of the duckbill valve housing 2782 to be positioned in the duckbill valve housing's 2782 inner passageway where the tool 2784 is configured to engage an inner surface of the duckbill valve housing 2782.

[0530] In an exemplary implementation, with the tool 2784 engaging the duckbill valve housing 2782, the tool 2784 is configured to be rotated relative to the inlet hosing 2758. The rotation of the tool 2784 is configured to rotate the duckbill valve housing 2782 and thus the duckbill valve 2750 attached to the duckbill valve housing 2782. The rotation of the duckbill valve housing 2782 relative to the inlet hosing 2758 is configured to cause the bayonet mechanism 2782a of the duckbill valve housing 2782 to become disengaged from the bayonet feature of the inlet hosing 2758. Thereafter, the tool 2784 is configured to be removed from the DWT 2702. The removal of the tool 2784 can be configured to also remove the duckbill valve housing 2782 and the duckbill valve 2750, or the duckbill valve housing 2782 and the duckbill valve 2750, rotated out of engagement with the inlet hosing's bayonet feature, can be manually removed from the DWT 2702 such as by pouring the duckbill valve housing 2782 and the duckbill valve 2750 out of the DWT's open proximal end 2786.

[0531] In another exemplary implementation, the duckbill valve housing 2782 includes at least one distal-facing shoulder in the inner passageway of the duckbill valve housing 2782. The distal end of the tool 2784 is configured to be inserted through the proximal opening 2782b of the duckbill valve housing 2782 in a first rotational orientation and to then be rotated relative to the duckbill valve housing 2782, e.g., rotated about a longitudinal axis of the tool 2784, to be in a second rotational orientation that is different from the first rotational orientation. For example, the first and second rotational orientations can be about 90 degrees apart. For another example, the first and second rotational orientations can be in a range of about 45 degrees to about 135 degrees. For yet another example, the first and second rotational orientations can be in a range of about 80 degrees to about 100 degrees. For still another example, the first and second rotational orientations can be in a range of about 85 degrees to about 95 degrees. With the tool 2784 in the first configuration and inserted through the proximal opening 2782b of the duckbill valve housing 2782, the tool 2784 is misaligned from the at least one distal-facing shoulder of the duckbill valve housing 2782. The rotation of the tool 2784 from the first rotational orientation to the second rotational orientation is configured to cause the tool 2784 to align with the at least one distal-facing shoulder. With the tool 2784 in the second configuration, the tool 2784 is configured to be pulled proximally to pull the

duckbill valve housing **2782**, and thus the duckbill valve **2750** attached to the duckbill valve housing **2782**, out of the DWT **2702** due. The duckbill valve housing **2782** proximal pulling of the tool **2784** is configured to cause the tool **2784** to abut the at least one distal-facing shoulder if not already abutting the at least one distal-facing shoulder when rotated to the second rotational orientation.

[0532] The tool **2784** is configured to reattach the duckbill valve housing **2782** and the duckbill valve **2750** to the DWT **2702** in a similar manner as discussed above but in reverse order. Instead of the same duckbill valve **2750** that was removed from the DWT **2702** being reattached to the DWT **2702**, a different, e.g., new, duckbill valve can be attached to the DWT **2702** either with the same duckbill valve housing **2782** that was removed from the DWT **2702** or with, a different, e.g., new, duckbill valve housing. In an exemplary implementation, a new duckbill valve and a new duckbill valve housing are attached to the DWT **2702**, which may improve user experience by not requiring a user to detach a duckbill valve from a valve housing or to attach a duckbill valve to a duckbill valve housing.

[0533] As in this illustrated implementation, the cap **2708** can be configured to be opened manually by a user to allow debris to exit the debris collection and storage cavity **2706** of the DWT **2702** without releasing the DWT **2702** from the main body **2704**. The cap **2708** can have a variety of configurations. As in this illustrated implementation, as shown in FIG. **63BB**, the cap **2708** can include a body **2708a**, a hook **2708b**, a plug **2708c**, and a pivot member **2708d**. The cap body **2708a** can have any of a variety of shapes.

[0534] The cap hook **2708a** extends from an inner surface **2708e** of the body **2708a**. The cap hook **2708a** is configured to engage a corresponding hook **2702a** of the DWT **2702**, e.g., of the DWT housing **2772**, to lock the cap **2708** in a closed position. FIG. **63F** shows the hooks **2708a**, **2704a** of the cap **2708** and the DWT **2702** engaged with one another. The handheld surface cleaner **2700** includes corresponding single hooks **2708a**, **2704a** in this illustrated implementation, but another number of hooks can be used, e.g., corresponding pairs of hooks, etc.

[0535] In other implementations, the DWT **2702** can include a groove configured to engage the cap hook **2708a**, or the cap **2708** can include the groove and the DWT **2702** can include the hook. Using engaging hooks **2708a**, **2704a** may help prevent accidental leakage of debris out of the debris collection and storage cavity **2706** through a groove formed in the DWT **2702** or the cap **2708** in the event that the cap **2708** is not fully engaged with the DWT **2702**.

[0536] The cap plug **2708c** extends from the inner surface **2708e** of the body **2708a**. The cap plug **2708c** is configured to plug a corresponding hole **2702b** formed through a wall of the DWT **2702**, e.g., of the DWT housing **2772**, and, if present, through the DWT lower cover **2710**. The hole **2702b** and the cap plug **2708c** have corresponding circular shapes in this illustrated implementation, but other shapes are possible, e.g., ovular, rectangular, etc. The hole **2702b** is in communication with the debris collection and storage cavity **2706**. With the cap plug **280c** seated in the hole **2702b**, the cap **2708** seals the hole **2702b** to prevent debris from exiting the debris collection and storage cavity **2706** through the hole **2702b** and to prevent debris from entering the debris collection and storage cavity **2706** through the hole **2702b**.

[0537] The cap's pivot member **2708d** is configured to engage a corresponding pivot member of the DWT **2702**, e.g., of the DWT housing **2772**. With the pivot members engaged, the cap **2708** is configured to pivot at the cap's pivot member **2708d** to move the cap **2708** between its open position, in which the hole **2702b** is not plugged closed, and closed position, in which the hole **2702b** is plugged closed.

[0538] In this illustrated implementation, the cap's pivot member **2708d** is fixedly attached to the DWT's pivot member. The cap **2708** is thus non-removable from the DWT **2702**, which may help prevent loss of the cap **2708**.

[0539] In other implementations, the cap **2708** can be removably attached to the DWT **2702**. The cap **2708** being removably attached to the DWT **2702** may facilitate replacement of the cap **2708**

should be cap **2708** be damaged, e.g., the cap plug **2708c** becoming worn due to repeated opening and closing of the cap **2708**, the cap plug **2708c** being damaged by rubbing against a surface during use of the handheld surface cleaner **2700**, the handheld surface cleaner **2700** being dropped accidentally, etc. For example, instead of having fixedly attached corresponding pivot members, the cap **2708** and the DWT **2702** can include releasably attached corresponding pivot members, the cap **2708** and the DWT **2702**.

[0540] In an exemplary implementation, the cap **2708** can include at least one C-shaped clip, and the DWT **2702** can include at least one pin configured to be releasably engaged by the at least one C-shaped clip. With the at least one C-shaped clip clipped to the at least one pin, the cap **2708** is configured to rotate open and closed. To remove the cap **2708** from the DWT **2702**, the cap **2708** can be rotated beyond a threshold amount to cause the C-shaped clips to pop off the pin. The cap **2708**, or a different, e.g., new, cap can be then reattached to the DWT **2702**. The at least one C-shaped clip can be rubber or other flexible material to facilitate release of the at least one C-shaped clip from the at least one pin.

[0541] In another exemplary implementation, the cap **2708** can include at least one magnet, and the DWT **2702** can include at least one magnet configured to magnetically engage the cap's at least one magnet. With the cap's at least one magnet magnetically engaged with the DWT's at least one magnet, the cap **2708** is configured to be selectively removed from the DWT **2702** by moving the cap **2708** to break the magnetic bond. The cap **2708**, or a different, e.g., new, cap can be then reattached to the DWT **2702**.

[0542] As discussed herein, the DWT **2702** and the main body **2704** are releasably attachable. As in this illustrated implementation, the DWT **2702** can be configured to be released completely from the main body **2704**. The releasable attachment can be achieved in any of a variety of ways.

[0543] The DWT **2702** can be configured to be attached to the main body **2704** using rotational motion only, translational motion (also referred to herein as “longitudinal motion”) only, or a selected one or both of rotational motion and translational motion. Achieving releasable attachment using only one of rotational motion and translation motion may ease user experience since attachment may be achieved using a single known movement. Achieving releasable attachment using rotational motion and/or translation motion may improve user experience by allowing the user to attach the DWT **2702** and the main body **2704** together using motion(s) most comfortable for the user and/or without the user becoming frustrated at a lack of attachment when a wrong of one of rotational motion and translation motion is used. In this illustrated implementation, as discussed further below, the DWT **2702** is configured to be attached to the main body **2704** using rotational motion and/or translational motion.

[0544] The DWT **2702** and the main body **2704** include corresponding attachment mechanisms configured to engage one another to releasably attach the DWT **2702** and the main body **2704**. The attachment mechanisms can have a variety of configurations. In general, the corresponding attachment mechanisms are configured to allow the DWT **2702** and the main body **2704** to be attached together using a selected one or more of rotational motion and translational motion.

[0545] As in this illustrated implementation, as shown in FIGS. **63G-63J**, the attachment mechanism of the DWT **2702** includes at least one opening **2788** and at least one bar **2790**, and the attachment mechanism of the main body **2704** includes at least one hook **2792**. The at least one hook **2792** is configured to be seated in the at least one opening **2788** to attach the DWT **2702** and the main body **2704**.

[0546] In this illustrated implementation, two openings **2788** and two hooks **2792** are used, but another corresponding number of openings **2788** and hooks **2792** can be used, e.g., one, three, etc. In other implementations, the DWT's attachment mechanism can include at least one hook and the main body's attachment mechanism can include at least one opening and at least one pivot bar. In this illustrated implementation, one bar **2790** is used, but another number can be used, such as one bar for each of the hooks.

[0547] The at least one hook **2792** is biased radially outward by at least one bias element **2794**. The radial outward direction is downward in the views of FIGS. **63D** and **63V**. As shown in FIG. **63CC**, the handheld surface cleaner **2700** in this illustrated implementation includes a pair of bias elements **2794**, one per hook **2792**, although another number of bias elements can be used. The at least one bias element **2794** includes a spring belt in this illustrated implementation but another type of bias element can be used.

[0548] In some implementations, the at least one bias element **2794** is omitted and the hook **2792** is not configured to be urged into engagement and is configured to remain disengaged. For example, the handheld surface cleaner **2700** can include a bump detent configured to keep the hook **2792** out of engagement until the hook **2792** is manually pressed over the bump detent.

[0549] The DWT **2702** and the main body **2704** are configured to be attached using longitudinal motion along the first longitudinal axis **A4** (see FIG. **63E**). One of the DWT **2702** and the main body **2704** can be stationary while the other of the DWT **2702** and the main body **2704** is moved longitudinally toward the one of the DWT **2702** and the main body **2704**, or both of the DWT **2702** and the main body **2704** can be moved longitudinally toward each other. When the DWT **2702** and the main body **2704** become sufficiently close, an exterior chamfered surface **2792b** of the at least one hook **2792** is configured to slide along the at least one bar **2790**, which causes the bias element force to be overcome and the at least one hook **2792** to pivot radially inward, e.g., upwardly, about a pivot axis **P** defined by a pivot pin **2796**. When the at least one hook **2792** has pivoted radially inward a sufficient amount, a tip **2792b** of the at least one hook **2792** is configured to clear the at least one bar **2790** to cause the at least one hook **2792** to be seated in the at least one opening **2988** by the at least one hook **2792** pivoting radially outward, e.g., downwardly, due to the biasing element force applied to the at least one hook **2792**. The main body **2704** and the DWT **2702** abut one another with the main body **2704** positioned over the DWT's open proximal end **2786**.

[0550] The DWT **2702** and the main body **2704** are also configured to be attached using rotational motion. The at least one hook **2792** is configured to be seated in the at least one opening **2988** with the main body **2704** angled relative to the DWT **2702**, e.g., with a longitudinal axis of the main body **2704** angled relative to a longitudinal axis of the DWT **2702** that is substantially parallel to the first longitudinal axis **A4**. With the at least one hook **2792** seated in the at least one opening **2988**, the main body **2704** is configured to be rotated in a counterclockwise direction **C1** (see FIG. **63D**) relative to the DWT **2702**. Alternatively, the DWT **2702** can be rotated in a clockwise direction relative to the main body **2704**, or the main body **2704** can be rotated in the counterclockwise direction **C1** simultaneously with the DWT **2702** being rotated in the clockwise direction. Once the DWT **2702** and/or the main body **2704** has rotated a sufficient amount, the main body **2704** and the DWT **2702** abut one another with the main body **2704** positioned over the DWT's open proximal end **2786**.

[0551] The DWT **2702** and the main body **2704** are configured to be detached using rotational motion in a similar manner as discussed above but in reverse order.

[0552] In this illustrated implementation, the handheld surface cleaner **2700** includes a lock configured to help secure the DWT **2702** to the main body **2704**. The lock in this illustrated implementation is the latch **2712**. The latch **2712** can have a variety of configurations. As shown in FIG. **63S**, the latch **2712** in this illustrated implementation includes a body **2712a**, at least one pivot pin **2712b**, and a finger **2712c**.

[0553] The finger **2712c** extends radially inward, e.g., downwardly in the view of FIGS. **63D** and **63V**, from an inner surface **2712d** of the body **2712a**. The finger **2712c** has a rectangular shape in this illustrated implementation, but other shapes are possible. The finger **2712c** is configured to engage the microswitch **2714**, as discussed further below. The latch **2712** includes a single finger **2712c** in this illustrated implementation to correspond to the handheld surface cleaner **2700** include a single microswitch **2714**.

[0554] The at least one pivot pin **2712b** is configured to be seated in a corresponding at least one

groove **2713** formed in the DWT **2702**, e.g., the DWT housing **2772**. The latch **2712** is configured to pivot about the at least one pivot pin **2712b** relative to the DWT **2702**, e.g., the DWT housing **2772**, and, with the DWT **2702** attached to the main body **2704**, relative to the main body **2704**. The latch **2712** includes a pair of opposed pivot pins **2712b** and the DWT **2702** includes a corresponding pair of grooves **2713** in this illustrated implementation, but another number of pivot pins and grooves is possible.

[0555] The latch **2712** is attached to the DWT **2702**, e.g., the DWT housing **2772**. As in this illustrated implementation, the latch **2712** can be pivotally attached to the DWT **2702**, e.g., the DWT housing **2772**.

[0556] The handheld surface cleaner **2700** includes a bias element **2715** (see FIG. **63T**) attached to the DWT **2702**, e.g., the DWT housing **2772**, and to the latch **2712**. The latch **2712** is thus attached to the DWT **2702**, e.g., the DWT housing **2772**, by the bias element **2715** in this illustrated implementation. The bias element **2715** is configured to bias the latch **2712** downwardly, e.g., toward the microswitch **2714** with the DWT **2702** attached to the main body **2704**. The bias element **2715** includes a single coil spring in this illustrated implementation but another number of bias elements is possible and another type of bias element is possible.

[0557] The latch **2712** is configured to move between a locked configuration, in which the latch **2712** prevents the DWT **2702** and the main body **2704** from being detached, and an unlocked configuration, in which the latch **2712** allows the DWT **2702** and the main body **2704** to be detached. FIGS. **63A**, **63B**, **63E**, **63F**, **63G**, **63H**, and **63AA** show the latch **2712** in the locked configuration.

[0558] The latch **2712** includes a stop surface **2712e** that faces proximally, e.g., away from the main body **2704** with the DWT **2702** and the main body **2704** being attached together. The latch's stop surface **2712e** includes a pair of opposed stop surfaces on either side of the finger **2712c** in this illustrated implementation, but another number and/or position of stop surfaces is possible. The latch's stop surface **2712e** is configured to abut a corresponding stop surface **2798a** of the main body **2704** with the DWT **2702** and the main body **2704** being attached together. The main body's stop surface **2798a** faces distally, e.g., away from the DWT **2702** with the DWT **2702** and the main body **2704** being attached together. The main body's stop surface **2798a** includes a pair of opposed stop surfaces on either side of the microswitch **2714** in this illustrated implementation, but another number and/or position of stop surfaces is possible. The main body's stop surface **2798a** is defined by a microswitch housing **2798** of the main body **2704**, which is discussed further below.

[0559] With the latch **2712** in the locked configuration, the main body's stop surface **2798a** abuts the latch's stop surface **2712e** to prevent detachment of the DWT **2702** and the main body **2704**.

[0560] The latch **2712** is configured to move automatically to the locked configuration in response to the DWT **2702** and the main body **2704** being attached together, whether the attachment is via longitudinal motion or rotational motion. A user thus does not have to take a separate action to lock the DWT **2702** and the main body **2704** together. The biasing of the latch **2712** by the bias element **2715** is configured to allow for the automatic movement of the latch **2712** to the locked configuration.

[0561] The latch **2712** is configured to be moved manually from the locked configuration to the unlocked configuration. The DWT **2702** and the main body **2704** may thus remain locked together until a user chooses to separate the DWT **2702** and the main body **2704**. The latch **2712**, e.g., an outer surface **2712f** of the body **2712a** (see FIG. **63F**), is configured to be pushed by a user over the bias element **2715** to move the latch **2712** from the locked configuration to the unlocked configuration. The pushing of the latch **2712** is configured to cause the latch **2712** to pivot the latch **2712** upwardly, e.g., in a direction away from the microswitch **2714**.

[0562] The handheld surface cleaner **2700** is configured to automatically detect whether the DWT **2702** is releasably coupled to the main body **2704**. Similar to that discussed above regarding the microswitch **40** of the handheld surface cleaner **10** of FIG. **1**, the activation of the microswitch

2714 of the handheld surface cleaner **2700** of FIG. **63A** indicates that the DWT **2702** is releasably coupled to the main body **2704**. Correspondingly, the microswitch **2714** not being activated indicates that the DWT **2702** is not releasably coupled to the main body **2704**.

[0563] With the latch **2712** in the locked position, the latch **2712**, e.g., the finger **2712c** of the latch **2712**, is configured to push the microswitch **2714** to activate the microswitch **2714**. With the latch **2712** in the unlocked configuration, the microswitch **2714** is configured to not be activated, e.g., the finger **2712c** of the latch **2712** does not push the microswitch **2714**.

[0564] As shown in FIG. **63T**, the microswitch **2714** is located in a recessed area **2798b** of the microswitch housing **2798**. The finger **2712c** of the latch **2712** has a size and shape configured to fit in the recessed area **2798b** of the microswitch housing **2798**.

[0565] The recessed area **2798b** has a width **2798c** that is large enough to allow the finger **2712c** of the latch **2712** to enter the recessed area **2798a** and push the microswitch **2714**. The width **2708c** of the recessed area **2798b** is too small for a finger of a user to enter the recessed area **2798a** to prevent pushing of the microswitch **2714** by a user, such as when the user is holding the main body **2704** without the main body **2704** being attached to the DWT **2702**. The microswitch housing **2798** is thus configured to help prevent accidental activation of the microswitch **2714** and falsely indicate that the DWT **2702** is attached to the main body **2704**. In an exemplary implementation, the width **2708c** of the recessed area **2798b** is in a range of about 1 mm to about 10 mm, e.g., about 1 mm to about 5 mm, a range of about 1 mm to about 3 mm, a range of about 1 mm to about 2 mm, a range of about 2 mm to about 3 mm, a range of about 3 mm to about 4 mm, a range of about 2 mm to about 4 mm, a range of about 3 mm to about 5 mm, a range of about 4 mm to about 5 mm, a range of a range of about 4 mm to about 6 mm, a range of about 5 mm to about 6 mm, a range of about 5 mm to about 7 mm, a range of about 6 mm to about 7 mm, a range of about 6 mm to about 8 mm, a range of about 7 mm to about 8 mm, a range of about 7 mm to about 9 mm, a range of about 8 mm to about 9 mm, a range of about 8 mm to about 10 mm, a range of about 9 mm to about 10 mm, about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 10 mm, etc.

[0566] The width **2708c** of the recessed area **2798b** is large enough for a tool to enter the recessed area **2798b** and push the microswitch **2714** without the DWT **2702** and the main body **2704** being attached, which may be desirable for, e.g., testing purposes. FIG. **63U** illustrates one implementation of a tool **2799** configured to enter the recessed area **2798b** and push the microswitch **2714**.

[0567] Instead of or in addition to including the microswitch **2714**, the handheld surface cleaner **2700** can include a switch in a circuit also including the separator motor **2718** and the power supply **2734**. The finger **2712c** or another protrusion of the DWT **2702** can be configured to push the switch, and thus activate the switch, similar to that discussed herein regarding the microswitch **2714**. With the switch activated, the switch is closed and the circuit is closed, which allows the power supply **2734** to supply power to the separator motor **2718**. With the switch deactivated, the switch is open and the circuit is open, which prevents the power supply **2734** from supplying power to the separator motor **2718**. The switch can be located, for example, at the main body **2704** similar to a location described herein for the microswitch **2714**.

[0568] The inlet **2736** at the distal end of the handheld surface cleaner **2700**, e.g., the distal end of the DWT **2702**, is configured to releasably couple to an attachment to help facilitate surface cleaning, such as any of the attachments **100, 200, 300, 310, 400** of FIGS. **19-23C** and **25A-29C**. As in this illustrated implementation, and as discussed further below, the handheld surface cleaner **2700** can be configured to prevent releasable coupling of an attachment to the inlet **2736** without the hair cage **2746** being coupled to the handheld surface cleaner **2700**, e.g., to the DWT **2702**.

[0569] A hair cage of a handheld surface cleaner can be a singular part or can include a plurality of parts. The hair cage **72** of FIG. **17** illustrates one implementation of a hair cage as a singular part. The hair cage **2746** of FIGS. **63P-63Q** illustrates one implementation of a hair cage that includes a

plurality of parts.

[0570] As shown in FIGS. 63P-63Q, the hair cage 2746 includes a first part 2746a and a second part 2746b. The first and second parts 2746a, 2746b of the hair cage 2746 are movably attached to one another, which may facilitate cleaning of the hair cage 2746, as discussed further below. The first and second parts 2746a, 2746b of the hair cage 2746 can be movably attached together in a variety of ways. As in this illustrated implementation, the first and second parts 2746a, 2746b of the hair cage 2746 can be hingedly attached together at a hinge 2746c on a bottom side of the hair cage 2746 that faces downwardly, although the first and second parts 2746a, 2746b can be attached elsewhere. The bottom of the hair cage 2746 is relative to an orientation of the hair cage 2746, with the downward direction being as shown in the views of FIGS. 63D, 63R, and 63V.

[0571] The hair cage 2746 includes a lock 2746d configured to lock the first and second parts 2746a, 2746b of the hair cage 2746 together. The lock 2746d is configured to move between a locked configuration, in which the hair cage 2746 is in a closed position and the lock 2746d prevents the first and second parts 2746a, 2746b from moving at the hinge 2746c, and an unlocked configuration, in which the hair cage 2746 is in an open position and the lock 2746d allows the first and second parts 2746a, 2746b to move at the hinge 2746c. With the hair cage 2746 in the open position, an interior of the hair cage 2746 is configured to help clean out debris because, in some instances, some debris may become dried to or otherwise stuck to the hair cage 2746.

[0572] The lock 2746d can have any of a variety of configurations. In this illustrated implementation, the lock 2746d includes a pair of interlocking lips. The hair cage's first part 2746a includes a first one of the interlocking lips, and the hair cage's second part 2746b includes a second one of the interlocking lips.

[0573] The hair cage lock 2746d is configured to be moved manually between the locked and unlocked configurations, which may allow the hair cage 2746d to remain locked while coupled to the handheld surface cleaner 2700, e.g., to the DWT 2702. As in this illustrated implementation, the hair cage 2746 can include a finger rest configured to facilitate manual handling of the hair cage 2746 for locking and unlocking. As in this illustrated implementation, the finger rest can include a first finger rest 2746e of the first part 2746a and a second finger rest 2746f of the second part 2746b. The first and second finger rests 2746e, 2746f are each configured to provide a surface on which a user can rest a finger to provide leverage for pushing the first and second parts 2746a, 2746b apart to move the hair cage 2746 from the locked configuration to the unlocked configuration or to push the first and second parts 2746a, 2746b together to move the hair cage 2746 from the unlocked configuration to the locked configuration.

[0574] A hair cage of a handheld surface cleaner can include a mesh filter. The mesh filter may help provide structural stability to the hair cage and/or may facilitate manufacturing of the part(s) of the hair cage. The hair cage 72 of FIG. 17 illustrates one implementation of a hair cage without a mesh filter. The hair cage 2746 of FIGS. 63P-63Q illustrates one implementation of a hair cage that includes a mesh filter 2746g. As in this illustrated implementation, the mesh filter 2746g can be located in an interior of the hair cage 2746.

[0575] As in this illustrated implementation, as shown in FIGS. 63P-63Q and 63DD, the hair cage 2746 can include a distal housing 2746h. The distal housing 2746h is configured to seat the inlet seal 2738, as shown in FIGS. 63R and 63DD. The distal housing 2746h includes a frame 2746i surrounding an opening in which the inlet seal 2738 is positioned. The distal housing 2746h includes an arm 2746j extending proximally from the frame 2746i of the distal housing 2746h.

[0576] As mentioned above, the hair cage 2746 can, as in this illustrated implementation, be configured to be removably coupled to the handheld surface cleaner 2700, e.g., to the DWT 2702. As also mentioned above, the handheld surface cleaner 2700 in this illustrated implementation is configured to prevent releasable coupling of an attachment to the inlet 2736 without the hair cage 2746 being coupled to the handheld surface cleaner 2700, e.g., to the DWT 2702.

[0577] As shown in FIG. 63DD, the handheld surface cleaner 2700, e.g., the DWT 2702, includes a

lockout housing **2747** configured to releasably seat the arm **2746j** of the hair cage **2746**. FIG. **63DD** shows the arm **2746j** seated in the locking housing **2747**.

[0578] A lockout member **2749** is movably attached to the lockout housing **2747**. The movable attachment is a hinged attachment in this illustrated implementation, but other attachments are possible. The lockout member **2749** can have a variety of configurations. As in this illustrated implementation, the lockout member **2749** can include a plate although other configurations are possible, such as a button or other member.

[0579] The lockout member **2749** is configured to move between a retracted configuration, in which the arm **2746j** is seated in the lockout housing **2747**, the lockout member **2749** allows distal housing **2745** to be attached to the handheld surface cleaner **2700**, and the lockout member **2749** allows an attachment to be attached to the handheld surface cleaner **2700**, and an extended configuration, in which the arm **2746j** is not seated in the lockout housing **2747**, the lockout member **2749** prevent the distal housing **2745** from being attached to the handheld surface cleaner **2700** and the lockout member **2749** prevent an attachment from being attached to the handheld surface cleaner **2700**. FIG. **63DD** shows the lockout member **2749** in the retracted configuration. FIGS. **63EE** and **63FF** show the distal housing **2745** as a standalone element.

[0580] The handheld surface cleaner **2700** includes a bias element **2751** is configured to bias the lockout member **2749** to the extended configuration. The bias element **2751** is a coil spring in this illustrated implementation, but other types of bias elements may be used.

[0581] The hair cage **2746** is configured to be pushed into the DWT **2702** in a proximal direction without the distal housing **2745** attached to the handheld surface cleaner **2700** and without an attachment attached to the handheld surface cleaner **2700**. The arm **2746j** is configured to move proximally into the locking housing **2747** during the proximal movement of the hair cage **2746**. The arm's movement into the locking housing **2747** is configured to cause the lockout member **2749** to move from the extended configuration to the retracted configuration by pivoting in response to the arm **2746j** pushing on the lockout member **2749**.

[0582] With the hair cage **2746** attached to the DWT **2702**, the distal housing **2745** is configured to be attached to the DWT **2702** by being pushed proximally onto the DWT **2702**. The handheld surface cleaner **2700** includes a distal housing lock **2753** configured to move from an unlocked configuration, in which the distal housing **2745** is not locked to the handheld surface cleaner **2700**, e.g., the DWT **2702**, and a locked configuration, in which the distal housing **2745** is locked to the handheld surface cleaner **2700**, e.g., the DWT **2702**. The distal housing lock **2753** is configured to move automatically from the unlocked configuration to the locked configuration in response to the distal housing **2745** being coupled to the handheld surface cleaner **2700**, e.g., the DWT **2702**, which may improve user experience by helping to ensure that the distal housing **2753** is locked to the handheld surface cleaner **2700**. The distal housing lock **2753** is configured to be manually actuated by a user to move the distal housing lock **2753** from the locked configuration to the unlocked configuration, which may help ensure that the distal housing **2753** remains attached to the handheld surface cleaner **2700** unless desired to be removed by a user, e.g., to gain access to the hair cage **2746** to remove the hair cage **2746** for cleaning or replacement.

[0583] The distal housing lock **2753** can have a variety of configurations. As in this illustrated implementation, the distal housing lock **2753** can include at least one hook **2753a** configured to engage, e.g., clip into, at least one groove **2745b** formed in the distal housing **2745**. The distal housing lock **2753** includes a pair of opposed hooks **2753a** in this illustrated implementation that are configured to engage a corresponding pair of grooves of the distal housing **2745**, but another number and/or positioning of the at least one hook **2753a** and the at least one groove is possible. FIGS. **63A** and **63B** show the at least hook **2753a** engaged with the at least one groove formed in the distal housing **2745**. The at least one hook **2753a** is configured to move automatically into the at least one groove of the distal housing **2745** in response to the distal housing **2745** being attached to the handheld surface cleaner **2700**, e.g., the DWT **2702**.

[0584] The distal housing lock **2753** includes at least one bias element **2753b** configured to bias the distal housing lock **2753**, and thus the at least one hook **2753a** downwardly. The at least one hook **2753a** will thus be urged into engagement with the distal housing **2745**. The downward direction is relative to an orientation of the distal housing lock **2753** and is as shown in the views of FIGS. **63D**, **63R**, and **63V**. The at least one bias element **2753a** includes a single coil spring in this illustrated implementation, but the at least one bias element **2753a** can include a different number and/or different type of bias element.

[0585] With the hair cage **2746** attached to the DWT **2702**, the lockout member **2749** is in the retracted configuration and does not impede the proximal movement of the distal housing **2745** into position to engage the distal housing lock **2753**. A shoulder **2745a** of the distal housing **2745** is spaced a distance away from the lockout member **2749**.

[0586] Without the hair cage **2746** attached to the DWT **2702**, and thus with the lockout member **2749** in the extended configuration, the lockout member **2749** is configured to abut the shoulder **2745a** of the distal housing **2745** as the distal housing **2745** is being attempted to be attached to the handheld surface cleaner **2700**. The abutting of the lockout member **2749** and the distal housing **2745**, e.g., the shoulder **2745a**, is configured to stop proximal movement of the distal housing **2745** and thus prevent attachment of the distal housing **2745** to the DWT **2702**.

[0587] With the distal housing **2745** attached to the DWT **2702**, an attachment is configured to be attached to the handheld surface cleaner **2700**. Correspondingly, without the distal housing **2745** attached to the DWT **2702**, an attachment cannot be attached to the handheld surface cleaner **2700**. Thus, the distal housing **2745** being configured to be locked out from attachment to the handheld surface cleaner **2700**, e.g., the DWT **2702**, unless the hair cage **2746** is attached to the handheld surface cleaner **2700**, e.g., the DWT **2702**, is configured to prevent attachment of an attachment to the handheld surface cleaner **2700**, e.g., the DWT **2702**, unless the hair cage **2746** is attached to the handheld surface cleaner **2700**. Debris larger than a certain size that would otherwise be trapped by the hair cage **2746** and prevented from flowing through the hair cage **2746** to the separator **2724** may thus be ensured of being trapped by the hair cage **2746** instead of flowing to the separator **2724**, where the larger-size debris may risk damage to and/or jamming of the separator **2724**.

[0588] A handheld surface cleaner can be configured to be removably attached to an attachment in a variety of ways. For example, in some implementations, the attachment can be configured to slide onto the handheld surface cleaner, e.g., by a rail and a cooperating track. For another example, in some implementations, the attachment can be configured to snap onto the handheld surface cleaner, e.g., by a hook and a cooperating groove or opening.

[0589] As shown in FIG. **63FF**, the distal housing **2745** in this illustrated implementation includes at least one hook **2745c**. The at least one hook **2745c** is configured to releasably engage an attachment, e.g., be releasably seated in at least one groove or opening in the attachment, to attach the attachment to the distal housing **2745** and thus to the handheld surface cleaner **2700** to which the distal housing **2745** is attached. The at least one hook **2745c** includes a pair of opposed hooks **2745c** in this illustrated implementation, but another number of hooks **2745c** is possible. FIG. **20** illustrates one implementation of at least one opening **100a** of an attachment **100** configured to be engaged by the at least one hooks **2745** (only one of the pair of openings **100a** is visible in the view of FIG. **20**). FIGS. **23A** and **23C** illustrate another implementation of at least one opening **200b** of an attachment **200** configured to be engaged by the hooks **2745c** (only one of the pair of openings **200a** is visible in the views of FIGS. **23A** and **23C**).

[0590] In another implementation, the distal housing includes the at least one groove, and the attachment includes the at least one hook.

[0591] As mentioned above, the handheld surface cleaner **2700** in this illustrated implementation includes a charging port **2756**. As shown in FIGS. **63C**, **63E**, and **63J**, the charging port **2756** in this illustrated implementation includes a magnetic port configured to magnetically mate with a corresponding charging contact of a charger, e.g., a charger **3100** of FIGS. **68A** and **68B** (discussed

further below) or another charger.

[0592] FIGS. **63GG** and **63HH** illustrate another implementation of the main body **2704** of FIG. **63A** as a main body **2704'** that includes a different type of charging port **2756'**. The charging port **2756'** in this illustrated implementation includes a DC jack configured to be plugged into by a DC charger cord, e.g., a DC charger cord **2757** of FIG. **63II** or other charger cord.

[0593] FIG. **63JJ** illustrates another implementation of the separator **2724**, the shaft **2740**, the centrifugal seal **2760**, the rim **2762**, the base **2763**, the plurality of movable covers **2764** of FIGS. **63M** and **63N** as a separator **2724'**, a centrifugal seal **2760'**, a rim **2762'**, a base **2763'**, and a plurality of movable covers **2764'** respectively. In this illustrated implementation, a seal **2765'** is provided between the rim **2762'** and a body **2768'** of the separator **2724'**, and the separator body **2768'** and the rim **2762'** are welded together.

[0594] FIG. **63KK** illustrates another implementation of the separator **2724**, the shaft **2740**, the centrifugal seal **2760**, the rim **2762**, the base **2763**, the plurality of movable covers **2764** of FIGS. **63M** and **63N** as a separator **2724''**, a centrifugal seal **2760''**, a rim **2762''**, a base **2763''**, and a plurality of movable covers **2764''** respectively. In this illustrated implementation, a seal **2765''** is provided between the rim **2762''** and a body **2768''** of the separator **2724''**, the rim **2762''** has a plurality of arms that extend proximally to a circumferential ring base of the rim **2762''** and are located outside of the separator body **2768''**, and the separator body **2768''** includes at least one hook **2767''** configured to attach to the rim **2762''** to attach the rim **2762''** and the separator body **2768''** together. The rim's arms extend along an entire length of the rim **2762''** in this illustrated implementation but can instead extend along only a partial length of the rim **2762''** in a proximal direction. In such an implementation, the circumferential ring base of the rim **2762''** is omitted.

[0595] A handheld surface cleaner (e.g., any of the handheld surface cleaners described herein) can be configured to be seated in a tray (also referred to herein as a “dock”). The tray is configured to provide convenient storage of the handheld surface cleaner. In some implementations, the handheld surface cleaner is configured to be recharged while seated in the tray, and/or the tray is also configured to seat one or more accessories configured for use with the handheld surface cleaner. Examples of the accessories include attachments and spray bottles.

[0596] A tray can have any of a variety of configurations. In general, the tray has a size and shape configured to complement a size and shape of the handheld surface cleaner and, if configured to seat one or more accessories, to complement a size and shape of each of the one or more accessories.

[0597] FIG. **64** illustrates schematically one implementation of a dock **2800** configured to seat a handheld surface cleaner. FIG. **64** shows the handheld vacuum cleaner **2350** of FIG. **57B** disposed within the dock **2800** although seating of another handheld surface cleaner is possible.

[0598] The dock **2800** in this illustrated implementation is configured as a charging dock configured to recharge one or more batteries of the handheld vacuum cleaner **2350** with the handheld vacuum cleaner **2350** seated in the dock **2800**. In some instances, as in this illustrated implementation, one or more of the separation assembly **2354** and/or the main body **2358** include charging contacts for recharging the one or more batteries. When the separation assembly **2354** includes charging contacts, the separation assembly **2354** included wiring and electrical connectors for electrically coupling the charging contacts to the one or more batteries. The electrical connectors are configured to electrically couple to corresponding electrical connectors on the main body **2358**.

[0599] In some instances, and as shown in this illustrated implementation, the dock **2800** covers at least a portion of the separation assembly **2354** (e.g., a dirty water tank) of the handheld vacuum cleaner **2350**.

[0600] The dock **2800** is configured to provide one or more maintenance functions to the handheld vacuum cleaner **2350**. For example, the dock **2800** can be configured to encourage a drying of one or more components of the handheld vacuum cleaner **2350** and/or to empty debris from the

separation assembly **2354**.

[0601] In the example of drying, a self-drying cycle can be configured to run to dry one or more components of the handheld vacuum cleaner **2350** by running the suction motor **2302** for a predetermined amount of time (e.g., ten seconds, thirty seconds, sixty seconds, ninety seconds, etc.) to draw air through the handheld vacuum cleaner **2350**. Some components, such as filter foam, may be more likely to not be dry than other components when suction stops in a cleaning operation. In an exemplary implementation, with the handheld vacuum cleaner **2350** docked a user can push a button (e.g., on the handheld vacuum cleaner **2350** or on the dock **2800**) to start the self-drying cycle. In some implementations, the self-drying cycle is locked out unless the handheld vacuum cleaner **2350**, e.g., a charging port of the cleaner **2350**, is coupled to a charger, e.g., the DC charger cord **2757** of FIG. **63HH**, the charger **3100** of FIGS. **67A** and **66B**, or other charger, which may help ensure that the suction motor **2302** receives sufficient power to run for the predetermined amount of time.

[0602] FIGS. **65A** and **65B** illustrate another implementation of a tray **2900** configured to seat a handheld surface cleaner. In particular, the illustrated tray **2900** of FIGS. **65A** and **65B** is configured to seat the handheld surface cleaner **10** of FIG. **1** and the attachments **100**, **200**, **400** of FIGS. **19-23C** and **29A-29C** although seating of another handheld surface cleaner and/or other attachment(s) is possible. The tray **2900** includes first, second, third, and fourth docking areas to seat the handheld surface cleaner **10** and the attachments **100**, **200**, **400**, respectively. In this illustrated implementation the docking areas include first, second, third, and fourth cavities **2902**, **2904**, **2906**, **2908** configured to seat the handheld surface cleaner **10** and the attachments **100**, **200**, **400**, respectively. The docking areas **2902**, **2904**, **2906**, **2908** may help keep the handheld surface cleaner **10** and the attachments **100**, **200**, **400** organized in the tray **2900** and/or may help a user easily identify which one or more of the handheld surface cleaner **10** and the attachments **100**, **200**, **400** are not stored in the tray **2900**. In some implementations, the tray **2900** can also include a cavity configured to store a spray bottle.

[0603] The tray **2900** in this illustrated implementation is configured as a caddy to facilitate carrying together all of the handheld surface cleaner **10** of FIG. **1** and the attachments **100**, **200**, **400** of FIGS. **19-23C** and **29A-29C** seated in the tray **2900**. The tray **2900** includes one or more handles **2910** to help a user hold the tray **2900**. The tray **2900** includes a pair of opposed cut-outs as the one or more handles **2910** in this illustrated implementation but can have other configurations, such as opposed hand grips attached to an exterior of the tray **2900**, a single arced handle attached to the tray **2900** similar to a bucket handle or picnic basket handle, or other configuration.

[0604] FIGS. **66A** and **66B** illustrate another implementation of a tray **3000** configured to seat a handheld surface cleaner. The tray **3000** includes first, second, third, and fourth docking areas to seat a handheld surface cleaner, a first attachment, a second attachment, and a spray bottle, respectively. In this illustrated implementation the docking areas include first and second cavities **3002**, **3004** configured to seat the handheld surface cleaner and a spray bottle, respectively, and first and second stands **3006**, **3008** configured to seat the first and second attachments, respectively. As shown in FIG. **66C**, the first docking area **3002** includes a sloped bottom surface **3002s** having a size and shape complementary to a corresponding surface of the handheld surface cleaner.

[0605] The tray **3000** in this illustrated implementation is configured as a caddy to facilitate carrying together all of the handheld surface cleaner, the spray bottle, and the first and second attachments seated in the tray **3000**. The tray **3000** includes one or more handles **3010** to help a user hold the tray **3000**. The tray **3000** includes a single arced handle **3010** attached to the tray **3000** in this illustrated implementation but can have other configurations, such as opposed hand grips attached to an exterior of the tray **3000**, a pair of opposed cut-outs, or other configuration.

[0606] The tray **3000** in this illustrated implementation is configured as a charging dock configured to allow the handheld surface cleaner to be charged while seated in the tray **3000**, e.g., seated in the first cavity **3002**. As shown in FIG. **66A**, the tray **3000** has a lowered sidewall **3000a** adjacent the

first cavity **3002**. The lowered sidewall **3000a** is configured to facilitate access to a charging contact of the handheld surface cleaner with the handheld surface cleaner seated in the first cavity **3002**.

[0607] In some implementations, a handheld surface cleaner is configured to run a cleaning cycle (also referred to herein as a “rinse operation”) while seated in a tray configured to seat the handheld surface cleaner. After the handheld surface cleaner is used to clean a surface, a small amount of debris may occasionally be in the handheld surface cleaner along a debris flow path of the handheld surface cleaner distal to an endpoint of the debris flow path defined by a collection area in the handheld surface cleaner where debris is configured to be collected prior to disposal, e.g., in a collection cavity of a dirty water tank of the handheld surface cleaner. For example, suction may be turned off before all debris that has entered an inlet of the handheld surface cleaner has traveled far enough along the debris flow path to be collected in the collection area in the handheld surface cleaner.

[0608] The cleaning cycle may help ensure that the flow path in the handheld surface cleaner is as clean as possible before a next use of the handheld surface cleaner to clean a surface and/or may help prevent debris from drying inside the handheld surface cleaner. Dried debris stuck to an interior surface of the handheld surface cleaner will be generally more difficult to draw into the handheld surface cleaner's debris collection area than debris that is not dried and stuck to an interior surface of the handheld surface cleaner. The cleaning cycle may be run automatically and not require a user to stop the cleaning cycle, which may help ensure that the full cleaning cycle is run and may improve user experience by not requiring the user to attend to the handheld surface cleaner during the cleaning cycle.

[0609] FIG. **67** illustrates another implementation of a tray **3020** configured to seat a handheld surface cleaner. In this illustrated implementation, the handheld surface cleaner is configured to run a cleaning cycle while seated in the tray **3020**.

[0610] The tray **3020** includes a first docking area **3022**, a second docking area **3024**, a third docking area (obscured in FIG. **67**), and a fourth docking area **3026** configured to seat a handheld surface cleaner, a first attachment, a second attachment, and a spray bottle, respectively. The tray **3020** of FIG. **67** is generally configured and used similar to the tray **3000** of FIGS. **66A-66C** except that, unlike the first docking area **3002** that is non-removable from the tray **3000** of FIGS. **66A-66C**, the first docking area **3022** of FIG. **67** is defined by a cup **3028** configured to be removably coupled to the tray **3020**. As shown in FIG. **67**, the first docking area **3022** includes a sloped bottom surface having a size and shape complementary to a corresponding surface of the handheld surface cleaner.

[0611] The cup **3028** being removable may facilitate cleaning of the first docking area **3028**. Debris may occasionally fall into the first docking area **3028** from a handheld surface cleaner seated in the first docking area **3028**, particularly if the handheld surface cleaner includes a hair cage or other filter proximal to an inlet of the handheld surface cleaner and distal to a separator of the handheld surface cleaner and the hair cage or other filter is not cleaned by a user prior to seating the handheld surface cleaner in the first docking area **3028**. Similar to the tray **3000** of FIGS. **66A-66C**, a handheld surface cleaner is configured to be seated in the first docking area **3028** with an inlet of the handheld surface cleaner facing downward.

[0612] The cup **3028** being removable may facilitate running a cleaning cycle and thus facilitate cleaning of a debris flow path of a handheld surface cleaner seated in the first docking area **3028**, e.g., seated in the cup **3028**. The cup **3028** includes a fill line configured to indicate a level where a cleaning liquid, e.g., clean water and/or other cleaning liquid, is configured to be filled in the cup **3028**, e.g., in the first docking area **3028**, prior to seating a handheld surface cleaner in the cup **3028**, e.g., in the first docking area **3028**. The fill line in this illustrated implementation is on an interior surface of the cup **3028**, e.g., printed on, etched on, embossed in, on a sticker applied to, or otherwise on the interior surface. The fill line is obscured in FIG. **67**. With the cup **3028** filled with

cleaning liquid up to the fill line and with the handheld surface cleaner seated in the cup **3028**, an inlet of the handheld surface cleaner is configured to be submerged in the cleaning liquid and spaced a distance away from the interior surface of the cup **3028** to allow cleaning liquid to enter the inlet upon application of a suction force.

[0613] With cleaning liquid in the cup **3028** up to the fill line and with the handheld surface cleaner seated in the cup **3028**, the handheld surface cleaner is configured to be activated for a cleaning cycle by a user, e.g., by pushing a cleaning cycle start button on the handheld surface cleaner or otherwise beginning the cleaning cycle. Activating the cleaning cycle is configured to cause the handheld surface cleaner to provide suction for a predetermined amount of time. The predetermined amount of time generally depends on one or more factors such as an amount of cleaning liquid configured to be held by the cup **3028** up to the fill line, a size of the handheld surface cleaner's collection cavity, a strength of the suction force, and/or other factor(s). The activated handheld surface cleaner draws the cleaning liquid from the cup **3028** into the handheld surface cleaner via a suction force according to the handheld surface cleaner's normal operation. The cleaning liquid flows along the debris flow path in the handheld surface cleaner and is collected in the handheld surface cleaner's collection area, cleaning the debris flow path as the cleaning liquid flows along the debris flow path. The collection area may then be emptied by a user. Although the suction force will tend to draw all the cleaning liquid from the cup **3028** into the handheld surface cleaner, a very small amount of cleaning liquid may remain in the cup **3028** when the cleaning cycle ends, and the cup **3028** may be emptied of this remaining cleaning liquid.

[0614] FIGS. **68A** and **68B** illustrate one implementation of a charger **3100** configured to be used in recharging a handheld surface cleaner. The charger **3100** includes a charging contact **3102**, a plug **3104** configured to be plugged into a wall socket, and a cord **3106** connecting the charging contact **3102** and the plug **3104**. The charging contact **3102** is configured to mate with a corresponding charging contact of a handheld surface cleaner. The charging contact **3102** in this illustrated implementation includes a magnetic port configured to magnetically mate with a corresponding charging port of a handheld surface cleaner.

[0615] The charger **3100** can be used, for example, to recharge a handheld surface cleaner seated in the tray **3000**, e.g., seated in the first cavity **3002**. The lower sidewall **3000a** is configured to ease access to the handheld surface cleaner's charging contact for mating with the charger's charging contact **3102** and help prevent the tray **3000** from snagging or interfering with the cord **3106** extending from the charging contact **3102**.

[0616] FIG. **69** illustrates another implementation of a tray **3200** configured to seat a handheld surface cleaner. The tray **3200** includes first, second, third, and fourth docking areas configured to seat a handheld surface cleaner **3202**, a first attachment **3204**, a second attachment **3206**, and a spray bottle **3208**, respectively. In this illustrated implementation the docking areas include first, second, third, and fourth stands configured to seat the handheld surface cleaner **3202**, the first and second attachments **3204**, **3206**, and the spray bottle **3208**, respectively.

[0617] FIGS. **70A** and **70B** illustrate another implementation of a tray **3300** configured to seat a handheld surface cleaner. The tray **3300** includes first, second, third, and fourth docking areas configured to seat a handheld surface cleaner **3302**, a first attachment **3304**, a second attachment **3306**, and a charger, respectively. In this illustrated implementation the docking areas include first, second, and third stands configured to seat the handheld surface cleaner **3302**, the first attachment **3304**, and the second attachment **3306**, respectively, and an opening **3308**, shown in FIG. **70C**, configured to seat the charger, e.g., seat the charging contact **3102** of the charger **3100** of FIGS. **68A** and **68B**.

[0618] The tray **3300** in this illustrated implementation is configured as a charging dock configured to allow the handheld surface cleaner **3302** to be charged while seated in the tray **3300**. FIGS. **70B** and **70C** show the charger **3100** of FIGS. **68A** and **68B** but another charger may be used. With the handheld surface cleaner **3302** seated in the first docking area **3302**, the charger is configured to

access the handheld surface cleaner's charging contact **3310** through the opening **3308**.
[0619] FIG. **71** illustrates another implementation of a tray **3400** configured to seat a handheld surface cleaner. The illustrated tray **3400** of FIG. **71** is configured to seat the handheld surface cleaner **2700** of FIG. **63A** although seating of another handheld surface cleaner is possible. The tray **3400** includes first, second, and third docking areas configured to seat a handheld surface cleaner, a first attachment, a second attachment, and a charger. In this illustrated implementation the docking areas include first, second, and third stands **3402**, **3404**, **3406** configured to seat the handheld surface cleaner, the first attachment, and the second attachment, respectively, and an opening **3408** configured to seat the charger, e.g., seat the charging contact **3102** of the charger **3100** of FIGS. **68A** and **68B**.

[0620] The tray **3400** in this illustrated implementation also includes a plurality of feet **3410** on a bottom side of a base **3412** of the tray **3400**. The tray **3400** includes two feet **3410** at a front of the base **3412** and two feet **3410** (obscured in FIG. **71**) at a back of the base **3412**, although another number and/or positioning of feet can be used.

[0621] The tray **3400** in this illustrated implementation is configured as a charging dock configured to allow the handheld surface cleaner to be charged while seated in the tray **3400**. With the handheld surface cleaner seated in the first docking area **3402**, the charger is configured to access the handheld surface cleaner's charging contact through the opening **3408**.

[0622] FIGS. **72A** and **72B** illustrate another implementation of a tray **3500** configured to seat a handheld surface cleaner, a spray bottle, and a plurality of attachments. The illustrated tray **3500** of FIGS. **72A** and **72B** is configured to seat the handheld surface cleaner **2700** of FIG. **63A** and the attachments **100**, **200**, **400** of FIGS. **19-23C** and **29A-29C** although seating of another handheld surface cleaner and/or other attachment(s) is possible. In this illustrated implementation, the handheld surface cleaner is configured to run a cleaning cycle while seated in the tray **3500** similar to that discussed above regarding the tray **3020** of FIG. **67**.

[0623] The tray **3500** includes a first docking area **3502**, a second docking area **3504**, a third docking area **3506**, a fourth docking area **3508**, and a fifth docking area **3510** configured to seat a handheld surface cleaner, a first attachment, a second attachment, a third attachment, and a spray bottle, respectively. The tray **3500** of FIGS. **72A** and **72B** is generally configured and used similar to the tray **3000** of FIGS. **66A-66C** except that, unlike the first docking area **3002** that is non-removable from the tray **3000** of FIGS. **66A-66C**, the first docking area **3502** of FIGS. **72A** and **72B** is defined by a cup **3512** configured to be removably coupled to the tray **3500**. FIGS. **72A** and **72B** show the cup **3512** coupled to the tray **3500**. FIGS. **72C** and **72D** show the cup **3512** as a standalone element. As shown in FIG. **72D**, the cup **3512** includes a sloped bottom surface having a size and shape complementary to a corresponding surface of the handheld surface cleaner.

[0624] The tray **3500** in this illustrated implementation also includes a plurality of feet **3514** on a bottom side of a base **3516** of the tray **3500**. The tray **3500** includes two feet **3514** at a back of the base **3516**, shown in FIG. **72B**, and two feet **3514** (obscured in FIGS. **72A** and **72B**) at a front of the base **3516**, although another number and/or positioning of feet can be used.

[0625] A handheld surface cleaner (e.g., any of the handheld surface cleaners described herein) can be used to clean a surface (e.g., a floor, upholstery, etc.) without prior application of a cleaning aid, such as a cleaning liquid or other cleaning aid, to the surface. A cleaning aid may not be pre-applied for any of one or more reasons, such as the surface being cleaned not being very soiled, a cleaning aid not being available to a user of the handheld surface cleaner, and/or other reason. However, a user may decide to pre-apply a cleaning aid to the surface to be cleaned using the handheld surface cleaner for any of one or more reasons, such as the surface being cleaned being very soiled and/or other reason.

[0626] A cleaning aid can have any of a variety of configurations (e.g., a liquid, a paste, etc.) and can be applied to a surface to be cleaned in any of a variety of ways. The applied cleaning aid can be recovered from the surface to be cleaned using a handheld surface cleaner.

[0627] In an exemplary implementation, a spray bottle is configured to apply a cleaning liquid to a surface to be cleaned with a handheld surface cleaner (e.g., any of the handheld surface cleaners described herein). The spray bottle can have a variety of implementations.

[0628] The spray bottles described herein are not limited to use with a handheld surface cleaner or in spraying cleaning liquid to a surface to be cleaned. For example, a spray bottle described herein can be configured to spray another type of liquid.

[0629] FIGS. 73-78 illustrate one implementation of a spray bottle **4000** for spraying cleaning liquid for use with a handheld surface cleaner, such as any of the handheld surface cleaners described herein. The spray bottle **4000** in this illustrated implementation includes a spray head **4002** (shown as a standalone unit in FIGS. 79 and 80) and a dual cleaning solution container **4004** (shown as a standalone unit in FIGS. 81-85) configured to removably attach to the spray head **4002**. In an exemplary implementation, the spray head **4002** is reusable and the dual cleaning solution container **4004** is disposable.

[0630] FIGS. 73, 75, 76, 79, and 80 show a hole **4006** on each of left and right sides of the spray head **4002** where a depressible button or other lock (not shown) is located. A user pressing both of the buttons (or other locks) is configured to allow the disposable dual cleaning solution container **4004** to be removed from the spray head **4002**.

[0631] The dual cleaning solution container **4004** includes two separate containers **4004a**, **4004b** attached together, such as via shrink wrap (not shown) and/or a band (not shown) wrapped around the two separate containers **4004a**, **4004b**. Pulling a trigger **4008** of the spray head **4002** is configured to draw liquid from each of the two containers **4004a**, **4004b** and sprays the two liquids mixed together from a spray nozzle **4010** of the spray head **4002**. The spray nozzle **4010** has a cover **4012** in this illustrated implementation that is shown in an open position in FIGS. 73-80. In an exemplary implementation, each of the containers **4004a**, **4004b** contains a different type of liquid therein, e.g., two different cleaning liquids such as a base cleaning solution in one of the container **4004a**, **4004b** and a peroxide boost cleaning solution in the other one of the containers **4004a**, **4004b**, and pulling the trigger **4008** causes a substantially same amount of each of the two liquids to be drawn out of the two separate containers **4004a**, **4004b**.

[0632] FIGS. 86A and 86B illustrate another implementation of a spray bottle **4100** for spraying cleaning liquid for use with a handheld surface cleaner, such as any of the handheld surface cleaners described herein. The spray bottle **4100** of FIGS. 86A and 86B is generally configured and used similar to the spray bottle **4000** of FIGS. 73-78.

[0633] FIGS. 87 and 88 illustrate another implementation of a spray bottle **4200** for spraying cleaning liquid for use with a handheld surface cleaner, such as any of the handheld surface cleaners described herein. The spray bottle **4200** of FIGS. 87 and 88 is generally configured and used similar to the spray bottle **1500** of FIGS. 50A-50D. FIGS. 89-91 show portions of the spray bottle **4200**.

[0634] FIG. 92A illustrates another implementation of a spray bottle **4300** for spraying cleaning liquid for use with a handheld surface cleaner, such as any of the handheld surface cleaners described herein. The spray bottle **4300** of FIG. 92A is generally configured and used similar to the spray bottle **4000** of FIGS. 73-78. FIG. 92B shows a portion of the spray bottle **4300**.

[0635] As shown in FIGS. 92A and 92B, the spray bottle **4300** in this illustrated implementation includes a spray head **4302** and a dual cleaning solution container **4304** configured to removably attach to the spray head **4302**. In an exemplary implementation, the spray head **4302** is reusable and the dual cleaning solution container **4304** is disposable.

[0636] The spray head **4302** includes a lock **4306** configured to allow the dual cleaning solution container **4304** to be removed from the spray head **4302**. The lock **4306** in this illustrated implementation includes a pair of buttons with one button on a left side of the spray head **4302** and one button on a right side of the spray head **4302**. A user pressing both of the buttons is configured to allow the disposable dual cleaning solution container **4304** to be removed from the spray head

4302.

[0637] A top of each of the containers **4304a**, **4304b** of the dual cleaning solution container **4304** includes a vented membrane **4308a**, **4308b**. The vented membranes **4308a**, **4308b** are enclosed within the spray head **4302** regardless of whether or not the spray head **4302** and the dual cleaning solution container **4304** are attached together. A portion of the spray head **4302** is omitted in FIG. **92B** to show the vented membranes **4308a**, **4308b**. Each of the vented membranes **4308a**, **4308b** is configured to allow venting of the cleaning solution in its associated one of the containers **4304a**, **4304b** as a safety feature.

[0638] FIG. **93** illustrates another implementation of a spray bottle **4400** for spraying cleaning liquid for use with a handheld surface cleaner, such as any of the handheld surface cleaners described herein. As shown, the spray bottle **4400** includes a supply tank **4402**, a spray nozzle **4404**, and an agitator **4406**. The agitator **4400** includes a brush in this illustrated implementation but can have another configuration, e.g., ribs, etc. The agitator **4406** is configured to be replaceable by a user (e.g., to enable a user to tailor the fluid applicator **4400** to a surface type of a surface to be cleaned), but in other implementations the agitator **4406** can be non-removable from the spray bottle **4400** and non-replaceable. In some implementations, the agitator **4406** is pivotable and configured to adjust an angle of the agitation brush relative to the surface to be cleaned, which may adjust an angle of a debris flow path, e.g., the flow path **2356** when entering the inlet **2214** of the handheld surface cleaner **2200** of FIG. **57A**.

[0639] The supply tank **4402** includes a first chamber **4408** and a second chamber **4410** that is fluidically separated from the first chamber **4408**. The first chamber **4408** is configured to store a first cleaning solution, and the second chamber **4410** is configured to store a second cleaning solution. When an actuator **4412** of the spray bottle **4400** is actuated, the first and second cleaning solutions are urged (e.g., using a manual or electrical fluid pump) from the first and second chambers **4408**, **4410**, respectively, and distributed to a surface to be cleaned via the spray nozzle **4404**. Once deposited on the surface to be cleaned, the cleaning solutions are configured to be agitated into the surface to be cleaned using the agitator **4406**.

[0640] In an exemplary implementation, the first cleaning solution is different from the second cleaning solution. In such an implementation, the first and second cleaning solutions may be configured to be reactive with each other upon mixing to enhance a cleaning performance.

[0641] FIGS. **94A** and **94B** illustrate another implementation of a spray bottle **4500** for spraying cleaning liquid for use with a handheld surface cleaner, such as any of the handheld surface cleaners described herein. The spray bottle **4500** of FIGS. **94A** and **94B** is generally configured and used similar to the spray bottle **4000** of FIGS. **73-78**. The spray bottle **4500** in this illustrated implementation includes a spray head **4502** and a dual cleaning solution container **4504** configured to removably attach to the spray head **4502**. The dual cleaning solution container **4504** includes two separate containers **4504a**, **4504b** attached together. FIGS. **94C-94E** show one of the containers **4504a** as a standalone element and is representative of both containers **4504a**, **4504b**.

[0642] The subject matter described herein can be implemented in analog electronic circuitry, digital electronic circuitry, and/or in computer software, firmware, or hardware, including the structural means disclosed in this specification and structural equivalents thereof or in combinations of them. The subject matter described herein can be implemented as one or more computer program products, such as one or more computer programs tangibly embodied in an information carrier (e.g., in a machine-readable storage device), or embodied in a propagated signal, for execution by, or to control the operation of, data processing apparatus (e.g., a programmable processor, a computer, or multiple computers). A computer program (also known as a program, algorithm, software, software application, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond

to a file. A program can be stored in a portion of a file that holds other programs or data, in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub-programs, or portions of code).

[0643] The processes and logic flows described in this specification, including the method steps of the subject matter described herein, can be performed by one or more programmable processors executing one or more computer programs to perform functions of the subject matter described herein by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus of the subject matter described herein can be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit).

[0644] Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processor of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for executing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. Information carriers suitable for embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, (e.g., EPROM, EEPROM, and flash memory devices). The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0645] The techniques described herein can be implemented using one or more modules. As used herein, the term “module” refers to computing software, firmware, hardware, and/or various combinations thereof. At a minimum, however, modules are not to be interpreted as software that is not implemented on hardware, firmware, or recorded on a non-transitory processor-readable recordable storage medium (i.e., modules are not software per se). Indeed “module” is to be interpreted to always include at least some physical, non-transitory hardware such as a part of a processor or computer. Two different modules can share the same physical hardware (e.g., two different modules can use the same processor). The modules described herein can be combined, integrated, separated, and/or duplicated to support various applications. Also, a function described herein as being performed at a particular module can be performed at one or more other modules and/or by one or more other devices instead of or in addition to the function performed at the particular module.

[0646] A feature described with respect to one implementation may also be incorporated into other implementations. For example, a handheld surface cleaner can include a duckbill valve upstream of the separator, e.g., similar to the duckbill valve **76** shown in FIGS. **2**, **3**, and **6**. For another example, a separator can include a rim, e.g., similar to the rim **1502** of FIGS. **50A-50C**. For yet another example, an illustrated implementation of a handheld surface cleaner that includes inner and outer separators can instead include a singular separator. For still another example, an illustrated implementation of a handheld surface cleaner that includes a singular separator can instead include inner and outer separators. For yet another example, a handheld surface cleaner can include a debris quantity sensor. For another example, a self-drying cycle can be run for a handheld surface cleaner.

[0647] One skilled in the art will appreciate further features and advantages of the devices, systems, and methods based on the above-described embodiments. Accordingly, this disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated herein by reference in their entirety for all purposes.

[0648] The present disclosure has been described above by way of example only within the context of the overall disclosure provided herein. It will be appreciated that modifications within the spirit

and scope of the claims may be made without departing from the overall scope of the present disclosure.

Claims

1. A cleaning system, comprising: a surface cleaner comprising: an inlet through which debris and air is configured to be suctioned into the surface cleaner, a separator proximal to the inlet, having an outlet opening, having a plurality of holes formed in a sidewall of the separator, and having a plurality of covers, each one of the plurality of covers being associated with at least one of the plurality of holes, and a motor configured to drive rotation of the separator; wherein the rotation of the separator is configured to: cause the plurality of covers to move from a closed position, in which each of the plurality of covers closes the associated at least one of the plurality of holes, to an open position, in which each of the plurality of covers does not close the associated at least one of the plurality of holes, separate the debris from the air by directing the debris radially outward through the plurality of holes, and direct the air out of the separator through the outlet opening; and the rotation of the separator stopping is configured to: cause the plurality of covers to move from the open position to the closed position.
2. The cleaning system of claim 1, wherein each of the plurality of covers in the closed position provides a fluid seal of the associated at least one of the plurality of holes.
3. The cleaning system of claim 1, wherein each of the plurality of covers is attached to the separator along a single edge of the cover.
4. The cleaning system of claim 1, wherein each of the plurality of covers is flexible.
5. The cleaning system of claim 4, wherein the separator is rigid.
6. The cleaning system of claim 1, wherein the outlet opening is at a proximal end of the separator; and the separator includes an inlet opening at a distal end of the separator, and the inlet opening is configured to allow the debris and air are allowed to enter the separator through the inlet opening.
7. The cleaning system of claim 1, wherein the sidewall of the separator extends from the inlet opening to the outlet opening.
8. The cleaning system of claim 1, wherein the surface cleaner further comprises a tank having a cavity configured to contain and store therein the debris that is directed radially outward through the plurality of holes.
9. The cleaning system of claim 8, wherein the surface cleaner further comprises an air exit hole configured to allow the air that is directed out of the separator through the outlet opening to exit to external atmosphere.
10. The cleaning system of claim 1, wherein the surface cleaner further comprises at least one of: a first filter upstream of the separator, and a second filter downstream of the separator.
11. The cleaning system of claim 1, wherein each one of the plurality of covers is associated with only one of the plurality of holes.
12. A separation apparatus, comprising: a separator comprising: a body; an inlet opening through which debris and air is configured to enter the separator, an outlet opening through which the air is configured to exit the separator, a plurality of holes formed in a sidewall of the body and through which the debris is configured to exit the separator, and a plurality of covers, each one of the plurality of covers being configured to move from a sealed configuration, in which each of the plurality of covers seals closed at least one of the plurality of holes, to an unsealed configuration, in which each of the plurality of covers does not seal closed the associated at least one of the plurality of holes.
13. The separation apparatus of claim 12, wherein: rotation of the separator is configured to cause the plurality of covers to move from the sealed configuration to the unsealed configuration; and the rotation of the separator stopping is configured to cause the plurality of covers to move from the unsealed configuration to the sealed configuration.

- 14.** The separation apparatus of claim 13, further comprising a motor configured to drive the rotation of the separator.
- 15.** The separation apparatus of claim 12, wherein each of the plurality of covers is attached to the body along a single edge of the cover.
- 16.** The separation apparatus of claim 12, wherein the separator further comprises a plurality of blades positioned downstream of the inlet opening and upstream of the outlet opening.
- 17.** The separation apparatus of claim 12, further comprising a surface cleaner that comprises the separator; the surface cleaner further comprises an inlet through which the debris and the air is configured to be suctioned into the surface cleaner; and the separator is downstream of the inlet.
- 18.** The separation apparatus of claim 17, wherein the surface cleaner further comprises a motor configured to drive rotation of the separator.
- 19.** The separation apparatus of claim 17, wherein the surface cleaner further comprises a tank having a cavity configured to contain and store therein the debris that enters the separator; with the plurality of covers in the sealed configuration, an interior of the separator is not in fluid communication with the cavity through the plurality of holes; and with the plurality of covers in the unsealed configuration, the interior of the separator is in fluid communication with the cavity through the plurality of holes.
- 20.** The separation apparatus of claim 17, wherein the surface cleaner is configured to be handheld.
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